

Bringing Services Closer to People? Distance to Administrators and the Reach of the State*

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Abstract

Looking to improve public service delivery, many developing countries have increased the number of subnational administrative units over the past three decades. A key dimension of this strategy is spatial: smaller administrative units reduce the physical distance between households and local administrators. Leveraging an episode of district proliferation in Uganda, this paper studies whether geographical proximity to local government administrative centers affects the provision of public goods and services. To establish causality, I exploit a policy implementation rule that introduced exogenous variation in households' proximity to new district headquarters. The main findings show that households living closer to new administrative centers have significantly better access to electricity, roads, health facilities, and water sources. These improvements are accompanied by a shift from contentious to participatory engagement with the state and lower direct experiences of corruption. Despite these gains, proximity does not translate into higher trust in government or increased democratic perceptions. These findings suggest that in hybrid regimes, bringing the state closer to citizens can improve services, engagement with the government, and accountability while leaving the underlying political relationship between citizens and the state unchanged, a pattern that holds even in politically aligned districts where engagement is stronger.

Keywords: Distance to Government, Public Service Delivery, Citizen Engagement, Decentralization, Administrative Unit Proliferation.

JEL codes: D72, D73, R12, O13, H77.

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1 Introduction

Governments across the globe face several challenges in providing public goods and services, particularly in developing countries where state capacity and monitoring are often weak (World Bank, 2016). Theories of decentralization stress that bringing the government closer to citizens can improve information about heterogeneous local preferences and foster engagement with the state (Tiebout, 1956; Oates et al., 1972; Alesina & Spolaore, 1997; Faguet, 2023). Over the past three decades, many developing countries have expanded the number of subnational administrative units through redistricting, explicitly aiming to improve public service delivery. The evidence suggests that these administrative splits can enhance local development by better capturing local preferences and improving resource allocation (Dahis & Szerman, 2024), particularly when smaller polities strengthen political accountability (Narasimhan & Weaver, 2024).

A central dimension of these reforms is spatial: decreasing the size of administrative units reduces the physical distance between citizens and local administrations. If geographical distance constrains interactions between citizens and the state (Asher et al., 2018), shorter distances may increase policymakers' responsiveness and the quality of public services. Yet the benefits of proximity may be offset if creating new units eliminates economies of scale, intensifies local elite capture (Oates et al., 1972), or creates administrative capacity constraints (Cohen, 2024). Moreover, in hybrid regimes decentralization can be used to extend central control rather than to empower citizens (Martinez-Bravo et al., 2022). Because the location of administrative offices is typically endogenous to historical settlement patterns, infrastructure, and preexisting services, isolating the causal impact of distance to administrative headquarters on public goods provision remains empirically challenging.

This paper studies whether geographical distance to local government administrative centers affects the provision of public goods and services in the context of administrative unit proliferation in Uganda. This country provides a relevant setting for several reasons. First, Uganda has gone through a large decentralization process, resulting in a rapid expansion of sub-national administrative units (e.g., the number of districts has grown from 38 in 1991 to 136 in 2020). Second, the national government established rules for the geographical location of new administrative headquarters, providing exogenous variation in distance to administrative centers. Third, the availability of geospatial data, combined with comprehensive data on households' access to and satisfaction with public goods provision, offers a unique opportunity to gain a broad overview of the effects of distance to administrators on public service delivery in a redistricting context.

To establish causality, I use an instrumental variable approach leveraging an implementation rule in the process of district creation, which introduced exogenous variation in distance to the new district headquarters. Specifically, governmental guidelines established that district headquarters should be located in the geographical middle of newly created districts (Green, 2010).

Building on [Campante & Do \(2014\)](#), I use the distance from each household's enumeration area to its district's centroid, interacted with an indicator for districts created between 1991 and 2020, as an instrument for the household's distance to the district capital. To implement this strategy, I construct a granular spatial dataset that combines *(i)* rich household microdata on access to public services from the 2021 National Service Delivery Survey (NSDS) and the 2024 Afrobarometer, *(ii)* geospatial data and administrative records identifying newly created districts between 1991 and 2020, and *(iii)* geocoded distances from each household's enumeration area to its district's headquarters and centroid. I complement the main analysis with the 2015 NSDS and the 2017 Afrobarometer to examine how the effects of proximity evolved over time, as new district institutions became more established. Together, these data allow me to study how distance to the district's administrative center affects households' access to public goods and services, as well as their relationship with the state.

The findings show that geographical proximity to new district headquarters significantly improves access to public goods and services. Households closer to the district capital have better access to electricity and paved roads, shorter travel times to health facilities, and higher access to drinking water. The effects are strongest for centrally provided services, reflecting both direct infrastructure investments near new headquarters and the coordinating role of district administrators in channeling national programs to nearby communities. Using two rounds of a national service delivery survey, I also document that these effects strengthened between 2015 and 2021. While only agricultural inputs responded to proximity in 2015, by 2021 significant gains had emerged for health, water, and electricity access, consistent with the gradual consolidation of administrative capacity in newly created districts.

Various tests provide empirical support for the identification strategy. First, I show that headquarters locations within new districts are not predicted by preexisting population density or night-time lights, alleviating concerns that they systematically target more developed areas. Second, I document that land-use dynamics in locations selected as new headquarters do not diverge from comparable locations within the same district prior to treatment, suggesting limited strategic placement. Third, I show that the instrument is not driven by proximity to large towns or major urban centers. Fourth, I confirm that proximity to district headquarters has no effect on households' knowledge of or distance to lower-tier local government institutions (LC1, LC2, LC3), establishing that the policy-induced variation is specific to the district level.

The results are also robust to alternative distance measures based on parish centroids, to a GIS based measure of households distance to district headquarters, to restricting the sample to newly created districts, to excluding households located near headquarters towns, and to placebo tests using outcomes unrelated to headquarters proximity. Finally, I address the concern of non random exposure to the instrument, since a household's fixed position within the geometry of its original district may partly predict its instrument value ([Borusyak & Hull, 2023](#)). Controlling for the expected instrument, computed across feasible counterfactual redistricting configu-

rations, leaves the estimates stable or larger. Randomization inference over 1,000 placebo configurations, further confirms the main service delivery results.

I leverage a unique dataset on service delivery to study two potential mechanisms behind these results: citizen engagement and accountability. On engagement, proximity increases state presence by raising citizens' knowledge of district level institutions and reducing reliance on traditional alternatives. As the state becomes more accessible, contentious engagement falls significantly: households closer to headquarters are less likely to join protests or contact the media. The evidence on participatory engagement is more suggestive, but the coefficients consistently point toward greater involvement in local decision making. Satisfaction with the Resident District Commissioner also increases significantly, suggesting that the decline in contentious engagement may partly reflect a shift toward formal participatory channels. On accountability, proximity reduces direct corruption experiences even as awareness of corruption remains high. This is consistent with a monitoring mechanism in which proximity makes public officials more observable, deterring corrupt behavior without eliminating awareness of its existence.

Despite these improvements, proximity does not translate into higher trust in the government or more favorable democratic perceptions. Satisfaction with democracy, perceived freedom to vote and organize, rule of law evaluations, and trust in local and national institutions are all largely unaffected. Interestingly, while political trust is unchanged, proximity does increase social capital. Households closer to headquarters are more likely to trust people of other religions and ethnicities and to engage in collective action to solve local problems, suggesting that the state getting closer builds community cohesion without generating political legitimacy.

Heterogeneity by political alignment between the elected district chairperson and the centrally appointed Resident District Commissioner sheds further light on these findings. In aligned districts, proximity delivers stronger service delivery gains, higher participatory engagement, and lower corruption experiences, yet democratic perceptions remain unaffected even in this more favorable environment. In non-aligned districts, proximity effects to new districts headquarters are more limited, and it is associated with declining trust in local government and the ruling party, suggesting that administrative proximity can reveal institutional frictions. Together, these results indicate that in hybrid regimes, bringing the state closer can improve services and accountability while leaving the underlying political relationship between citizens and the state unchanged, and could undermine it when local political incentives are misaligned.

Finally, I quantify the fiscal cost of this proliferation strategy. I aggregate intergovernmental transfers to fixed territories defined by the 1991 district borders and estimate an staggered difference in differences model showing that real transfers rise by roughly 14 percent once a territory is first split. This increase is concentrated in duplicated administrative payroll rather than in conditional service delivery grants. The composition of these additional resources suggests that the service gains are not mechanically driven by larger delivery budgets. At the

same time, it reveals the limit of the strategy. Each split expands the administrative apparatus, so bringing the state closer becomes increasingly costly with every additional district.

This paper contributes to three strands of literature. First, it adds to the emerging literature on administrative unit proliferation. Seminal contributions in this area study why central governments create new subnational units during decentralization processes, emphasizing motives such as patronage (Green, 2010), gerrymandering (Awortwi & Helmsing, 2014), parliamentary and legislative control (Resnick, 2017; Green, 2010), and recentralization (Lewis, 2018). This work highlights that episodes of administrative unit proliferation reflect a confluence of national executives' objectives and local elites' and citizens' demands (Grossman & Lewis, 2014; Pier-skalla et al., 2019). A smaller set of studies examines the consequences of proliferation for service delivery and finds mixed effects. Cross-country panel evidence shows that more subnational units are on average associated with higher service provision, with decreasing returns as the number of units grows (Grossman et al., 2017). Country-level difference-in-differences analyses report that administrative splits increased service delivery in Brazil (Dahis & Szerman, 2024) and service delivery inputs in Uganda (Cohen, 2024), but had adverse effects in Burkina Faso (Billing, 2019). This paper contributes by exploiting a natural experiment to estimate the causal effect of geographical distance to administrative centers, within a context of rapid district proliferation, on the provision of public goods and services. Unlike existing work, I focus on distance to newly created district headquarters and explore citizen engagement with government and accountability as mechanisms in Uganda, a setting where redistricting is widespread and policy-relevant for many African countries.

Second, the paper relates to the literature on spatial inequality and local development. This work has emphasized the role of proximity to urban centers and markets in shaping labor outcomes (Fafchamps & Shilpi, 2005; Fafchamps & Wahba, 2006), specialization (Fafchamps & Shilpi, 2005), and economic growth (Jacoby, 2000; Michaels et al., 2012; Storeygard, 2016). In contrast, the effects of proximity to administrative centers have received less attention. A notable exception is Asher et al. (2018), who document that greater distance to administrative headquarters reduces access to public goods and worsens economic outcomes in India. I add to this literature by studying the effects of proximity to newly created local government centers in the context of redistricting. Given the rapid expansion of local government units in Uganda, the findings shed light on the importance of administrative centers in initially remote rural areas.

Lastly, this paper contributes to the literature on decentralization and the provision of public goods and services, with a particular focus on hybrid regimes. Existing work emphasizes that greater local autonomy can help tailor policies to heterogeneous local preferences and improve service delivery (Oates et al., 1972; Alesina & Spolaore, 1997; Gadenne & Singhal, 2014; Faguet, 2004; Galiani et al., 2008; Kis-Katos & Sjahrir, 2017; Canavire-Bacarreza et al., 2020). More recent research shows that, in non-democratic or hybrid settings, decentralization reforms can be strategically designed to extend central control and weaken local accountability

rather than empower citizens (Martinez-Bravo et al., 2022), and that citizens in such contexts respond less to government performance in terms of trust in the government (Besley et al., 2026). This paper contributes by showing that decentralization-induced increases in access to public goods and services do not necessarily translate into higher trust in the government or increased democratic perceptions in hybrid regimes.

The remainder of the paper is organized as follows. Section 2 describes the institutional background. Section 3 summarizes the conceptual framework guiding the empirical analysis. Section 4 details the data construction and displays descriptive statistics for the main variables. Section 5 outlines the empirical framework. Section 6 presents the main results, while sections 7 – 11 detail the mechanisms, effects on democratic perceptions, heterogeneities, dynamic effects, and fiscal costs, respectively. Section 12 concludes.

2 Institutional Background

The decentralization process in Uganda began in 1987 with the establishment of a five-tier administrative structure consisting of villages (LC1), parishes (LC2), sub-counties (LC3), counties (LC4), and districts (LC5), with districts being the largest administrative unit. While each level has distinct operational responsibilities, local government councils, which serve as the highest political authority within their jurisdictions, operate specifically at the district and sub-county levels. These councils typically consist of a chairperson, vice-chairperson, and five councilors, though additional members may be included, such as representatives of women, youth, and people with disabilities. They are responsible for planning and implementing public service-related projects within their jurisdictions. At the district level, two key figures share administrative responsibilities: the Local Council V (LCV) chairperson, who is directly elected by local voters and leads the district council, and the Resident District Commissioner (RDC), who is appointed by the President and represents the central government within the district. While the LCV is accountable to local constituents, the RDC is accountable upward to the national government, and the degree of political alignment between these two figures can shape how effectively district resources are coordinated and services delivered.

The constitutional framework for decentralization and devolution of powers in Uganda was established in the 1995 Ugandan Constitution and the 1997 Local Governments Act. Notably, the 1995 constitution granted the central government the authority to create sub-national units. Article 179 states that: “Parliament may: (a) alter the boundaries of districts; and (b) create new districts [...] Any measure for the alteration of the boundaries of the creation of districts or administrative units shall be based on the necessity for effective administration and the need to bring services closer to the people” (GoU, 1995). This marked a turning point in the decentralization policy, as, since 1997, district creation increased substantially from 39 districts in 1994 to 112 in 2014, reaching a total of 136 districts in 2020.¹

¹The literature does not have a consensus on the underlying motives for splitting a district. While Green (2010)

The 1997 Local Governments Act define the division of service delivery responsibilities between district governments and the central state (GoU, 1997). Districts are mandated to provide primary education, primary health care through health centers and district hospitals, rural water supply for domestic use, agricultural extension services, district and feeder roads, and community development services (Ahmad et al., 2009). The central government is responsible for the provision of services with national reach such as the national and trunk-road network, electricity transmission and rural electrification, regional and national referral hospitals, and large-scale water infrastructure (Mushemeza, 2019). This institutional structure motivates the categorization of outcomes by the level of government primarily responsible for delivery, distinguishing between services provided by district governments and those under central mandate.

The legal procedure for creating local government units involves several stages. First, local residents, through their council representative, present a request to the district council. If approved, a resolution to create a new district is submitted to the Ministry of Local Government. After approval, the proposal is referred to the Cabinet for its evaluation. If the Cabinet supports the request, it is then submitted to Parliament for final approval (Awortwi & Helmsing, 2014). Upon successful completion of the process, the district issuing the request is split, allowing the splitter district to establish a new administrative system. Once a district is approved, the new administrative unit is authorized to recruit over 200 technical and administrative staff, a process that has been linked to patronage networks (Green, 2010) and electoral incentives, including vote-buying (Awortwi & Helmsing, 2014). Importantly, the governmental guidelines establish that new districts should locate their headquarters in the geographical middle of the newly created district (Green, 2010).²

According to the Ugandan Constitution, the main objectives of creating new administrative units are political devolution and improving public service delivery. Political devolution aims to empower leaders at lower administrative levels, enabling them to manage local affairs more effectively (Mushemeza, 2019). This process is legitimized by local elections held every four years, which have witnessed significant participation in recent periods. Simultaneously, the creation of new administrative units seeks to enhance public service delivery through a bottom-up approach, where local governments provide opportunities for local participation, leading to better utilization of available resources (Mushemeza, 2019).

3 Conceptual Framework

As described in the previous section, the creation of new districts in Uganda introduces two administrative figures with different accountability relationships operating simultaneously at

argues that there is no concrete evidence that districts have been created in deprived areas, Grossman & Lewis (2014) suggests that districts with politically marginalized minorities were more likely to be split.

²Fieldwork interviews with district officials across twelve districts confirm that the centrality norm is a consistent practice. Officials in every district visited stated that the council deliberately chose a central location to guarantee equal access for local leaders and citizens attending meetings at the headquarters.

the district level: the elected LCV chairperson, who is accountable to local voters, and the centrally appointed RDC, who represents the national government. The geographical placement of the new district headquarters determines the physical distance between households and these administrators. This section summarizes the conceptual framework that guides the empirical analysis, describing how proximity to the administrative center and political alignment between the LCV and RDC jointly shape service delivery, citizen engagement, and accountability. A formal model developing these predictions is presented in Appendix B.³

Provision of public goods and services. Administrative effort translates into household-level access to public goods and services. Administrators choose an overall level of investment for the district, but the household-level realization of that investment depends on distance to the headquarters. Infrastructure radiates outward from the administrative center: electricity grids, paved roads, health facilities, and water sources are more accessible near the headquarters. While the marginal welfare gains from public goods may be largest in remote, underserved areas, the spatial decay of delivery means these gains are the hardest to realize. This argument motivates creating new headquarters closer to previously remote areas.

The degree of political alignment between the LCV and RDC affects the total effort directed toward locally valued services. When both administrators share political affiliation, their incentives are complementary: the elected LCV channels local preferences into service delivery decisions while the appointed RDC coordinates the allocation of centrally financed programs, and both pull in the same direction. When they are not aligned, the RDC's career concerns and political loyalty rewards may be larger than local welfare, reducing the effort the RDC devotes to services valued locally. Importantly, distance and alignment interact. Proximity amplifies the returns to any effort directed at local services, and alignment ensures that a larger share of total effort is so directed. This complementarity predicts that the effects of proximity on service delivery should be stronger in politically aligned districts.

Citizen engagement. The quality of public service delivery depends not only on administrative effort but also on the information that administrators receive about local needs. A central insight of the decentralization literature is that citizens have an informational advantage regarding local conditions and the performance of local officials (Björkman & Svensson, 2009). How effectively citizens can exercise this advantage depends on how they choose to engage with the state, which is shaped by the costs they face.

Citizens can engage through participatory channels, such as attending district council meetings, meeting the RDC, or using formal feedback mechanisms, which transmit detailed information about local priorities directly to decision-makers but require traveling to the headquarters. Alternatively, they can engage through contentious channels, like joining protests, contacting the media, or organizing collective pressure campaigns, which are noisier but can

³The theoretical model follows the model presented in the working paper version of Martinez-Bravo et al. (2022).

be pursued locally at costs that do not depend on distance. This cost structure implies that proximity reduces the relative cost of participatory engagement, shifting citizens from contentious to formal channels. The pattern is one of substitution: proximity does not suppress citizen voice but rather redirects it toward channels that are more informative for administrators. Political alignment further shapes this choice. When the RDC internalizes local welfare, the state is more responsive to citizen input, increasing the returns to participatory engagement. In non-aligned districts, the lower responsiveness of the state reduces these returns, and citizens may not shift away from contentious channels even when the headquarters is nearby. Together, these mechanisms predict that proximity should reduce contentious engagement and increase participatory engagement, and that this shift should be more pronounced in aligned districts.

Accountability. A further channel through which proximity affects service delivery is the monitoring of local officials. Citizens who live closer to the headquarters interact more frequently with district institutions and are better positioned to observe and report misconduct, raising the expected cost of corruption for administrators. Two forces reinforce each other in deterring corrupt behavior: external monitoring by citizens, which increases with proximity, and internal alignment with local welfare, which increases with political congruence between the LCV and RDC. In aligned districts, the RDC internalizes the welfare cost of resource diversion, making corruption less attractive even when citizen monitoring is imperfect. These mechanisms predict that proximity should reduce corruption experiences and that the reduction should be stronger in politically aligned districts.

Taken together, the framework identifies three reinforcing channels (provision, engagement, and accountability) through which proximity to the administrative center improves public service delivery. All three channels are amplified by political alignment between the elected and appointed district leaders. The main results in Section 6 test the distance predictions, while Sections 7 examine the engagement and accountability channels. The alignment predictions are tested in the heterogeneity analysis of Section 9, which splits the sample by political congruence between the LCV chairperson and the RDC.

4 Data

To study the effects of the geographical distance between citizens and public administrators in the Ugandan redistricting context, I combine various sources of geolocalized secondary data, as well as earth observation data. This section describes the data sources and construction as well as the main variables of interest.

4.1 Sources and Dataset Construction

4.1.1 Household Survey Data

The main data source is the 2021 National Service Delivery Survey (NSDS), conducted by the Uganda Bureau of Statistics on behalf of the Ministry of Public Services to assess the efficiency and effectiveness of service delivery from the perspectives of both providers and recipients (UBOS, 2022). The survey covers 10,246 households across 1,088 enumeration areas in all 136 districts as of 2020, of which 9,313 form the sample for analysis after excluding households with insufficient geolocalization data and households belonging to Kampala and its metropolitan area Wakiso.⁴ The survey provides rich household and community-level information on education, health, water, electricity, roads, and agricultural inputs, covering services provided by both district and central governments. I complement the analysis using the 2015 wave of the NSDS to examine how the effects of proximity evolve over time. Detailed variable definitions and index construction are described in Appendix A.

Measures of citizen engagement, corruption experiences, and democratic perceptions come from the tenth round of the Afrobarometer survey, conducted in Uganda in 2024. The Afrobarometer is a nationally representative survey covering 2,400 respondents in more than 270 enumeration areas, providing detailed information on citizens' interactions with government institutions, experiences of bribery and petty corruption, satisfaction with democracy, perceptions of the rule of law, and trust in local and national institutions. While the NSDS offers broader household coverage and richer service delivery measures, the Afrobarometer provides attitudinal outcomes that are not available in the NSDS, making the two surveys complementary.

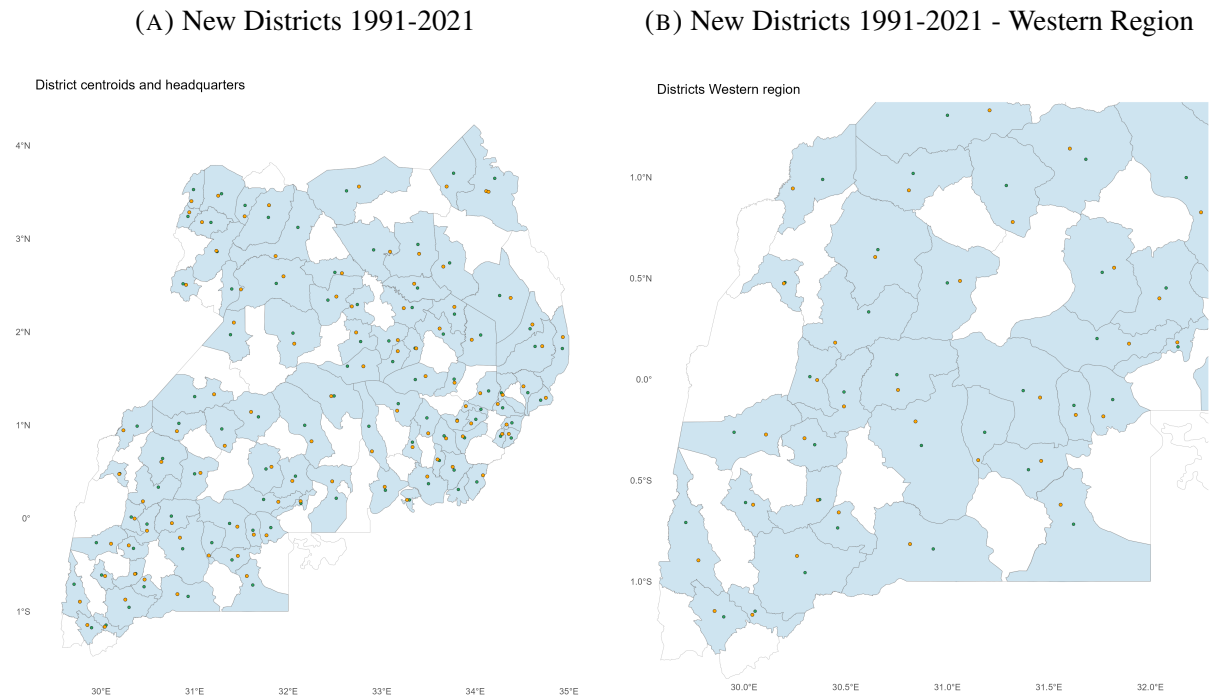
4.1.2 Distance Variables

To assess the impact of a household's proximity to the district's administrative center on public service delivery, I leverage a government policy mandating district headquarters to be located near the district's geographical center. However, in practice, headquarters placement may be influenced by additional factors such as road access, proximity to other constituencies, and availability of infrastructure. These factors could introduce bias when analyzing the effects of proximity on service provision. To address this, I compute the straight-line distance from each household's enumeration area to the district's centroid, which serves as an instrument for

⁴These households are excluded as the capital city service delivery is governed by a separate institutional framework under the Kampala Capital City Authority. Additionally, as noted by Cohen (2024), Kampala is an extreme outlier across a variety of service access indicators. In addition, for eleven enumeration areas the survey's administrative district label disagrees with the district containing the enumeration area's coordinates. For nine of these, the coordinates fall inside a neighboring district carved from the same 1991 mother district, and the spatial assignment is adopted; the remaining two, whose coordinates lie implausibly far (over 100 kilometers) from the labeled district, are dropped.

household proximity to the district headquarters.⁵

FIGURE 1: SPATIAL DISTRIBUTION OF NEW DISTRICTS' HEADQUARTERS



Notes: Panel (A) displays the distribution of new districts and old districts (both mother and never split districts) for the period 1991-2020. Panel (B) shows the spatial correlation between the district's town councils and the district's centroids for the Western region.

Two key proximity variables are constructed from the household questionnaire. *SR Proximity Household – District Headquarters* is defined as the negative log of the NSDS self-reported distance (in kilometers) from the household to the district capital. *Proximity Household – District Centroid* is the negative log of the straight-line distance between the household's enumeration area geocoordinates and the district's centroid. Relying on proximity calculations using log distance throughout ensures that higher values of all proximity variables correspond to households located closer to the reference point, so that a positive coefficient in the regression analysis is consistent with the expected beneficial effects of administrative proximity.

4.1.3 Evolution of Districts

I combine geospatial data with national district registries to document geographical changes associated with decentralization and district splits between 1991 and 2020. The geospatial data, obtained from UBOS, includes the geographical borders of districts for different years, enabling the identification of district splits and distinguishing between mother and splitter districts. The geographical data is complemented with administrative records on the creation of districts for

⁵Figure 1a maps the headquarters and centroids of new districts, while Figure 1b illustrates district centroids in the Western region.

years in which district's borders are not available. This dataset enables the construction of the variable *NewDistrict*_{1991–2020}, an indicator variable that takes the value of one for newly created districts between 1991 and 2020.

4.1.4 Remote Sensing Data

I combine household and community survey data with three remote sensing products to capture dynamic development patterns across land use, economic activity, and population over time. First, I use land classification data from the MODIS Land Cover Type product, which provides annual observations at a 500-meter resolution from 2001 to 2023.⁶ From these data I construct parish-level variables capturing the share of land under cultivated cropland and cropland mixed with natural vegetation. Second, to proxy for local economic activity and urbanization, I use the harmonized DMSP-OLS/VIIRS nighttime lights for 2020. I aggregate mean radiance to the parish level. Third, I measure local population using the Gridded Population of the World (GPW v4.11) from CIESIN/SEDAC, which provides population count and density estimates at a 30-arc-second (1 km) resolution for 2020. These parish-level datasets are used as evidence of non-strategic placement of district headquarters.

4.1.5 Qualitative Fieldwork

The empirical analysis is further supported by fieldwork that was conducted in Uganda from July to August 2024. The fieldwork covered twelve districts spanning both old (created in or before 2010) and newly created (after 2010) administrative units. In each district, I conducted semi structured interviews with district-level officials, including Chief Administrative Officers, district planners, finance officers, and Resident District Commissioners (RDCs), as well as with Local Council 1 (LC1) leaders and households in towns located both near to and far from district headquarters. This design comparing officials and citizens across old and new districts and close and far towns was intended to capture qualitative variation along the dimensions exploited in the empirical analysis. The evidence gathered informs the institutional background (Section 2), motivates several robustness checks (Section 6.4), and informs the analysis of the engagement, accountability, and political alignment channels.

⁶The MODIS product classifies land cover using the IGBP scheme into 17 categories. I focus on cultivated cropland (>60% cultivated) and cropland mixed with natural vegetation (40–60% cultivation), as these classes together account for the majority of rural land cover in Uganda and provide the most informative measure of agricultural intensification and local economic activity around newly established administrative centers. Changes in built-up area are studied separately using the JRC Global Human Settlement Layer built-up surface product (GHS-BUILT-S), available at 100-meter resolution in five-year epochs from 1990 to 2025, whose long span provides genuine pre-periods for the headquarters cohorts.

4.2 Descriptive Statistics

Table 1 contains summary statistics for the final sample, detailing information on distance variables, the main service delivery-related outcomes, and mechanisms.

On average, households are 16 kilometers from their district headquarters, while the average distance from the enumeration area to the district headquarters is approximately 12 kilometers. The distance from the enumeration area to the district centroid is 14.6 kilometers, closely aligning with the self-reported measure. Under the 1991 district boundaries, these same households would have been, on average, 25.5 kilometers from their district headquarters (approximately 60 percent farther). Finally, nearly half of the households in the sample belong to districts created between 1991 and 2020, reflecting the substantial redistricting that took place over this period.

Turning to the main outcomes, access to services remain limited. While 71 percent of households with primary school age children report school attendance, and 86 percent of households that vaccinated their children in the past year did so at a government facility, infrastructure access is considerably lower. Only 16 percent of households have access to improved roads and 47 percent have electricity. Access to improved water sources during the dry season reaches 83 percent, with households traveling 0.8 kilometers on average to their water source. The outcome indices used in the main analysis aggregate these and other related measures. They are centered at zero by construction, with standard deviations between 0.41 and 0.78, reflecting meaningful variation in service access across households.⁷ Among the mechanism variables, 25 percent of households report involvement in LC5 meetings and 10 percent report having met the RDC, while direct experiences of corruption are relatively low: 3 and 7 percent of households report being victims of nepotism and absenteeism respectively. Satisfaction with the RDC is reported by 33 percent of households, and 38 percent of households know their magistrate court, suggesting moderate but heterogeneous awareness of district-level institutions.

In the Afrobarometer sample, 36 percent of respondents report engaging in collective civic action, while protest participation is considerably lower at 4 percent. Bribery for medical care is reported by 41 percent of respondents who sought health services, pointing to substantial corruption in healthcare access. Trust in local government stands at around 50 percent, while trust in the ruling party is somewhat higher at 59 percent. Finally, 74 percent of households in the sample are located in politically aligned districts, where the LCV chairperson and RDC share the same political affiliation.

⁷Appendix Table A1 lists the components included in each index.

TABLE 1: Summary Statistics

	N	Mean	SD	Min	Max
<i>Panel A: Proximity Variables</i>					
Distance HH – District Headquarters (km)	8640	16.03	14.59	0.10	90.00
Distance HH – District Headquarters 1991 (km)	8640	25.51	22.45	0.04	105.02
Distance EA – District Headquarters (km)	9315	12.31	10.45	0.05	84.90
Distance EA – District centroid (km)	9315	14.58	8.30	0.28	57.07
New district (1991–2020)	9503	0.50	0.50	0.00	1.00
<i>Panel B: Main Outcomes (NSDS 2021)</i>					
School attendance (prim. age)	4726	0.71	0.41	0.00	1.00
Gov. health facility (immunization)	2361	0.86	0.34	0.00	1.00
Improved water source (dry s.)	8738	0.83	0.37	0.00	1.00
Water distance (dry s., km)	7474	0.79	0.92	0.00	5.00
Improved roads	8731	0.16	0.37	0.00	1.00
HH has electricity	8752	0.47	0.50	0.00	1.00
Education index (district)	7811	0.00	0.73	-2.53	0.92
Health index (district)	8400	-0.01	0.49	-3.37	2.21
Water index (district)	9472	0.00	0.47	-2.29	1.19
Roads index (district)	9471	0.00	0.41	-1.42	2.26
Agricultural inputs index (district)	8752	0.03	0.63	-0.47	2.40
Education index (central government)	6009	0.00	0.73	-2.26	0.83
Health index (central government)	9315	0.03	0.78	-0.89	2.43
Water index (central government)	9436	-0.01	0.53	-7.89	0.62
Roads index (central government)	9471	-0.06	0.73	-2.52	1.08
Electricity index (central government)	9482	-0.03	0.61	-2.26	1.92
<i>Panel C: Mechanisms (NSDS 2021)</i>					
Involved in LC5 meetings	8752	0.25	0.43	0.00	1.00
Household meets with RDC	8752	0.10	0.30	0.00	1.00
Victim of nepotism	8752	0.03	0.18	0.00	1.00
Victim of absenteeism	8752	0.07	0.26	0.00	1.00
Satisfied with RDC performance	8752	0.33	0.47	0.00	1.00
Family uses community services	8728	0.06	0.23	0.00	1.00
Knows magistrate court	8729	0.38	0.48	0.00	1.00
<i>Panel D: Mechanisms (Afrobarometer Round 10)</i>					
Civic and collective action	2182	0.36	0.48	0.00	1.00
Engagement in protest	2184	0.04	0.19	0.00	1.00
Bribe for medical care	1604	0.41	0.49	0.00	1.00
Trust in local government	2132	0.51	0.50	0.00	1.00
Trust in ruling party	2184	0.59	0.49	0.00	1.00
<i>Panel E: Heterogeneity</i>					
Aligned in 2016 - 2021	9524	0.74	0.44	0.00	1.00

Notes: Summary statistics for the main variables used in the analysis. Panels A–C use NSDS 2021 data. Panel D uses Afrobarometer Round 10 data.

5 Empirical Framework

The impacts of geographical proximity to governmental administrative centers on public service delivery are challenging to measure due to the endogenous decisions on the geographical placement of administrative offices. The location of bureaucratic centers is generally determined based on various criteria such as population density, regional economic activity, urbanization, access to services, and road availability, among others. In addition, a household's remoteness is generally likely to be correlated with many unobserved household and contextual characteristics, which themselves could affect the outcomes of interest. The resulting challenge is that the distance to bureaucratic centers could typically capture many different factors unrelated to the accessibility of public service delivery. To address these challenges, this paper leverages quasi-random variation in headquarters placement resulting from the redistricting process in Uganda, which is governed by strict rules regarding the placement of new district headquarters, combined with various sources of geolocalized secondary data.

Throughout the analysis, the key variable of interest is *proximity* to the district headquarters. In this sense, a positive estimate implies that households located closer to the district capital experience better public service delivery outcomes, consistent with the government's stated objective of bringing services closer to citizens through district creation.

The conceptual framework in Section 3 distinguishes between two administrators with different accountability structures operating at the district level: the elected LCV chairperson, responsible for coordinating district-mandated services, and the appointed RDC, responsible for channeling centrally financed programs. To map this distinction to the empirical analysis, I construct outcome indices separately for services under district mandate (primary education, district health facilities, rural water supply, district and feeder roads, and agricultural extension and inputs) and services under central government mandate (secondary education, national and referral hospitals, large-scale water infrastructure, the national roads network, and electricity). This separation allows us to further understand the role of the distance gradient in service provision at different levels, as well as heterogeneities in service provision for both levels by political alignment.

The identification strategy exploits a natural experiment triggered by governmental guidelines that specify the location of new district headquarters. According to these guidelines, the headquarters should be located in the geographical middle of newly created districts (Green, 2010). Consequently, the creation of a district exogenously changes the proximity to the district's headquarters for households belonging to the splitter district, as the geographical shape of the new district determines the new location of its headquarters. Following Campante & Do (2014), a household's proximity to the district's centroid can be used as an instrument. Specifically, I interact this exogenous variation in proximity to the district's centroid with an indicator variable for newly created districts between 1991 and 2020 to account for confounding effects arising from pre-split remoteness. Formally, the instrumental variable first-stage and

second-stage are given by:

$$\begin{aligned} ProximityHH_HQ_{i,d} = & \gamma_0 + \gamma_{DN}(ProximityHH_DC \times NewDistrict_{1991-2020})_{i,d} \\ & + \gamma_D ProximityHH_DC_i + \gamma_N NewDistrict_{1991-2020,d} + \mathbf{X}'_{i,d} \gamma_X + v_{i,d} \end{aligned} \quad (1)$$

$$\begin{aligned} Y_{i,d} = & \alpha + \beta \widehat{ProximityHH_HQ}_{i,d} + \beta_D ProximityHH_DC_i \\ & + \beta_N NewDistrict_{1991-2020,d} + \mathbf{X}'_{i,d} \beta_X + \varepsilon_{i,d}, \end{aligned} \quad (2)$$

where $ProximityHH_DC_i$ denotes a household's enumeration area proximity to the district's centroid in logarithms, the variable $NewDistrict_{1991-2020,d}$ indicates if district d has been created between 1991 and 2020, $(ProximityHH_DC \times NewDistrict_{1991-2020})_{i,d}$ is the excluded instrument, $ProximityHH_HQ_{i,d}$ represents the self-reported proximity to the district's headquarters for household i of district d , $\mathbf{X}'_{i,d}$ is a vector that denotes a set of controls.⁸ Finally, $v_{i,d}$ is the error term clustered at the parish level.

The second-stage estimates, as shown in Equation 2, capture the causal effects of households' proximity to the district's headquarters on a wide variety of public service delivery indicators, including education, health, water, electricity, roads, and agricultural inputs denoted by $Y_{i,d}$. The coefficient of interest, β has a straightforward interpretation: a positive estimate indicates that households located closer to the district capital have better public service delivery outcomes, consistent with the hypothesis that administrative proximity enhances the availability and quality of public services.

For instrumental variables estimates to be interpreted causally, three main assumptions must hold: relevance, independence, and exclusion restriction. The relevance assumption holds as long as policymakers comply with the governmental guidelines. If this is the case, households' proximity to a district's centroid should increase the likelihood of being closer to the district's headquarters. This assumption can be tested through the first-stage estimations.

The independence assumption is based on the arbitrariness of the new district's centroid. As the location of a district's centroid depends exclusively on the geographical shape of the new administrative unit, it is unlikely to be related to unobservable characteristics that determine public service delivery outcomes, conditional on the set of controls. In other words, the new districts' centroids are located as good as randomly since they only depend on the respective district's shape, which is unlikely to be related to potential confounders.

For the exclusion restriction to hold, the interaction between a household's distance to the

⁸Controls include district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's distance to district's capital as of 1991 (in log.), presence of lakes other than Victoria, sub-county mean elevation (in m.a.s.l.), four major regions.

district's centroid and newly created districts between 1991 and 2020, conditional on the set of controls, should affect public service delivery outcomes only through the household's distance to the district headquarters. This assumption is likely to be fulfilled as it appears plausible that a household's distance to the new district's centroid does not affect any relevant outcomes in and of itself (Campante & Do, 2014) once the distance to the old district centroid is controlled for. The use of distance variation exclusively arising from new districts addresses potential problems associated with pre-existing remoteness before the splits (i.e., houses that were remote both before and after belonging to a new district). However, the exclusion restriction could be violated if the district-splitting procedure is not considered, as this process is unlikely to be performed at random. For this reason, all specifications include controls for split districts between 1991 and 2020. Additionally, controlling for the presence of water bodies, elevation, and macro-regions capture any potential confounders coming from geographical factors. As the size of the district could also indirectly affect the decision of splitting the administrative unit, introducing a potential violation of the exclusion restriction, and affecting the distance to the district's centroid, specifications control for the size of the district at the beginning of the period (as of 1991). Similarly, controlling from the households distance to districts capitals as of 1991 captures potential confounders coming from differential proximity to urban centers.

While it is plausible that the exclusion restriction holds, there are some latent threats to the instrument's validity. For instance, if the geographical shape of a new district is designed to favor a specific group by locating its centroid strategically, the instrument is no longer valid. However, this situation is unlikely for several reasons. First, district creation follows a complex legal procedure involving multiple negotiation stages, reducing the possibility of manipulating the geographical shape, especially the centroid location (see Section 2). Second, the borders of newly created districts are determined by established county borders, which were drawn during colonial times, further limiting the ability to shape boundaries to favor a particular central location. Third, the composition of new districts depends on local council representatives' cooperation during the initial petition, reducing the probability of Parliament strategically selecting sub-national units to benefit specific constituencies. Fourth, section 6.4.1 shows that sub-county population density and night-lights within districts created after 2010 does not predict future headquarters placement. In other words, newly established headquarters are not strategically located in the most populated or most developed area of the new districts. This is consistent with qualitative evidence that headquarters placement is constrained jointly by the centrality norm and the availability of public land, neither of which is systematically correlated with local economic activity. In line with these results, section 6.1 shows similar trends in land-use variables in parishes before the creation of the headquarters. Finally, anecdotal evidence suggests that some new districts' headquarters offices are often built from scratch, reflecting the arbitrariness of their location (Green, 2010). As the new district creation involves several negotiation steps with different actors having conflicting incentives, it is implausible that newly created districts are established to benefit a particular group located in its geographical middle.

6 Results

6.1 Land Use Dynamics After Headquarters' Establishment

Before turning to the main results, I first examine development patterns in parishes before and after they become new district headquarters. Since the cross sectional nature of the NSDS 2021 dataset does not allow me to study these dynamics directly, I complement the analysis with remote sensing data. Specifically, I use MODIS land classification data from 2001 to 2023, available at a 500m resolution, to measure changes in land use around the designation of new district headquarters. This section therefore documents how land use evolves in parishes before and after they acquire district headquarter status.⁹

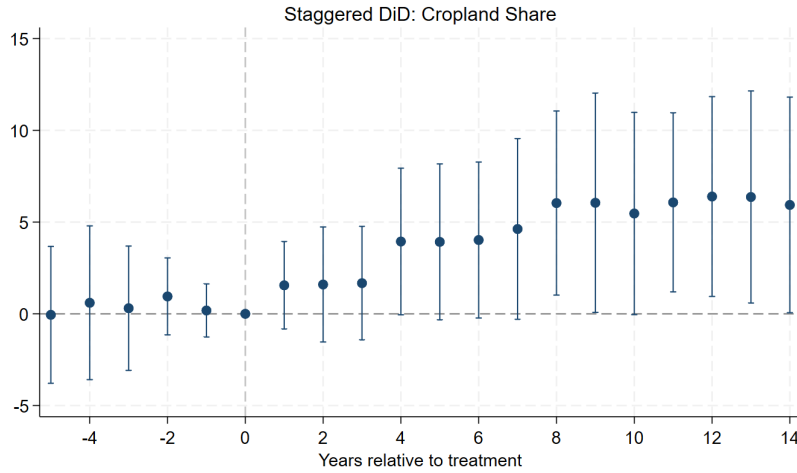
To account for heterogeneous treatment effects over time, I follow [De Chaisemartin & d'Haultfoeuille \(2020\)](#)'s staggered difference-in-differences estimator at the parish level. Figure 2 presents an event study graph showing the effects of hosting a new district headquarters on the cropland share in each parish, defined as having more than 60 percent cultivated cropland. The results indicate that before the establishment of new headquarters, cropland trends in these parishes were evolving similarly. However, following their designation, a continuous increase in the area devoted to cultivated cropland is observed. These effects persist over time, even 14 years after the headquarters' creation.

This expansion is accompanied by a decline in the share of cropland with natural vegetation, a land class that includes pixels with 40–60 percent cultivation and natural vegetation (Figure A1). Taken together, these findings suggest that parishes designated as district headquarters become more land-intensive for farming, potentially leading to higher agricultural production, suggesting more local economic activity.¹⁰

⁹The control group is based in 79 not-yet-treated parishes which became district headquarters between 2001 and 2023.

¹⁰While the MODIS resolution does not allow for detailed assessments of urban sprawl, Figure A2 in the appendix studies built-up area directly using the JRC Global Human Settlement Layer built-up surface product (GHS-BUILT-S), which spans 1990 to 2025 in five-year epochs and therefore provides genuine pre-periods for the headquarters cohorts. The built-up share of treated parishes rises steadily and monotonically after headquarters creation, by roughly 1.1 percentage points over twenty years—on the order of 40 percent of the baseline built-up share (joint nullity $p = 0.003$)—with no pre-existing upward trend, reinforcing the interpretation that headquarters designation spurs local urbanization.

FIGURE 2: EVENT STUDY: DISTRICT HEADQUARTER CREATION ON PARISH CROPLAND SHARE



Notes: Event study graph of the estimated coefficients of the De Chaisemartin & d’Haultfoeuille (2020) staggered difference in difference estimator of vegetation share on headquarters creation.

6.2 First Stage

Ensuring the relevance of the instrument is key before proceeding with the second-stage estimates. While it is possible that policymakers did not comply with the governmental rules to locate the district’s headquarters in the geographical middle of a new district, the spatial correlation between the district’s centroid and headquarters suggests compliance with the rules (see Figure 1a).

The first stage establishes the relationship between a household’s proximity to the district’s capital and its proximity to the district’s centroid. Specifically, the household’s self-reported proximity to the district headquarters is estimated using the household’s enumeration area proximity to the district centroid, an indicator variable for newly created districts between 1991 and 2020, and the instrument formed by the interaction of the former two. The first-stage results, presented in Table 2, show that the instrument has a statistically significant coefficient, confirming its predictive power. Consistent with the governmental placement rules, the findings reveal a positive relationship between a household’s proximity to the district’s centroid and its proximity to the new district’s capital for districts established between 1991 and 2020. Taken together, the coefficients indicate that a 10 percent increase in proximity between the household and the district centroid (a 10 percent reduction in distance) corresponds to a 4.29 percent increase in the household’s proximity to the district’s capital. The F-statistic exceeds 10 for both the NSDS and the Afrobarometer, reflecting the strength of the instrument.

TABLE 2: FIRST-STAGE: PROXIMITY HOUSEHOLD - DISTRICT'S HEADQUARTERS

	Log SR Proximity Household - District Headquarters	
	NSDS 2021	Afrobarometer R10
Log Proximity Household - District's centroid $\times$ New District 91-20	0.429*** (0.080)	0.474*** (0.107)
Log Proximity Household - District's centroid	-0.051 (0.063)	0.231*** (0.064)
New district between 1991-2020	1.696*** (0.220)	2.099*** (0.322)
Observations	8,417	2,184
Number of Parishes	967	271
IV F-stat Kleibergen-Paap rk Wald	28.75	19.45
Controls	Yes	Yes

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Regression of self-reported household's distance to the district's capital on household's distance to the district's centroid, an indicator for new districts between 1991-2020, and the interaction between the former two. Both distance variables are in logarithms. Controls include: district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), four major regions. The instrument for Household's distance to the district's capital is the interaction of distance from the household to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

Figure 3 displays the identifying variation behind these estimates. After residualizing on the full set of controls, the figure plots binned means of the household's self-reported proximity to the district headquarters against its proximity to the district centroid, separately for households in new and old districts. The first-stage gradient is strong and positive in districts created between 1991 and 2020, where the placement rule binds, while households in old districts display no positive gradient. The corresponding reduced-form relationships for the main outcome indices, which mirror this pattern, are presented in Appendix Figure A3.

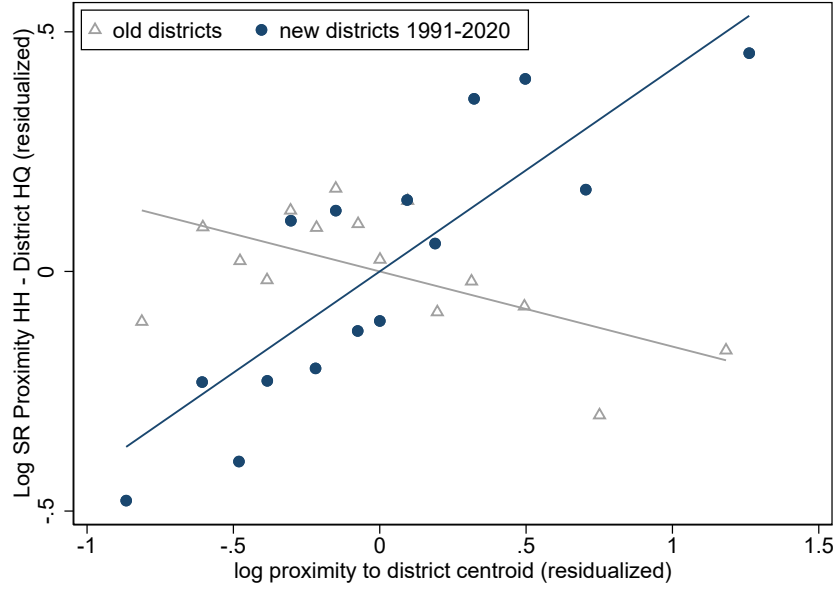


FIGURE 3: First stage binned scatter of residualized self-reported proximity to the district headquarters against residualized proximity to the district centroid for both households in new districts (1991–2020) and old districts. Both variables are residualized on the main specification’s control set (survey-weighted). Each dot is a ventile of the residualized centroid proximity.

6.3 Main Results

6.3.1 Effects on Public Service Delivery

The National Service Delivery Survey offers a rich set of variables on public goods and services, allowing the analysis of a broad range of outcomes. As predicted by the conceptual framework (Proposition 1), proximity to district headquarters should improve access to public goods and services. However, testing each variable individually raises concerns about false rejections of true null hypotheses. To address this, I follow [Anderson \(2008\)](#) and construct summary indices for major outcome groups. Given that public goods and services are provided by both district and central governments, the indices are estimated separately for these two levels of government.

The indices cover education, health, water, and roads (each constructed separately for district and central government provision), while electricity is uniquely provided by the central government and agricultural inputs are supplied only by districts. To create each index, I standardize the component variables to have a mean of 0 and a standard deviation of 1, and compute the index as the average of the standardized components available for each household. The components entering each index are listed in Appendix Table [A1](#).

Table 3 presents the causal effects of household proximity to the district headquarters on ten summary indices: (i) Education District, (ii) Health District, (iii) Water District, (iv) Roads Dis-

trict, (v) Agricultural Inputs District, (vi) Education Central Government, (vii) Health Central Government, (viii) Water Central Government, (ix) Roads Central Government, (x) Electricity Central Government, where each label refers to the level of government responsible for provision in that domain. The table also reports Anderson's sharpened False Discovery Rate (FDR) q-values to correct for multiple hypothesis testing across outcome groups.

TABLE 3: Effects on Services and Proximity to District Headquarters (2SLS)

	School (1)	Health (2)	Water (3)	Roads (4)	Agr inputs (5)
<i>Panel A: NSDS 2021 District services</i>					
Log Proximity to District HQ	0.282 (0.214)	0.387*** (0.121)	0.279* (0.157)	0.089 (0.110)	-0.085 (0.145)
p-val	[0.187]	[0.001]	[0.076]	[0.416]	[0.556]
q-val	{0.234}	{0.008}	{0.179}	{0.452}	{0.452}
Observations	7137	7595	8417	8417	8417
Number of clusters	958	951	967	967	967
Mean of DV	-0.003	-0.016	-0.004	0.008	0.030
IV F-stat (KP rk Wald)	22.243	33.349	28.745	28.745	28.745
	School (1)	Health (2)	Water (3)	Roads (4)	Electricity (5)
<i>Panel B: NSDS 2021 Central govt services</i>					
Log Proximity to District HQ	0.151 (0.192)	0.233 (0.201)	0.372** (0.146)	0.706*** (0.261)	0.428*** (0.137)
p-val	[0.432]	[0.248]	[0.011]	[0.007]	[0.002]
q-val	{0.209}	{0.142}	{0.015}	{0.014}	{0.009}
Observations	5553	8417	8417	8417	8417
Number of clusters	951	967	967	967	967
Mean of DV	-0.001	0.013	-0.017	-0.070	-0.043
IV F-stat (KP rk Wald)	22.621	28.745	28.745	28.745	28.745

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Coefficients from regressions of outcome index variables on self-reported household's distance to the district's capital. The indices are constructed to have a mean of 0 and a standard deviation of 1. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis. Unadjusted p-values are in brackets. Anderson's sharpened False Discovery Rate (FDR) q-values are in curly brackets.

The results show that proximity to new district headquarters improves access to several key public services, with effects that are stronger and more precisely estimated for centrally provided services. Among district-provided services (Table 3, Panel A), households living

closer to the district headquarters have significantly better health access (Column 2), and water (Column 3). The indices for education and roads point in the expected direction but are not statistically significant. The agricultural inputs index shows a small negative and insignificant coefficient, which likely reflects the fact that input provision operates primarily at the sub-county level, making it less sensitive to district-level administrative proximity.

The effects are most consistent and precisely estimated for centrally provided services (Table 3, Panel B). Households closer to district headquarters have significantly better access to water (Column 3), roads (Column 4), and electricity (Column 5). These significant results are robust to multiple hypothesis testing corrections, with FDR q-values below 0.10. Disaggregated results for the individual components underlying each index are presented in Appendix Tables A14–A19. The individual tables show that the improvements detected in the indexes correspond to increases in several indicators and are not driven by specific components. In addition, Table A13 contains results for service availability from the Afrobarometer adding evidence to the higher service access in areas proximate to district headquarters.

Complementing the service delivery findings, Table 4 shows that households living closer to new district headquarters report higher satisfaction with local administrators. While all three types of leaders (the elected district politician (Local Council V), the centrally appointed Resident District Commissioner (RDC), and civil servants) show positive coefficients, only satisfaction with the RDC is statistically significant. This result is consistent with the stronger effects found for centrally provided services. Since the RDC is the main link between the district and the central government, s/he plays a central role in coordinating resource allocation and program implementation from above. Households living closer to the headquarters may therefore not only benefit more from these services, but also be more likely to recognize the RDC’s coordinating role. In terms of magnitude, a 10 percent increase in proximity raises the likelihood of being satisfied with the RDC’s performance by 1.2 percentage points, equivalent to 3.8 percent of the sample mean.

TABLE 4: Satisfaction with Local Leaders and Proximity to District Headquarters (2SLS)

	LCV performance (1)	RDC satisfaction (2)	Civil servants performance (3)
<i>Panel A: NSDS 2021</i>			
Log Proximity to District HQ	0.084 (0.076)	0.122** (0.059)	0.103 (0.085)
Observations	8417	8417	7451
Number of clusters	967	967	966
Mean of DV	0.428	0.329	0.547
IV F-stat (KP rk Wald)	28.745	28.745	30.047

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Coefficients from regressions of satisfaction with local leaders' variables on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

Taken together, the main results describe a consistent pattern. Reducing the distance between households and their district administrative center improves access to public goods and services and raises satisfaction with appointed local leaders, who are responsible for coordinating centrally financed programs. The following section assesses the robustness of these findings before turning to the mechanisms: citizen engagement with the government and accountability.

6.4 Robustness Checks

This section studies the robustness of the results by providing evidence regarding the exogeneity of the instrument, conducting placebo tests, and showing the consistency of the results to alternative samples and distance measures.

6.4.1 Instrument Exogeneity

This section provides further empirical evidence in support of the exclusion restriction, complementing the analysis on land use dynamics presented earlier. In particular, I evaluate whether new headquarters' placements are determined by pre-existing local development of parishes within the same district.

Headquarter's placement

One of the main threats to the exclusion restriction is the endogenous placement of the new district headquarters. Particularly, if the borders of new districts are drawn to benefit a specific centrally located group, this could invalidate the instrument, as the distance to the centroid would no longer be an arbitrary geometric point.

This section analyzes the potential endogenous placement of new headquarters by investigating whether new headquarters are located in the most densely populated or more developed sub-county of the new district. To do this, I evaluate whether sub-county population density in 2010¹¹ and sub-county night lights in 2013¹² predict the likelihood that the sub-county will become the headquarters of newly created districts between 2014 and 2020.¹³ Table A2 shows that neither population density nor night lights at the sub-county level predict future headquarters locations, whether pooling all the sub-counties that belong to new districts (Columns 1 and 3) or focusing only on sub-counties within newly created districts (Columns 2 and 4). These findings indicate that the placement of district headquarters is not driven by a strategic focus on the most developed areas, as they were not systematically located in sub-counties with the highest population density or the greatest levels of economic development, as proxied by night light intensity.

As an example, figure 4 illustrates this pattern for the mother district of Kabarole. Before the split, both night lights and population density were concentrated in the pre-split headquarters town (star), the most developed sub-county of the district. However, when the district was partitioned, the new headquarters (triangle) was placed in a sparsely lit, low-density sub-county near the geographic centre of the newly created district, rather than on the pre-existing economic centre.

6.4.2 Non-random exposure and design-based inference

A second concern is that even if the district’s headquarters placement itself is as good as random, households’ exposure to it may not be. For instance, a household located in the geometric interior of its mother district mechanically receives a high instrument value under any way the district could have been partitioned, and interior location may correlate with unobservables generating non-random exposure to the shock (Borusyak & Hull, 2023). To address this, I follow Borusyak & Hull (2023) and enumerate all feasible contiguous re-partitions of each splitting 1991 mother district into its realized number of 2020 districts, using predetermined 1991 county borders as building blocks, and compute each household’s expected instrument μ_i across these counterfactual redistrictings. Three results, presented in Appendix F, support the identification strategy. First, adding the expected instrument (together with the expected proximity and new-district terms) as a control leaves the estimates stable or slightly larger, with first-stage F-statistics between 10 and 25, indicating that the results are not an artifact of predictable exposure to the redistricting process. Second, weak-instrument-robust Anderson–Rubin confidence sets for this specification exclude zero for the health (district and central government), water (district and central government), roads (central government), and

¹¹Data obtained from the Gridded Population of the World Database constructed by NASA.

¹²The Annual global VIIRS nighttime lights come from the Earth Observation Group, Payne Institute for Public Policy, Colorado School of Mines.

¹³There are 24 newly created districts between 2014 and 2020. I restrict the analysis to this period as 2010 and 2013 are the earliest years with consistent population density and nightlight data available for Uganda.

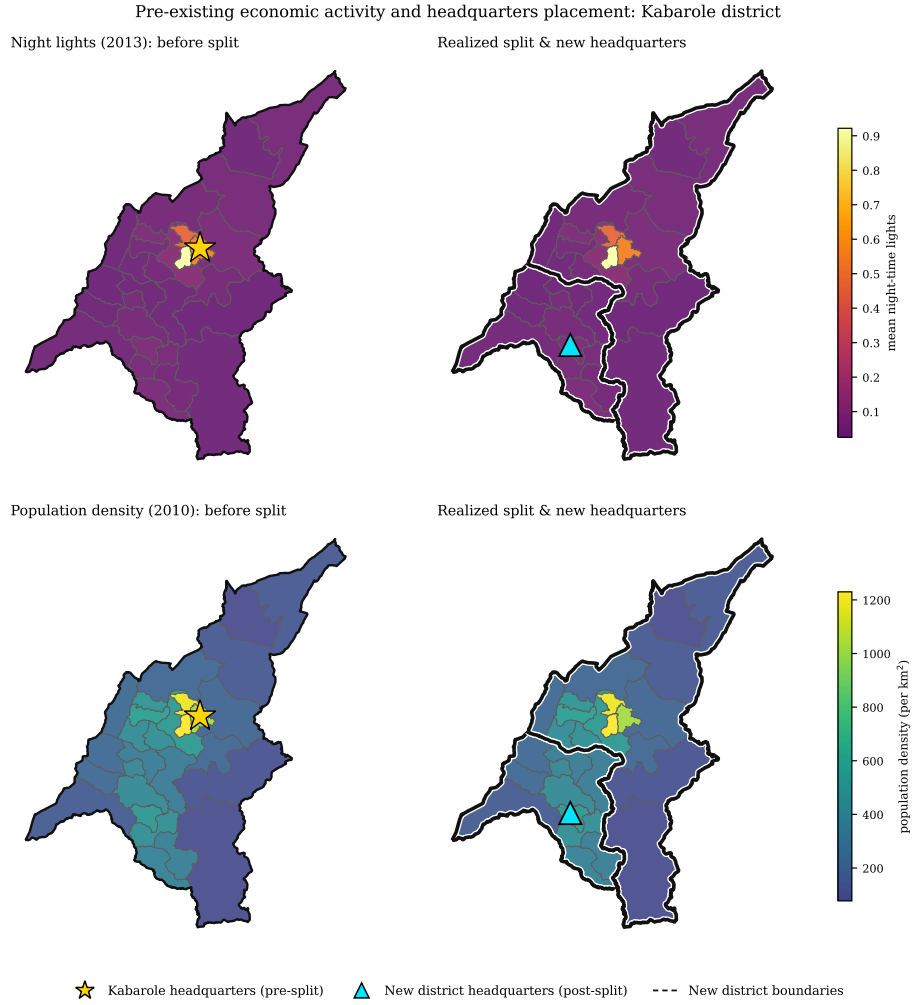


FIGURE 4: Pre-existing economic activity and headquarters placement. The panels show the sub-counties of the pre-split mother district of Kabarole, shaded by night-time lights (2013, top) and population density (2010, bottom). Left panels mark the pre-split district headquarters (star); right panels overlay the realized new-district boundaries and the new headquarters (triangle). The new headquarters is located away from the brightest and most densely populated sub-counties, consistent with the placement rule and with the null estimates in Table A2.

electricity indices. Third, I conduct randomization inference over 1,000 counterfactual national configurations: the realized redistricting is an extreme draw, with a first-stage randomization p-value of 0.001 and reduced-form randomization p-values at or below 0.02 for nine of the ten service indices.

6.4.3 Distance to large cities

A second potential concern is that the instrument captures households' proximity to large urban centers rather than to administrative headquarters. If distance to the district centroid simply reflects distance to large cities, it would not provide independent variation in proximity to administrative centers. To address this, I include three alternative measures of urban proximity

as mediators in the first stage: the household's distance to the closest large city (defined as the top 20 most populous cities in Uganda), the household's distance to the closest large city within its region, and the household's distance to Kampala. Table A3 shows that the instrument remains strong and stable across all specifications. The coefficient on the interaction term ranges from 0.373 to 0.437 and remains highly significant throughout, while the F-statistic stays above 19 in all columns.

6.4.4 Placebo tests

I perform placebo regressions to test whether the variable of interest is capturing some unobserved geographical features. This test examines whether household proximity to the district capital has predictive power for variables that are plausibly unrelated to administrative proximity. If the placebo regressions show a significant effect, it could suggest that the instrument captures unobservable characteristics unrelated to proximity to the district capital, indicating potential omitted variable bias. Table A4 presents 2SLS estimates for three placebo outcomes: households' use of cookstoves, use of air transportation, and involvement in the mining sector. The estimates are close to zero and statistically insignificant across all three outcomes, providing reassuring evidence against omitted variable bias.

As a further placebo test, I study whether proximity to district headquarters affects households' knowledge of and distance to lower-tier local government institutions (LC1, LC2, and LC3). As these operate below the district level, these variables should not be directly affected by the creation of new district headquarters. Table A5 shows that proximity to district headquarters has no systematic effect on these outcomes: eleven of the twelve coefficients are statistically indistinguishable from zero at the 5 percent level, and the single exception (the reported distance to the LC2 office) does not survive correction for multiple hypothesis testing (sharpened FDR q-value of 0.27). This confirms that the variation in proximity induced by the policy is specific to the district level and does not reflect a general reduction in distance to all levels of government.

6.4.5 Distance to the Headquarters Town

Another potential concern is that the estimated effects reflect the growth of the headquarters town itself rather than improved administrative access for the surrounding population. If the results were driven entirely by households located in the immediate vicinity of the new headquarters, they would speak to urban agglomeration around the administrative center rather than to the reach of the state. To address this, Table A8 re-estimates the main specification excluding households progressively closer to their district headquarters: those in the same parish as the headquarters point, and those within 2 and 5 kilometres. The electricity results are essentially unchanged across all three donuts, and the roads coefficients become larger the more

of the headquarters' surroundings are excluded, indicating that these effects are not an artifact of headquarters-town growth. The health index attenuates by roughly a fifth when households within 5 kilometres are excluded but remains significant at the 10 percent level, suggesting that part (though not most) of the health effect is concentrated near the headquarters town, consistent with the location of district health facilities.

6.4.6 Alternative Distance Variable: GIS Distance

While the household's self-reported distance to the district headquarters arguably captures effective proximity (it reflects not only straight-line distance but also the available transportation network) it may suffer from over- or under-reporting. To address this, Table A9 replaces the self-reported measure with the GIS straight-line proximity from the household's enumeration area to the headquarters point, keeping the instrument and all controls unchanged. The two measures are highly but imperfectly correlated (0.80; Appendix Figure A4), consistent with noisy but informative self-reports. The 2SLS estimates with the GIS measure are statistically significant for all three headline outcomes and, if anything, larger in magnitude, while the first-stage F-statistics remain above 20 for both measures. OLS estimates with either measure are three to six times smaller than the 2SLS estimates, a gap discussed in the next subsection.

6.4.7 OLS Estimates

Appendix Table A10 compares the 2SLS estimates with their OLS counterparts for the ten service indices. The OLS coefficients are positive and statistically significant for most outcomes, confirming the descriptive association between proximity to headquarters and service access. However, for the outcomes where the 2SLS estimates are significant, the OLS coefficients are three to five times smaller. This pattern is informative as it goes against the most direct selection concern. If headquarters had been systematically placed in already better served locations, the naive OLS estimates would likely be larger than the 2SLS estimates. Instead, OLS is substantially smaller, being consistent with two explanations. First, both the self reported and GIS based measures are noisy proxies for effective proximity, so OLS is likely affected by attenuation bias, which the instrument helps correct. Second, the 2SLS estimates identify local average treatment effects for compliers, that is, households whose proximity to headquarters was shifted by redistricting. These households were previously more remote, and the marginal returns to administrative access are plausibly larger for them. The recentering exercise points in the same direction: correcting for predictable exposure to the redistricting process moves the estimates away from zero rather than toward it (Appendix F). This suggests that naive comparisons understate, rather than overstate, the effect of proximity.

6.4.8 Alternative Instrument: Parish-Level Distance

As an additional robustness check, I replace the enumeration area geocoordinates with parish centroid distances to construct an alternative version of the instrument. Table A6 shows that the results are similar to the ones obtained in Table 3. If anything, several coefficients are larger in magnitude and more precisely estimated under this alternative, while the first stage remains strong with F-statistics exceeding 23 across all specifications. These results show that the findings are robust to the choice of geocoordinates used to compute household distances.

6.4.9 Alternative Sample: Only New Districts

The main specification identifies the effect of proximity from the differential relationship between centroid proximity and headquarters proximity in new versus old districts, interacting centroid proximity with the new-district indicator and using old districts as a comparison group. Because the government placement rule applies only to newly created districts, an alternative is to restrict the sample to these districts, where the location of the centroid directly determines the location of the headquarters. In this subsample the interaction is no longer necessary: the household's proximity to the district centroid can be used on its own as an instrument for its proximity to the district capital. Because the placement rule ties the centroid to the headquarters mechanically within new districts, the instrument is considerably stronger here, with first-stage F-statistics ranging from 29 to 49, well above conventional thresholds.

Table A7 reports the results. All ten point estimates are positive, and the two most robust findings of the main specification survive: proximity significantly improves access to district-provided health services (0.317, significant at the 1 percent level) and to centrally provided electricity (0.149, significant at 5 percent). District roads (0.143) and secondary education (0.255) become marginally significant, while water and roads under central government retain the expected positive sign but lose statistical significance. The loss of precision reflects the fact that restricting the sample to new districts drops almost half of the observations and removes the identifying variation coming from the comparison with old districts. The magnitudes are also not directly comparable to the main estimates, since the within new district design identifies effects from a different mix of compliers. Overall, this exercise indicates that the main results are not an artifact of the new-versus-old district contrast, as the central findings are recovered using only the variation generated within newly created districts, where the placement rule directly applies.

7 Mechanisms

To investigate the underlying mechanisms behind the main findings, this study focuses on two potential channels through which the distance to new district headquarters may impact

the provision of public goods and services. First, the level of citizen engagement with local governments could influence policymakers' awareness of the most pressing issues within the community. Second, individuals' perceptions of corruption may reflect problems associated with corruption in public service delivery, potentially leading to increased demands for accountability.

7.1 Citizens' Engagement with the Local Government

One of the main objectives behind the creation of new administrative units was to empower local populations and enhance their participation in decision-making. As discussed in the conceptual framework, reducing the geographic gap between citizens and government lowers the cost of formal engagement with district institutions, shifting citizens from contentious to participatory channels (Propositions 4–5). By minimizing information and communication frictions between administrators and citizens, closer proximity to administrative centers may foster greater awareness of state institutions, shift citizens away from reliance on traditional alternatives, and ultimately translate into more effective engagement with formal government. This section studies whether proximity to new district headquarters influences these dimensions of citizen-state relations.

Before turning to direct measures of citizen engagement, I first analyze whether households living closer to district headquarters are more aware of district-level institutions and whether proximity reshapes the balance between formal state institutions and traditional ones. Table 5, Panel A shows that proximity to headquarters significantly increases citizens' knowledge of important state institutions. Households living closer are more likely to know about magistrate courts, the police, the Office of the Auditor General, and the judiciary system, reflecting that the creation of new headquarters is strengthening state presence and legitimacy in previously remote areas. Additionally, households closer to headquarters are more likely to have a family member employed as a civil servant, suggesting a broader deepening of state-society linkages (Panel B, Column 1).

TABLE 5: Institutional Knowledge and State-Traditional Substitution (2SLS)

	Knows magistrate court (1)	Knows police (2)	Knows land office (3)	Knows OAG (4)	Knows judiciary (5)
<i>Panel A: NSDS 2021</i>					
Log Proximity to District HQ	0.273*** (0.090)	0.088* (0.053)	0.096 (0.063)	0.088** (0.039)	0.162** (0.080)
Observations	8413	8415	8413	8417	8417
Number of clusters	967	967	967	967	967
Mean of DV	0.375	0.930	0.156	0.092	0.384
IV F-stat (KP rk Wald)	28.592	28.639	28.592	28.745	28.745
	Family conflict (1)	Resolved by govt (2)	Resolved by trad (3)	Contacted MP (4)	Contacted trad leader (5)
<i>Panel B: NSDS 2021 and Afrobarometer R10</i>					
Log Proximity to District HQ	0.077** (0.031)	0.096 (0.144)	-0.080 (0.110)	0.076 (0.071)	-0.181* (0.106)
Observations	8414	2131	2131	2178	2146
Number of clusters	967	271	271	271	271
Mean of DV	0.057	0.556	0.211	0.168	0.277
IV F-stat (KP rk Wald)	28.655	19.705	19.705	19.462	19.819

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Coefficients from regressions of state/traditional institutions variables on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

Panel B further reveals that this increased awareness of formal institutions is accompanied by a substitution away from traditional ones. While the coefficients on conflict resolution and MP contact are positive but insignificant, households living closer to headquarters are significantly less likely to contact traditional leaders, pointing to a shift in which institutions citizens turn to when seeking redress or representation. Together, these results indicate that administrative proximity makes the state more visible and accessible, gradually displacing traditional institutions.

Table 6 presents indicators of citizen participation with district-level governmental actors, differentiating between participatory and contentious forms of engagement. The results show that increased proximity reduces contentious engagement (Panel B). A 10 percent increase in proximity to district headquarters significantly reduces the likelihood of participating in protests and contacting the media by 0.7 and 1.2 percentage points, respectively (equivalent to 19 and 13 percent of their respective sample means). Together with the results on higher state presence above, these results are consistent with citizens having less need to engage in protests to raise their worries. Adding evidence to this interpretation, all coefficients on participatory engagement in Panel A are positive, indicating that proximity increases formal engagement with district officials, even if the individual coefficients are not statistically significant.¹⁴

¹⁴Importantly, Appendix Table A5 confirms that proximity to headquarters does not systematically affect engagement with LC1 leaders (no coefficient is significant at the 5 percent level), reinforcing that the effects documented here are specific to institutions belonging to the districts.

TABLE 6: Citizen Engagement and Proximity to District Headquarters (2SLS)

	Involved LC5 (1)	Met RDC (2)	RDC Baraza (3)
<i>Panel A: NSDS 2021</i>			
Log Proximity to District HQ	0.086 (0.066)	0.045 (0.043)	0.033 (0.027)
Observations	8417	8417	8417
Number of clusters	967	967	967
Mean of DV	0.255	0.102	0.052
IV F-stat (KP rk Wald)	28.745	28.745	28.745
	Collective civic (1)	Protest (2)	Contacted media (3)
<i>Panel B: Afrobarometer R10</i>			
Log Proximity to District HQ	-0.176 (0.112)	-0.071** (0.034)	-0.116** (0.058)
Observations	2182	2184	2184
Number of clusters	271	271	271
Mean of DV	0.360	0.037	0.093
IV F-stat (KP rk Wald)	19.413	19.446	19.446

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Coefficients from regressions of citizens' involvement with the government variables on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

Additional evidence suggests that this shift toward formal engagement is translating into improved decision-making outcomes. Appendix Table A20 illustrates the relationship between proximity to headquarters and the likelihood that households have benefited from a public investment in a sector they consider a priority. Households closer to district headquarters are significantly more likely to have benefited from a new school in a sector they prioritize, and the coefficients for classrooms, health, and water investments are positive though not statistically significant. This pattern suggests that greater interaction with district officials helps channel local preferences into tangible projects.

Taken together, these results support the engagement channel described in the conceptual framework (Proposition 4) by showing that increasing proximity between citizens and the district administrative center increases state presence and institutional knowledge, shifts citizens away from traditional institutions toward formal ones, reduces the need for contentious political action, and increases participatory engagement.

7.2 Corruption Perceptions

A further potential mechanism through which proximity affects public service delivery is accountability. The conceptual framework predicts that proximity strengthens citizen monitoring, deterring corruption (Proposition 6). Table 7 presents two sets of measures: Panel A present results on households' corruption awareness and their direct experience of victimization, while Panel B contains bribery experiences across different service contexts.

TABLE 7: Corruption and Proximity to District Headquarters (2SLS)

	Nepotism (1)	Embezzlement (2)	Absenteeism (3)	Victim nepotism (4)	Victim embezzlement (5)	Victim absenteeism (6)
<i>Panel A: NSDS 2021</i>						
Log Proximity to District HQ	0.084 (0.082)	0.044 (0.077)	-0.011 (0.081)	-0.031* (0.019)	-0.029 (0.025)	-0.086* (0.045)
Observations	8412	8412	8413	8417	8417	8417
Number of clusters	967	967	967	967	967	967
Mean of DV	0.494	0.647	0.487	0.034	0.038	0.075
IV F-stat (KP rk Wald)	28.613	28.613	28.656	28.745	28.745	28.745
	Bribe health (1)	Bribe police (2)	Bribe avoid police (3)	Bribe ID (4)		
<i>Panel B: Afrobarometer R10</i>						
Log Proximity to District HQ	-0.287** (0.118)	-0.274 (0.183)	-0.151 (0.166)	0.092 (0.151)		
Observations	1604	345	904	396		
Number of clusters	271	196	257	208		
Mean of DV	0.409	0.771	0.523	0.487		
IV F-stat (KP rk Wald)	20.664	19.947	12.130	17.949		

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of corruption variables on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

The results reveal that proximity to district headquarters is associated with slightly higher awareness of nepotism, embezzlement, and absenteeism, though none of these coefficients is statistically significant. This suggests that households living closer to administrative centers are not necessarily exposed to more corruption. Importantly, when examining direct victimization, the results show that households closer to headquarters are significantly less likely to have personally experienced nepotism and absenteeism, with a 10 percent increase in proximity reducing the probability of experiencing nepotism or absenteeism by approximately 0.3 and 0.8 percentage points respectively, equivalent to 9 and 11 percent of their sample means. Consistent with these findings, Panel B reveals that proximity significantly reduces the likelihood of paying a bribe to access health services by approximately 3 percentage points (about 7 percent of the sample mean).

These results suggest that administrative proximity strengthens accountability by reducing direct corruption experiences even as awareness of corruption remains high or increases slightly. The combination of high awareness, together with lower victimization, is consistent with a monitoring mechanism. Households closer to district headquarters engage more frequently with local officials and are better able to observe and report misconduct, which in turn

may deter corrupt behavior.

8 Democratic perceptions

The evidence so far suggests that bringing the administration closer changes what the state delivers and how citizens interact with it. This section examines whether these changes also affect political attitudes, moving from what the state does to how citizens perceive it. The theoretical case for decentralization has been based on a sequence running from proximity to performance to legitimacy: government that is closer is easier to observe and hold accountable, performs better as a result, and thereby earns citizens' trust and support for the political system (Oates et al., 1972; Bardhan, 2002; Faguet, 2023). The first links of this chain have empirical support. Decentralization can improve delivery and responsiveness (e.g. (Faguet, 2004)), as the previous sections confirm for Uganda. However, the final link is often assumed rather than established. Credible evidence is scarce because political attitudes and state presence are jointly determined. Governments may expand where support is high or where contestation is politically costly, and communities that trust the state may both demand and receive more of it. As a result, observed correlations between government performance and political trust have limited causal content. Moreover, in hybrid regimes, there are reasons to expect this link to fail. Decentralization can be designed to extend central control rather than empower citizens (Martinez-Bravo et al., 2022), and improvements in government performance need not translate into political trust where democratic institutions are weak (Besley et al., 2026). The setting of this paper helps address this gap. The proximity induced by redistricting provides an exogenous shock to citizens' exposure to the state, one that improved services, engagement, and accountability. This section studies whether these improvements translate into democratic attitudes, using the same specification as before.

Table 8 details the relationship between proximity to district headquarters and a broad set of democratic perception indicators (Panel A), rule of law perceptions (Panel B), and trust in government (Panel C). The results show that across all fourteen outcomes (including satisfaction with democracy, perceived freedom to vote and organize, evaluations of judicial independence, official accountability, and trust in local government, parliament, the president, the ruling party, the opposition, and the courts) the estimated coefficients are small and statistically insignificant. This pattern is robust across specifications and holds for both local and national institutions, suggesting that the gains in service delivery and accountability documented above do not translate into measurable shifts in how citizens evaluate the democratic system or the state more broadly.¹⁵

¹⁵Appendix Table A22 shows that the same null pattern holds in the 2017 Afrobarometer round. The six trust in government coefficients are small and statistically insignificant, suggesting that the null is not driven by political trust responding more slowly than service delivery, as it holds both before and after the service effects documented in Section 10 materialized.

TABLE 8: Democratic Perceptions and Proximity to District Headquarters (2SLS)

	Is democracy (1)	Satisf democracy (2)	Free vote choice (3)	Free join orgs (4)		
<i>Panel A: Democratic perceptions</i>						
Log Proximity to District HQ	-0.012 (0.053)	0.072 (0.085)	-0.015 (0.044)	-0.064 (0.060)		
Observations	2079	2108	2159	2151		
Number of clusters	271	271	271	271		
Mean of DV	0.882	0.593	0.934	0.847		
IV F-stat (KP rk Wald)	19.352	19.744	20.486	21.665		
	President ignores law (1)	Unequal law (2)	Judges influenced (3)	Officials impunity (4)		
<i>Panel B: Rule of law perceptions</i>						
Log Proximity to District HQ	-0.009 (0.098)	0.020 (0.101)	-0.100 (0.108)	-0.029 (0.111)		
Observations	1859	2112	1980	2075		
Number of clusters	271	271	271	271		
Mean of DV	0.366	0.640	0.601	0.607		
IV F-stat (KP rk Wald)	22.695	20.535	19.684	20.123		
	Trust localgov (1)	Trust president (2)	Trust parliament (3)	Trust ruling party (4)	Trust opposition (5)	Trust courts (6)
<i>Panel C: Trust in government</i>						
Log Proximity to District HQ	0.029 (0.079)	-0.066 (0.084)	0.043 (0.099)	0.044 (0.084)	0.017 (0.086)	0.024 (0.083)
Observations	2132	2116	2184	2184	2184	2087
Number of clusters	271	271	271	271	271	271
Mean of DV	0.507	0.695	0.503	0.589	0.315	0.526
IV F-stat (KP rk Wald)	20.692	19.058	19.446	19.446	19.446	21.146

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of democratic perception variables on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

Interestingly, while trust in government remains unchanged, Appendix Table A21 reveals that proximity does affect other dimensions of trust and collective capacity. Households closer to new district headquarters are significantly more likely to trust people of other religions and other ethnicities, and are more likely to engage in collective action to solve local problems. These findings suggest that administrative proximity generates an environment that foster social capital even if political trust remains unaffected.

9 Heterogeneous Effects by Local Political Alignment

The conceptual framework predicts that political alignment between the elected LCV and the appointed RDC amplifies the effects of proximity on service delivery (Proposition 3), participatory engagement (Proposition 5), and accountability (Proposition 7). This section tests these predictions. As described in Section 2, the LCV is accountable to local voters while the RDC is accountable to the central government. When these two figures share political affiliation, their incentives are more likely to be complementary, easing coordination on resource allocation and program implementation. When they are not aligned, frictions may emerge as

each administrator responds to different principals with potentially different priorities. This section studies whether the effects of proximity to district headquarters vary with the degree of political alignment between appointed and elected local leaders.

9.1 Aligned Districts

As predicted by the model, in districts where the elected and the appointed politician have the same political affiliation, the effects of proximity are consistently stronger. Table A24 shows that proximity significantly improves access to centrally provided roads and electricity, and to district-provided health services (the latter at the 10 percent level), with larger coefficients than in the pooled specification.¹⁶ This is consistent with the idea that political alignment facilitates coordination between both district leaders for the implementation of public goods and services, particularly when communities are within closer reach of the administrative center.

The engagement and accountability results follow a similar pattern. Proximity in aligned districts increases involvement in local decision-making (significant at the 10 percent level) and more sharply reduces contentious engagement through protests, media contact, and collective civic actions to raise problems to the state (Table A25); the corresponding results for institutional knowledge and traditional institutions are reported in Table A26. On corruption, proximity increases awareness while the point estimates on victimization remain negative, in line with the accountability channel described above (Table A28). Satisfaction with the RDC is also higher in aligned districts (Table A27), suggesting a more responsive administrative environment when political coordination costs are low.

The stronger effects on services, engagement, and accountability in aligned districts are consistent with the complementarity between proximity and alignment predicted by the conceptual framework (Propositions 3, 5, and 7). However, despite these improvements across services, engagement, corruption, and satisfaction, proximity in aligned districts produces no change in democratic perceptions or trust in government. Satisfaction with democracy, perceived political freedoms, rule of law evaluations, and trust in both local and national institutions all remain unchanged (Table A29). This result shows that even in the most favorable political environment, where the state is functioning better and citizens recognize it, these gains do not translate into deeper democratic legitimacy.

9.2 Non-Aligned Districts

In non-aligned districts the results are mixed. Table A30 shows some evidence that proximity continues to improve health access, but effects on other service domains are smaller and largely insignificant, suggesting that political frictions between the appointed and elected politicians limit the extent to which administrative proximity translates into broader service delivery gains.

¹⁶Splitting the sample reduces the first-stage F-statistics to between 12 and 17. Appendix Table A11 therefore complements the subsample estimates with weak-instrument-robust Anderson–Rubin confidence sets. In aligned districts these exclude zero for electricity, roads, and health.

The engagement and accountability channels are no longer present. Proximity has no significant effect on participatory engagement, contentious mobilization, corruption experiences, or satisfaction with local leaders (Tables [A31–A34](#)).

The democratic perception results point to a different direction. Table [A35](#) shows that proximity in non-aligned districts significantly reduces perceived freedom to vote, increases the likelihood that people had a negative perception about the rule of law, and lowers trust in both local government and the ruling party. Taken together, these results suggest that citizens closer to headquarters in these districts appear to observe the lack of coordination more directly. While the administrative center is more accessible, citizens are more likely to experience political conflict and therefore stop engaging. Therefore, rather than building state legitimacy bringing service in non-aligned districts, it seems to decrease it.

Overall, the alignment results help explain both the main findings and the limits of decentralization in hybrid regimes. When political incentives are aligned, administrative proximity improves services, engagement, and accountability, but does not deepen democratic support. When they are not, the benefits of proximity are more limited and political trust falls, suggesting that the governance environment shapes whether bringing government closer to citizens works as intended.

10 Dynamics

So far the results analyzed documented the effects of distance to headquarters on public service delivery from a static perspective. However, it is likely that these effects vary depending on the stage of development of the administrative centers. This section investigates the potential dynamic effects of distance on public service delivery using data from the NSDS 2015 and the NSDS 2021.

Given the cross-sectional nature of the NSDS, the different rounds of data collection do not capture information from the same households. Nevertheless, comparing cross-sectional coefficients of the impacts of distance to headquarters on public service delivery within a 5-year time span in the same districts can provide valuable insights of the evolution of these effects over time. To do so, I focus on the evolution of the importance of the distance to administrative headquarters for households belonging to districts created between 1991 and 2014 (one year before the earliest available dataset). To guide the analysis, I take the NSDS 2015 as the benchmark and analyze how the magnitude and significance of the distance to administrative headquarters change in both samples 2015 and 2021.

Since the NSDS 2015 does not include household geo-coordinates nor consistent parish names, I approximate a household's location using the household's sub-county centroid as its geocoordinate. Thus, I calculate the distance to the district's centroid by using the distance between the household's sub-county centroid and the district's centroid. Although this measure

may contain some measurement error, its primary purpose is to capture the exogenous variation in distance introduced by the creation of a new district.

10.1 Dynamics - Main results

Table 9 presents results for the major outcome indices in districts created between 1991 and 2014, using data from the NSDS 2015 (Panel A) and the NSDS 2021 (Panel B). Comparing the two panels reveals an informative evolution in the effects of proximity over time.

TABLE 9: DYNAMICS IMPACT OF HOUSEHOLD'S DISTANCE TO HEADQUARTERS ON INDICES OF MAJOR OUTCOMES

	Education district (1)	Education central gov (2)	Health district (3)	Water district (4)	Water central gov (5)	Electricity central gov (6)	Roads district (7)	Agr inputs district (8)
<i>Panel A. NSDS 2015 - Period 1991-2014</i>								
Log SR Proximity HH - District HQ	0.155	0.063	0.014	0.067	-0.115	-0.172	0.103	1.274***
SE	(0.143)	(0.122)	(0.157)	(0.197)	(0.200)	(0.154)	(0.197)	(0.391)
p-value	[0.280]	[0.605]	[0.929]	[0.734]	[0.565]	[0.264]	[0.603]	[0.001]
sharpened q-value	{1.000}	{1.000}	{1.000}	{1.000}	{1.000}	{1.000}	{1.000}	{0.010}
Mean of Dependent Variable	-0.017	-0.006	0.104	0.004	0.010	-0.019	0.005	0.026
Observations	5,271	4,048	8,918	8,934	8,928	8,940	8,940	8,940
Number of Parishes	946	938	953	953	953	953	953	953
IV F-stat Kleibergen-Paap rk Wald	16.620	18.056	17.799	17.512	17.598	17.505	17.505	17.505
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Panel B. NSDS 2021 - Period 1991-2014</i>								
Log SR Proximity HH - District HQ	0.262	0.173	0.397***	0.295**	0.377***	0.346***	0.085	-0.093
SE	(0.193)	(0.171)	(0.107)	(0.131)	(0.129)	(0.116)	(0.097)	(0.129)
p-value	[0.174]	[0.313]	[0.000]	[0.025]	[0.004]	[0.003]	[0.382]	[0.468]
sharpened q-value	{0.162}	{0.264}	{0.002}	{0.033}	{0.009}	{0.009}	{0.280}	{0.306}
Mean of Dependent Variable	-0.016	0.004	-0.035	-0.093	-0.036	-0.060	-0.018	0.087
Observations	6,555	5,148	6,959	7,752	7,752	7,752	7,752	7,752
Number of Parishes	885	879	878	894	894	894	894	894
IV F-stat Kleibergen-Paap rk Wald	27.382	28.469	40.570	36.997	36.997	36.997	36.997	36.997
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. All regressions control for district ever split between 1991-2014, lake Victoria district, districts' area in Km² as of 1991, household's parish centroid distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2014. Standard errors clustered at the Parish level are in parenthesis. Unadjusted p-values are in brackets. Anderson's sharpened False Discovery Rate (FDR) q-values are in curly brackets.

In 2015, proximity to district headquarters had limited effects on most service domains. The only statistically significant result is a large positive effect on agricultural inputs, robust to multiple hypothesis testing corrections. All remaining indices in education, health, water, electricity, and roads are small and insignificant. By 2021, this pattern changes considerably. Proximity is no longer affecting the agricultural inputs index, while it now significantly improves access to health services, water at both district and central government levels, and electricity.

These dynamics suggest that the relevance of administrative proximity for public service delivery strengthened considerably between 2015 and 2021, and that the domains in which it matters shifted over time. Two complementary forces could explain this pattern. First, the gradual consolidation of administrative capacity in newly created districts: the rapid pace of district creation between 2005 and 2010 imposed significant burdens on government capacity, generating delays in the effective establishment of district headquarters (Green, 2015; Cohen,

2024), so that many districts surveyed in 2015 may still have been in the early stages of administrative development. Second, the differential timeline of public investments: a new administration can quickly influence access to fast-turnaround services such as the distribution of agricultural inputs through existing extension services structures, but infrastructure investments require several years from planning and procurement through to completion. The pattern in Table 9 is consistent with this. While in 2015 the only significant effect was on agricultural inputs, an outcome that responds quickly to administrative capacity, by 2021 infrastructure investments near new headquarters materialized, producing the significant effects on health, water, and electricity access documented in the main results. While the evidence is suggestive, the combination of consolidating administrative capacity and the timeline of infrastructure provision goes in line with the main results and with existing research of the administrative constraints facing newly created districts in Uganda (Cohen, 2024).

To complement the NSDS results, I use Afrobarometer to provide further evidence on this consolidation pattern. The Afrobarometer records service availability at the enumeration area level, based on direct observation by enumerators rather than household self reports, and uses identical questions in the 2017 and 2024 rounds (Appendix Table A23). The results show that in 2017, proximity to district headquarters has no significant effect on five of the six availability indicators, with only a marginal effect on piped water. By 2024, the effects on piped water, paved roads, and police stations are large and statistically significant. The evidence coming from the Afrobarometer mirrors the evolution observed in the NSDS, reinforcing the interpretation that the returns to administrative proximity increased as new district institutions became more established.

11 The Fiscal Cost of District Creation

The former sections establish that proximity to new district headquarters improves access to public goods and services, increases citizen participation, and strengthens accountability. While these benefits are important, it is also necessary to understand the costs associated with district creation. This section provides evidence from government budgets to examine the fiscal implications of creating new districts. In particular, each new district requires its own administrative apparatus, while the intergovernmental transfer system finances a fixed administrative cost for every district. District creation therefore generates a growing fiscal burden that constrains how far the strategy of bringing services closer to citizens can be efficiently extended.

To estimate the cost of district creation, I use a panel of approved local-government transfers from the Ministry of Finance, Planning and Economic Development (MoFPED), covering fiscal years 2012/13 to 2023/24. These data allow me to construct a detailed decomposition of each district's annual transfer grants. All amounts are expressed in real fiscal year 2017/18 shillings,

deflating nominal budgets with the Uganda consumer price index.¹⁷ For each of the 24 districts created during the sample period (calendar years 2015–2020), I build an event-time fiscal accounting around the split. Specifically, I compare the mother district’s average budget in the two fiscal years before the split with its average budget in the two years after the split. I then add the child district’s average budget during its first two years of operation to obtain the net change in total spending across the territory. The wage component of the unconditional block grant (which finances the LC5 chairperson, chief administrative officer, district headquarters staff, and other political officeholders) provides the cleanest measure of duplicated administrative payroll costs.

Table 10 presents the dynamics of fiscal cost reallocation. Panel A shows an inverse relationship between the total transfers received by a district and the relative importance of unconditional grants. Smaller districts allocate a substantially larger share of their central government transfers to the unconditional block grant. This pattern is evident in both the wage and non wage components, while the share allocated to conditional grants, which are tied to service delivery formulas, shows no relationship with district fiscal size. Similarly, Panel B shows that recently created districts devote roughly two percentage points more of their budget to unconditional grants than established districts, with most of the difference concentrated in the wage component. In contrast, the share allocated to conditional grants remains flat. Because conditional grants scale with service needs, while unconditional grants finance the fixed administrative apparatus required to operate as a district, this contrast suggests that the larger administrative share observed in small and newly created districts reflects a fixed administrative cost rather than differences in service provision.

TABLE 10: Administrative overhead share and recently-created districts

	Wage bill (1)	Uncond. (admin) (2)	Uncond. wage (3)	Uncond. non-wage (4)	Cond. (services) (5)
<i>Panel A. Cross-sectional scale (log of total transfers)</i>					
ln(transfers_full)	0.0168 (0.0129)	-0.0539*** (0.0053)	-0.0419*** (0.0042)	-0.0120*** (0.0015)	0.0064 (0.0103)
Observations	1,468	1,468	1,468	1,468	1,468
R-squared	0.387	0.584	0.485	0.650	0.516
<i>Panel B. Recently-created district premium (created within 3 FYs)</i>					
Created within last 3 FYs	-0.0108 (0.0162)	0.0202*** (0.0060)	0.0156*** (0.0052)	0.0046*** (0.0015)	-0.0050 (0.0176)
Observations	1,468	1,468	1,468	1,468	1,468
R-squared	0.383	0.255	0.140	0.536	0.515

Notes: * p<0.1, ** p<0.05, *** p<0.01. Each column is a separate regression whose dependent variable is the share of the district’s total transfers going to that grant category; unconditional grants (Columns 2–4) finance the fixed administrative apparatus, while conditional grants (Column 5) are tied to service-delivery formulas. Panel A regresses each share on the log of the district’s total transfers (its fiscal size); Panel B, on an indicator for districts created within the last 3 fiscal years. Column 5 shares the same denominator as the overhead measures and serves as a placebo: its flat relationship with size rules out a mechanical correlation between the overhead share and total transfers. All columns include Financial Year fixed effects; standard errors clustered by district in parentheses. Sample: district-FY observations FY2012/13–FY2023/24, excluding KAMPALA, WAKISO, and Municipal/City Councils.

Finally, it is important to address whether changes of budget are a zero-sum exercise. Appendix Table A36 therefore traces the fiscal accounting across the 24 splits in real terms

¹⁷World Development Indicators series FP.CPI.TOTL; each fiscal year is assigned the average of the two overlapping calendar years.

and benchmarks each split against the contemporaneous budget growth of never-split 1991 territories. Relative to this counterfactual, total spending in the territory increases by 5.37 billion shillings on average, equivalent to 19 percent of the mother district's pre-split budget (12 percent at the median), with a positive excess in 19 of the 24 splits.

To complement this analysis, I aggregate the panel to fixed territories defined by the 1991 district borders. This means that transfers to a mother district and to any districts later carved out of it are added within the same geographic unit, netting out reallocations between old and new districts. I then estimate a staggered event study of log real territory transfers using the estimator of [de Chaisemartin & D'Haultfoe uille \(2020\)](#). A territory is classified as treated in the first fiscal year in which a newly created district carved out of it receives its own budget vote. Of the 37 territories, 18 experience such a split during the sample period, while the remaining 19 either are never split or were already subdivided before the panel begins, and therefore serve as controls. Figure 5 shows that real transfers to treated territories increase by approximately 14 percent after the first split (corresponding to an average total effect of 0.135 log points, with a standard error of 0.042). The effects grow from 9 to 17 percent over the following four years, with no evidence of differential pre trends (with a joint placebo p value of 0.73). The composition of this increase mirrors the allocation mechanism documented in Table 10. The wage component of the unconditional grant, which captures the duplicated administrative payroll, rises by 26 percent, compared with only 7 percent for conditional service delivery grants (Appendix Figure A7).¹⁸ Consistently, Table 11 shows that, holding the territory fixed, the share of transfers devoted to the administrative payroll increases with the number of districts into which the territory is divided.

¹⁸The joint placebo test for the unconditional wage component is significant, with a p value of 0.02.

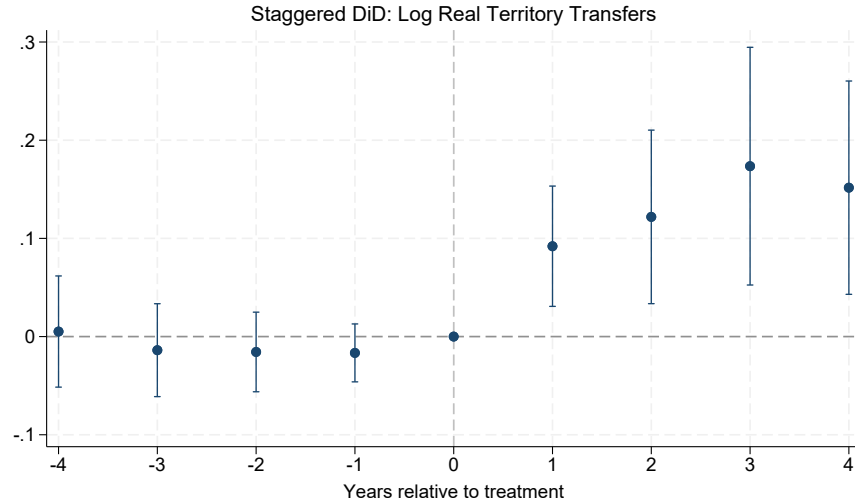


FIGURE 5: Event study: log real territory transfers around the first district split. Balanced panel of 37 1991-mother territories over fiscal years 2012/13–2023/24 (real FY2017/18 shillings); 18 territories treated at the first fiscal year in which a newly created district receives its own budget vote, and 19 control territories with no split during the sample window (their subdivision, where any, was completed before FY2012/13). Estimates from the heterogeneity-robust estimator of [de Chaisemartin & D’Haultfoe uille \(2020\)](#) with four pre- and four post-periods; the joint test of the placebo estimates has a p-value of 0.73. Kampala, Wakiso, and municipal/city councils excluded.

TABLE 11: Administrative Overhead and Territory Fragmentation

	UCG-wage share (1)	Unconditional share (2)
Log number of districts in territory	0.021*** (0.004)	0.020*** (0.007)
Territory FE	Yes	Yes
FY FE	Yes	Yes
Observations	444	444
Territories	37	37

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Balanced panel of 37 1991-mother territories over fiscal years 2012/13–2023/24, aggregating the approved transfers of all constituent districts. The dependent variables are the shares of the territory’s total approved transfers allocated to the wage component of the unconditional block grant (Column 1) and to the unconditional block grant as a whole (Column 2). The regressor is the logarithm of the number of district votes in the territory in each fiscal year. Kampala, Wakiso, and municipal/city councils are excluded throughout. Standard errors clustered at the territory level are in parenthesis.

These results show that there is a limit to the strategy of bringing government closer to citizens through district creation. Each new district requires its own administrative apparatus, so the costs of further subdivision increase with every split, while the additional gains from proximity are likely to diminish. Importantly, these fiscal costs are different from the mechanism identified in the main analysis. The estimated benefits of proximity are not driven by higher service delivery budgets in newly created districts, since the additional resources are concentrated in administrative transfers rather than conditional grants. The evidence therefore suggests that district creation can improve service access, engagement, and accountability, but

does so through a fiscally costly expansion of the state apparatus.

12 Conclusions

This paper provides empirical evidence on the impact of proximity to administrative centers on public service delivery and state-citizen relations in the context of Uganda's redistricting process. By leveraging a governmental rule governing the placement of new district headquarters through an instrumental variable approach, this study identifies the causal effects of households' proximity to new administrative centers using granular georeferenced survey data combined with detailed geographical information on districts.

The findings show that proximity to administrative centers significantly improves access to public goods and services. Households closer to new district headquarters have better access to electricity and high-quality roads, shorter travel times to health facilities, and improved access to water sources. The stronger effects for centrally provided services reflect not only direct infrastructure investments but also the coordinating role of district administrators in channeling and implementing national programs for nearby communities. At the same time, proximity increases engagement through formal institutions, reduces direct corruption experiences, and raises satisfaction with appointed local leaders, consistent with stronger monitoring and accountability when citizens and officials are within closer reach. Comparing survey rounds, these effects strengthen as new administrations consolidate, becoming more pronounced by 2021 than in the years immediately following district creation.

Despite these improvements, proximity to new district headquarters does not translate into higher trust in government or more favorable democratic perceptions. Satisfaction with democracy, perceived political freedoms, rule of law, and trust in both local and national institutions remain largely unchanged. Heterogeneity by political alignment between the elected district chairperson and the centrally appointed RDC sheds further light on this finding. In aligned districts, proximity is associated with stronger service delivery gains, higher participatory engagement, and greater satisfaction with local leaders, with corruption victimization also pointing lower. However, democratic perceptions remain unaffected even in this more favorable environment. In non-aligned districts, the higher access to public goods and services are more limited, and closer distance to headquarters is associated with declining trust in local government and the ruling party, suggesting that proximity can reveal institutional dysfunction. Together, these results suggest that in hybrid regimes, bringing the state closer to citizens can simultaneously improve services and accountability while leaving the underlying political relationship between citizens and the state unchanged.

These benefits come at a fiscal cost. Each new district requires its own administrative apparatus, and the intergovernmental transfer system finances a fixed administrative core regardless of district size. As a result, district creation increases total public spending across the original territory by around 14 percent. This additional spending is concentrated in administrative

overhead, especially the duplicated wage bill of headquarters staff and political officeholders, rather than in service delivery grants. The service improvements are therefore not driven by proportionally larger service budgets. Instead, they appear to reflect a more effective deployment of resources when administration is closer. At the same time, administrative duplication implies diminishing returns to further subdivision, since each additional district absorbs a growing share of transfers into the fixed costs of administration.

One of the main limitations of the instrumental variable approach is that the estimates provide a Local Average Treatment Effect, focusing on compliers — in this context, households whose proximity to the new district's centroid influenced their likelihood of living closer to the new district's capital. While these estimates offer valuable insights into the effects of proximity in newly created districts that followed governmental placement rules, they do not speak to the effects of distance to headquarters in non-split or mother districts.

The findings have several policy implications. Reducing the geographical distance between citizens and administrators can be an effective way to improve service delivery, particularly when accompanied by institutional arrangements that facilitate meaningful participation in local decision-making. Proximity may also strengthen accountability and reduce inefficiencies in public spending. However, the alignment results caution that the governance environment matters: the benefits of administrative proximity are larger and more consistent when elected and appointed local leaders share political incentives. The fiscal evidence adds a further caution: because each new district carries a fixed administrative cost, improving proximity through subdivision faces diminishing returns, and its benefits must be weighed against a growing administrative burden. Finally, while redistricting can be an effective tool for bridging the gap between government and citizens, the evidence that democratic perceptions remain unresponsive suggests that administrative decentralization alone is unlikely to foster deeper democratic legitimacy in hybrid regimes.

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Appendix

A Variables construction

Education. The first set of outcomes relates to education. Like many other public services, education in Uganda is provided at both the central and district levels. While districts oversee primary education, secondary education falls under the jurisdiction of the central government. Therefore, the analysis includes indicators for both education levels.

Although the NSDS collects school attendance data, it does not specify the grade level of children. To address this limitation, I construct two school attendance variables based on schooling age brackets. *School attendance (prim. age)* measures attendance for 4,726 households with children aged 6–12, while *School attendance (second. age)* captures attendance for 3,663 households with children aged 13–18.

Following the same criteria, I construct variables for *Distance to school*, *School travel times*, and measures of perceived quality including *High quality of teachers*, *High quality of facilities*, and *School has an auditor*. The variable *Public school* is an indicator variable that takes the value of one if the household has any school-age children attending a public institution. *School travel time* measures the amount of time, in minutes, that families spend traveling to school.

Health. Healthcare provision in Uganda is divided between the central government and district governments, each responsible for different levels of service. Districts manage health centers, whereas the central government operates hospitals.

At the district level, I analyze several indicators, including *Government health center*, which measures access to public health centers for 2,361 households that vaccinated their children in the past year. *HH member sick/injured* captures the proportion of household members who reported illness or injury within the last 30 days. *Health center travel time* records the number of minutes taken to reach a health center, considering only households that visited a health facility in the last 30 days. At the community level, I assess access to Government health centers, focusing on their proximity *Government health center is within walking distance*, staffing (availability of *Full-time staff*), resources (*Ambulance presence*), management challenges (*Absenteeism concerns*), and oversight (*District-level supervision on a monthly basis*).

For central government provision, the analysis focuses on three indicators: *Government hospital*, *Government hospital within walking distance*, and *Insufficient PHC funds*. The latter captures whether central government funds for primary health care are insufficient in the community.

Water. The third set of outcomes examines water services, which are also provided by both district and central governments. District governments are responsible for supplying water for household consumption, while the central government provides water for productive activities.

Key district-level water indicators include *Improved water source*, which identifies households with access to safe drinking water in both dry and wet seasons, following World Health Organization guidelines (WHO, 2017). *Water distance* records the self-reported distance (in kilometers) from households to their water source. *Water committee* captures whether a functioning water committee exists within the household's sub-county. Additionally, community-level indicators include access to *Piped water*, *Boreholes*, and *Protected wells*.

For central government-provided water, the dataset includes indicators for *Improved water* and *Water distance* for sources used for production (e.g., agriculture, livestock, and fisheries) in both wet and dry seasons. Additional community-level indicators assess access to *Water for crop production*, *Water for cattle production*, and *Water for fish production* within walking distance.

Electricity. Electricity provision falls under the jurisdiction of the central government. The analysis includes *HH has electricity*, which measures household access to electricity, as well as *HH connected to grid* and *HH interested in grid*, which indicate household access to and demand for grid electricity. Community-level indicators include *Community has electricity* and *Community connected to the grid*, measuring electrification at the community level.

Roads. The fifth set of outcomes examines road infrastructure, which is provided at both the central and district levels. The central government focuses on inter-district connectivity, while district governments are responsible for roads linking sub-counties and villages.

Although the NSDS does not explicitly classify roads by provider, the provider type can be inferred from road quality. Central government roads are typically paved (*Tarmac*), whereas district roads are predominantly unpaved (*Murram*). Additionally, district roads are categorized as *Feeder* or *Communal* roads. These household-level information is complemented by community-level information on community leaders' perceptions of *Road quality* and the extent of *Road maintenance* by different levels of government.

Agricultural Inputs. The final set of outcomes assesses the availability and use of agricultural inputs, including *Hybrid seeds*, *Planting materials*, *Pesticides*, and *Fertilizer*. Although the main national input distribution programmes, the National Agricultural Advisory Services (NAADS) and Operation Wealth Creation (OWC), are coordinated by the Ministry of Agriculture, Animal Industry and Fisheries, agricultural extension is a district mandate under the Local Governments Act, and inputs are distributed through district production departments and sub-county extension officers. Household-level variables measure whether these inputs were used in the past year and their availability in the household's sub-county. Additional community-level indicators capture access to *Agricultural markets*, *Livestock markets*, *General markets*, *Agricultural extension services*, *Maize/rice and coffee mills*, and *Milk cooling systems*.

Institutional Knowledge and Traditional Institutions. To measure households' awareness of district-level state institutions, I construct indicator variables from the NSDS capturing whether

the household knows the location of the *Magistrate court*, the *Police*, the *District land office*, the *Office of the Auditor General*, and the *Judiciary*. *Family civil servant* indicates whether a household member is employed as a civil servant. I complement these with measures from the Afrobarometer to capture the balance between formal and traditional institutions. *Resolved by government* and *Resolved by traditional* indicate whether households resolve disputes through government or traditional institutions, respectively. *Contacted MP* and *Contacted traditional leader* indicate whether the respondent contacted a member of parliament or a traditional leader at least once during the past year.

Citizen Engagement. Participatory engagement with the district government is measured with three NSDS indicators. *Involved LC5* captures whether the household is involved in LC5 decision-making and the management of district resources. *Met RDC* indicates whether the household has ever attended an activity organized by the Resident District Commissioner, and *RDC Baraza* whether the RDC has held a baraza (a public accountability meeting) in the community. Contentious and collective forms of engagement come from the Afrobarometer: *Collective civic action* indicates whether the respondent joined others to request government action, while *Protest* and *Contacted media* indicate participation in a demonstration or protest march and contacting the media about politics or community affairs, each measured as any occurrence during the past year.

Satisfaction with Local Administrators. Three NSDS indicators capture households' evaluations of district leaders: *LCV performance*, which indicates a positive evaluation of the elected LCV chairperson's performance; *RDC satisfaction*, which indicates that the household is satisfied or very satisfied with the RDC's work; and *Civil servants performance*, which captures a positive evaluation of district civil servants.

Corruption. Corruption outcomes distinguish awareness from direct experience. From the NSDS, *Nepotism*, *Embezzlement*, and *Absenteeism* indicate whether the household has heard of each form of corruption, while *Victim nepotism*, *Victim embezzlement*, and *Victim absenteeism* indicate whether the household has been a victim of the corresponding practice. Bribery experiences come from the Afrobarometer: *Bribe health*, *Bribe police*, *Bribe avoid police*, and *Bribe ID* indicate whether the respondent paid a bribe at least once during the past year to obtain medical care, to obtain police assistance, to avoid problems with the police, or to obtain an identity document, in each case conditional on having sought the corresponding service.

Democratic Perceptions and Trust. All variables in this group come from the Afrobarometer. *Is democracy* indicates that the respondent considers Uganda a democracy (whether full or with problems), and *Satisf democracy* that the respondent is fairly or very satisfied with the way democracy works in Uganda. *Free vote choice* and *Free join orgs* indicate that the respondent feels somewhat or completely free to choose whom to vote for and to join any political organization. The rule of law perceptions (*President ignores law*, *Unequal law*, *Judges influenced*, and *Officials impunity*) take the value of one when the respondent believes the corresponding

situation occurs often or always. The trust variables (*Trust localgov*, *Trust president*, *Trust parliament*, *Trust ruling party*, *Trust opposition*, and *Trust courts*) indicate that the respondent trusts the corresponding institution somewhat or a lot. The social capital variables in Appendix Table A21 follow the same construction for trust in relatives, neighbours, other citizens, people of other religions, and people of other ethnicities.

Political Alignment. A district is classified as aligned when the elected LCV chairperson and the centrally appointed RDC share political affiliation. Since RDCs are appointed by the president, alignment is measured with an indicator taking the value of one when the district's LCV chairperson elected in the 2016 local elections belongs to the ruling party (NRM). Districts created after 2016 are assigned the affiliation of the LC5 chairperson recorded at their creation.

Summary Indices. Table A1 lists the components entering each of the ten service indices used in the main analysis. Components are standardized before averaging, and variables for which higher values indicate worse access enter with reversed sign, so that higher values of every index correspond to better service access.

TABLE A1: Components of the Summary Indices

Index	Components
Education (district)	School attendance (prim. age); distance to school (rev.); school travel time (rev.); high quality of teachers; high quality of facilities; school has an auditor — all for primary-age children
Education (central gov.)	Same six components, for secondary-age children
Health (district)	Gov. health facility for immunization; HH member sick/injured in last 30 days (rev.); health center travel time (rev.); full-time staff at health center; absenteeism in health center (rev.); motorized ambulance in health center; monthly district supervision of health center
Health (central gov.)	Government hospital available; gov. hospital within walking distance; PHC funds not enough (rev.)
Water (district)	Improved water source (dry and wet seasons); water distance (dry and wet seasons, rev.); water committee; piped water available; borehole available; protected well available
Water (central gov.)	Improved water source for productive use (dry and wet seasons); water distance for productive use (dry and wet seasons, rev.); water for crop, cattle, and fish production within walking distance
Electricity (central gov.)	HH has electricity; HH connected to grid; HH interested in grid; community has electricity
Roads (district)	Sub-county has murram road; sub-county has feeder road; sub-county has communal road (rev.); high quality murram, feeder, and community roads; murram, feeder, and community roads mainly repaired by the district
Roads (central gov.)	Sub-county has tarmac road; high quality tarmac road; tarmac roads mainly repaired by the central government
Agricultural inputs (district)	Use and sub-county availability of hybrid seeds, planting materials, pesticides, and fertilizer

Notes: Components of each Anderson (2008) summary index. Each component is standardized to mean zero and standard deviation one before averaging; components marked (rev.) enter with reversed sign so that higher values of every index indicate better service access. Household-level components come from the NSDS household questionnaire and community-level components (e.g., walking-distance and availability measures) from the community questionnaire.

B Conceptual Framework

This section develops a reduced-form model of public good provision with two local administrators who differ in their accountability relationships and citizens who choose how to engage with the state. The model formalizes how geographical distance to the administrative center and political alignment between local leaders shape service delivery, citizen engagement, and accountability (proxied by corruption).

B.1 Setup

Consider a district with two administrators: an elected Local Council V (LCV) chairperson, indexed by L , who is directly accountable to local voters, and a centrally appointed Resident District Commissioner (RDC), indexed by R , who represents the national government within the district.¹⁹ A household i is located at distance $d_i \geq 0$ from the district headquarters.

Public good production and spatial delivery. Each administrator simultaneously chooses effort $e_j \geq 0$, $j \in \{L, R\}$. Effort translates into higher public goods and services for the district. However, the benefits of this effort are decreasing with distance to the headquarters. The public good available to household i at distance d_i from the headquarters is

$$g_i = (e_L + \alpha e_R) \phi(d_i), \quad (3)$$

where $\alpha \in (0, 1]$ is a *political alignment* parameter governing the share of RDC effort that translates into local public goods, and $\phi(d_i)$ is a *spatial attenuation function* satisfying $\phi(0) = 1$, $\phi'(d_i) < 0$, and $\phi''(d_i) \leq 0$.

The spatial attenuation function $\phi(d_i)$ captures the fact that infrastructure coverage, service quality, and administrative reach decline with distance from the district headquarters. This reflects the empirical patterns that electricity grids and national roads are less available farther from administrative centers, health workers tend to concentrate near headquarters, and oversight of service delivery weakens at the periphery (Asher et al., 2018). Importantly, the attenuation does not reflect a deliberate choice by administrators to favor nearby households. Instead, it reflects the spatial structure of service delivery: administrators choose effort once for the district, and the household-level realization of that effort depends on where the household is located relative to the headquarters.

The alignment parameter α captures the weight the RDC places on local welfare relative to a fixed outside payoff from central loyalty. When the LCV chairperson and the RDC share

¹⁹The distinction between elected and appointed local officials builds on Besley & Coate (2003), who study the trade-offs between these two selection mechanisms. A related dual-principal structure is detailed in the working paper version of Martinez-Bravo et al. (2022) with village chiefs in rural China being simultaneously accountable to villagers and to the autocratic central government.

political affiliation (α close to 1), local welfare dominates the RDC's payoff, and the RDC exerts high effort on locally valued services. When they are not aligned (α is small), the fixed outside payoff (which captures career concerns, patronage rents, or political loyalty rewards) dominates the RDC's payoff, and effort on local services falls accordingly.²⁰

Administrator preferences. Administrators choose effort to maximize their respective payoffs. The LCV, being accountable to local voters, maximizes a payoff that is increasing in aggregate household welfare and decreasing in effort cost:

$$U_L = (e_L + \alpha e_R) \bar{\phi} - \frac{1}{2} e_L^2, \quad (4)$$

where $\bar{\phi} \equiv \int \phi(d) dF(d)$ denotes the average spatial attenuation across the district population. The LCV cares about the welfare of the representative constituent, whose access to public goods depends on $\bar{\phi}$. The RDC internalizes local welfare with weight α and receives a fixed outside benefit $(1 - \alpha)\bar{u}$ from serving the central government's agenda, capturing career concerns, patronage rents, or political loyalty rewards:

$$U_R = \alpha(e_L + \alpha e_R) \bar{\phi} + (1 - \alpha) \bar{u} - \frac{1}{2} e_R^2. \quad (5)$$

B.2 Equilibrium Provision

Administrators choose effort simultaneously in a Nash equilibrium. The first-order conditions are:

$$\text{LCV: } e_L^* = \bar{\phi}, \quad (6)$$

$$\text{RDC: } e_R^* = \alpha^2 \bar{\phi}. \quad (7)$$

The LCV's optimal effort equals $\bar{\phi}$, the average spatial return to effort in the district. The RDC's effort scales with $\alpha^2 \bar{\phi}$. Substituting into (3), the equilibrium public good available to household i is

$$g_i^* = (1 + \alpha^3) \bar{\phi} \phi(d_i). \quad (8)$$

The equilibrium expression separates two components. The term $\bar{\phi}$ captures the district-level supply of public goods: it determines how much effort administrators exert and reflects the average accessibility of the population. The term $\phi(d_i)$ captures the household-level realization: it determines what share of the district's effort effectively reaches household i . Together, they imply that a household benefits from both living in a compact district (high $\bar{\phi}$) and from being

²⁰This specification follows [Martinez-Bravo et al. \(2022\)](#) in modeling local officials under dual accountability, where preference congruence between vertically and locally accountable principals shapes governance outcomes, abstracting from the within-agent reallocation of effort across competing tasks between the central and the local government

located close to the headquarters (high $\phi(d_i)$). In the context of redistricting, the creation of a new district with a centrally located headquarters raises $\bar{\phi}$ (the average distance falls for the splitter district population) and, for many households, also increases $\phi(d_i)$ (the household is now closer to its headquarters). These two forces reinforce each other.

Proposition 1 (Provision decreasing in distance). *Equilibrium public good provision for household i is strictly decreasing in its distance to the headquarters:*

$$\frac{\partial g_i^*}{\partial d_i} = (1 + \alpha^3) \bar{\phi} \phi'(d_i) < 0.$$

Households located closer to the district headquarters have better access to public goods and services. The result follows directly from the spatial attenuation of service delivery: infrastructure and administrative capacity radiate outward from the administrative center, and households farther away receive less of the public good regardless of the effort level chosen by administrators. This prediction is consistent with the government's stated objective of bringing services closer to citizens through district creation.

Proposition 2 (Alignment premium). *Equilibrium provision is strictly increasing in political alignment:*

$$\frac{\partial g_i^*}{\partial \alpha} = 3\alpha^2 \bar{\phi} \phi(d_i) > 0.$$

When the RDC and LCV share political incentives, a greater share of RDC effort is channeled toward locally valued services, raising total provision. The convexity in α implies that the marginal return to alignment is largest when alignment is already high, reflecting the compounding nature of incentive congruence between the two administrators.

Proposition 3 (Complementarity between proximity and alignment). *The marginal benefit of proximity is strictly increasing in alignment:*

$$\frac{\partial}{\partial \alpha} \left(-\frac{\partial g_i^*}{\partial d_i} \right) = -3\alpha^2 \bar{\phi} \phi'(d_i) > 0.$$

This cross-derivative establishes that distance and alignment are complements in the production of local public goods. The intuition is that proximity amplifies the returns to any effort directed at local services, and alignment ensures that a larger share of total effort is so directed. In non-aligned districts, even nearby households benefit less from proximity because RDC effort is partly diverted to central objectives. This predicts that the effects of proximity on service delivery should be stronger in politically aligned districts.

B.3 Citizen Engagement

The improvements in service delivery documented above depend not only on administrative effort but also on the quality of information that administrators receive about local needs.

This section models how proximity shapes citizens' decisions to engage with the state and, consequently, the information available to administrators.

Citizen preferences. A citizen located at distance d_i from the headquarters derives utility from the public good g_i^* and may choose to engage with the state. Engagement takes one of two forms: *participatory* (attending district council meetings, meeting the RDC, using formal complaint and feedback channels) or *contentious* (joining protests, contacting the media, organizing collective pressure campaigns). The citizen's payoff is:

$$V_i = g_i^* + \max \left\{ \underbrace{b_P(\alpha) - c_P(d_i)}_{\text{participatory}}, \underbrace{b_C - \bar{c}_C}_{\text{contentious}}, 0 \right\}, \quad (9)$$

where $b_P(\alpha)$ is the benefit of participatory engagement, $c_P(d_i)$ is its cost, b_C is the benefit of contentious engagement, and $\bar{c}_C$ is the cost of contentious action. The citizen chooses whichever form of engagement yields the highest net benefit, or does not engage at all.

Two features of the cost structure are important. First, the cost of participatory engagement is increasing in distance ($c'_P(d_i) > 0$): attending a district council meeting, visiting the RDC's office, or filing a formal complaint all require traveling to the headquarters, with time and transport costs that rise with d_i . Second, the cost of contentious engagement $\bar{c}_C$ is independent of distance to the headquarters (e.g. protests can be organized locally).

The benefit structure includes the role of alignment. Participatory engagement is assumed to be more effective at influencing policy outcomes ($b_P(\alpha) > b_C$ for sufficiently high α) because it transmits detailed information about local priorities directly to decision-makers, while contentious action is a noisier signal. Additionally, $b_P(\alpha)$ is increasing in alignment: when the RDC internalizes local welfare, the state is more responsive to citizen input, making participatory engagement more rewarding. In non-aligned districts, citizens who engage through formal channels may find their input disregarded by an RDC pursuing central directives, reducing the returns to participation.

Engagement equilibrium. Given these assumptions, there exists a distance threshold $\hat{d}(\alpha)$, defined by $b_P(\alpha) - c_P(\hat{d}) = b_C - \bar{c}_C$, such that citizens at $d_i < \hat{d}(\alpha)$ choose participatory engagement, while citizens at $d_i > \hat{d}(\alpha)$ choose contentious engagement or do not engage at all. Since $b'_P(\alpha) > 0$, the threshold $\hat{d}(\alpha)$ is increasing in alignment: in aligned districts, participatory engagement dominates over a wider range of distances.

Proposition 4 (Proximity shifts engagement from contentious to participatory). *A reduction in distance d_i increases participatory engagement and decreases contentious engagement:*

$$\frac{\partial \Pr(\text{participatory})}{\partial d_i} < 0, \quad \frac{\partial \Pr(\text{contentious})}{\partial d_i} > 0.$$

The mechanism is that proximity reduces the cost of formal engagement with district institutions. When the headquarters is within reach, citizens can attend council meetings and communicate their preferences directly to local administrators, making contentious action unnecessary. When the headquarters is distant, the cost of formal engagement rises and citizens resort to protests and media contact as lower-cost alternatives for voicing demands. The overall pattern is one of substitution: proximity does not suppress citizen voice but rather redirects it from contentious to participatory channels.

Proposition 5 (Alignment amplifies participatory engagement). *The range of distances over which participatory engagement dominates is increasing in alignment: $\hat{d}'(\alpha) > 0$. In aligned districts, proximity produces stronger shifts toward participatory engagement.*

Proof. Differentiating $b_P(\alpha) - c_P(\hat{d}) = b_C - \bar{c}_C$ with respect to α yields $b'_P(\alpha) = c'_P(\hat{d}) \hat{d}'(\alpha)$. Since $b'_P(\alpha) > 0$ and $c'_P(\hat{d}) > 0$, it follows that $\hat{d}'(\alpha) > 0$. $\square$

When the state is responsive to citizen input (high α), the returns to participatory engagement are larger, making citizens willing to bear higher travel costs to engage formally. This predicts that the reduction in contentious engagement and the increase in participatory engagement from proximity should be more pronounced in politically aligned districts. In non-aligned districts, the lower responsiveness of the state limits the returns to formal engagement, and citizens may not shift away from contentious channels even when the headquarters is nearby.

B.4 Corruption and Monitoring

A further channel through which proximity affects public service delivery is accountability. After effort choices are made, each administrator can divert a fraction $\kappa_j \in [0, \bar{\kappa}]$ of resources for private gain, yielding private benefit $b \cdot \kappa_j$. Citizens monitor administrators with intensity

$$\mu(d_i), \quad \text{with } \mu'(d_i) < 0 \text{ and } \mu(0) = \bar{\mu} \leq 1, \quad (10)$$

so that closer households are better able to observe and report misconduct. Detection triggers a penalty $P > 0$. The monitoring-through-proximity mechanism builds on [Campante & Do \(2014\)](#), who show that isolated capital cities are associated with higher corruption, and with [Narasimhan & Weaver \(2024\)](#), who show that polity size can improve public service delivery under political accountability mechanisms. In the Ugandan context, proximity to the headquarters determines how effectively citizens can exercise this informational advantage.²¹

The RDC's net payoff from corruption is $b \kappa_R - \mu(d_i) P \kappa_R - \alpha \lambda \kappa_R$, where $\lambda > 0$ is the marginal welfare cost of diversion, internalized at rate α . Since the payoff is linear in κ_R , the

²¹Experimental evidence from Uganda supports the accountability channel modeled here. [Björkman & Svensson \(2009\)](#) show that community-based monitoring significantly improves the quality of health service delivery in Ugandan health facilities, suggesting that citizen oversight can meaningfully discipline local officials.

RDC diverts resources if and only if $b - \alpha \lambda > \mu(d_i) P$. Two forces discipline the administrator: external monitoring by citizens ($\mu(d_i) P$) and internal alignment with local welfare ($\alpha \lambda$). Both are stronger when the household is closer and when the district is politically aligned.

Proposition 6 (Proximity deters corruption). *There exists a distance threshold $\hat{d}_c$ defined by $b - \alpha \lambda = \mu(\hat{d}_c) P$ such that corruption is deterred for $d_i < \hat{d}_c$ and occurs for $d_i > \hat{d}_c$. Since $\mu'(d_i) < 0$, households closer to the headquarters experience less corruption.*

Proposition 7 (Alignment amplifies the accountability effect of proximity). *The corruption threshold $\hat{d}_c(\alpha)$ is increasing in alignment: $\partial \hat{d}_c / \partial \alpha > 0$. In aligned districts, corruption is deterred over a wider range of distances.*

Proof. Implicit differentiation of $b - \alpha \lambda = \mu(\hat{d}_c) P$ with respect to α gives $-\lambda = \mu'(\hat{d}_c) P \frac{\partial \hat{d}_c}{\partial \alpha}$. Since $\lambda > 0$, $\mu'(\hat{d}_c) < 0$, and $P > 0$, it follows that $\frac{\partial \hat{d}_c}{\partial \alpha} = \frac{\lambda}{-\mu'(\hat{d}_c) P} > 0$. $\square$

Alignment introduces an additional deterrent beyond citizen monitoring: an aligned RDC internalizes the welfare cost of corruption, making diversion less attractive even at moderate distances from the headquarters. Taken together, these two propositions predict that corruption should be lower in areas closer to the administrative center, and that this reduction should be more pronounced in politically aligned districts.

Discussion. The model predicts that proximity to district headquarters improves public service delivery, shifts citizen engagement from contentious to participatory channels, and deters corruption. Political alignment between the elected LCV and the appointed RDC amplifies each of these effects. The main results in Section 6 test these predictions.

C Proofs of Propositions

C.1 Derivation of the Equilibrium

The administrators choose effort simultaneously in a Nash equilibrium. Each administrator takes the other's effort as given and maximizes their own payoff.

LCV's problem. The LCV chooses e_L to maximize (4). The first-order condition is:

$$\frac{\partial U_L}{\partial e_L} = \bar{\phi} - e_L = 0 \implies e_L^* = \bar{\phi}. \quad (11)$$

RDC's problem. The RDC chooses e_R to maximize (5). The first-order condition is:

$$\frac{\partial U_R}{\partial e_R} = \alpha^2 \bar{\phi} - e_R = 0 \implies e_R^* = \alpha^2 \bar{\phi}. \quad (12)$$

Equilibrium public good. Substituting $e_L^* = \bar{\phi}$ and $e_R^* = \alpha^2 \bar{\phi}$ into (3):

$$\begin{aligned} g_i^* &= (e_L^* + \alpha e_R^*) \phi(d_i) \\ &= (\bar{\phi} + \alpha \cdot \alpha^2 \bar{\phi}) \phi(d_i) \\ &= \bar{\phi} (1 + \alpha^3) \phi(d_i). \end{aligned} \tag{13}$$

C.2 Proof of Proposition 1: Provision Decreasing in Distance

Statement. Equilibrium public good provision for household i is strictly decreasing in its distance to the headquarters: $\frac{\partial g_i^*}{\partial d_i} < 0$.

Proof. Differentiating (13) with respect to d_i :

$$\frac{\partial g_i^*}{\partial d_i} = (1 + \alpha^3) \bar{\phi} \phi'(d_i).$$

Since $\alpha \in (0, 1]$, the term $(1 + \alpha^3) > 0$. Since $\bar{\phi} > 0$ by construction. Since $\phi'(d_i) < 0$ by assumption. Therefore:

$$\frac{\partial g_i^*}{\partial d_i} = \underbrace{(1 + \alpha^3)}_{>0} \underbrace{\bar{\phi}}_{>0} \underbrace{\phi'(d_i)}_{<0} < 0.$$

C.3 Proof of Proposition 2: Alignment Premium

Statement. Equilibrium provision is strictly increasing in political alignment: $\frac{\partial g_i^*}{\partial \alpha} > 0$.

Proof. Differentiating (13) with respect to α :

$$\frac{\partial g_i^*}{\partial \alpha} = \bar{\phi} \phi(d_i) \frac{\partial}{\partial \alpha} (1 + \alpha^3) = 3\alpha^2 \bar{\phi} \phi(d_i).$$

Since $\alpha \in (0, 1]$, $\bar{\phi} > 0$, and $\phi(d_i) > 0$ for finite d_i :

$$\frac{\partial g_i^*}{\partial \alpha} = \underbrace{3\alpha^2}_{>0} \underbrace{\bar{\phi}}_{>0} \underbrace{\phi(d_i)}_{>0} > 0.$$

C.4 Proof of Proposition 3: Complementarity Between Proximity and Alignment

Statement. The marginal benefit of proximity is strictly increasing in alignment: $\frac{\partial}{\partial \alpha} \left(-\frac{\partial g_i^*}{\partial d_i} \right) > 0$.

Proof. The marginal benefit of proximity is the negative of the distance derivative (since lower d_i means higher proximity):

$$-\frac{\partial g_i^*}{\partial d_i} = -(1 + \alpha^3) \bar{\phi} \phi'(d_i).$$

This expression is positive since $\phi'(d_i) < 0$. Differentiating with respect to α :

$$\frac{\partial}{\partial \alpha} \left(-\frac{\partial g_i^*}{\partial d_i} \right) = -3\alpha^2 \bar{\phi} \phi'(d_i).$$

Since $\alpha^2 > 0$, $\bar{\phi} > 0$, and $\phi'(d_i) < 0$:

$$\frac{\partial}{\partial \alpha} \left(-\frac{\partial g_i^*}{\partial d_i} \right) = - \underbrace{3\alpha^2}_{>0} \underbrace{\bar{\phi}}_{>0} \underbrace{\phi'(d_i)}_{<0} > 0.$$

The positive cross-derivative establishes that distance and alignment are complements: proximity amplifies the returns to any effort directed at local services, and alignment ensures that a larger share of total effort is so directed.

C.5 Proof of Proposition 4: Proximity Shifts Engagement from Contentious to Participatory

Statement. A reduction in distance d_i increases participatory engagement and decreases contentious engagement.

Setup. The citizen at distance d_i chooses among three options: participatory engagement (net benefit $b_P(\alpha) - c_P(d_i)$), contentious engagement (net benefit $b_C - \bar{c}_C$), or no engagement (net benefit 0). The citizen chooses the option with the highest net payoff.

Participatory engagement dominates contentious engagement if and only if:

$$b_P(\alpha) - c_P(d_i) > b_C - \bar{c}_C. \quad (14)$$

Rearranging:

$$c_P(d_i) < b_P(\alpha) - b_C + \bar{c}_C.$$

Define the threshold $\hat{d}(\alpha)$ as the value of d_i at which the citizen is indifferent between the two forms of engagement:

$$c_P(\hat{d}(\alpha)) = b_P(\alpha) - b_C + \bar{c}_C. \quad (15)$$

Proof. Since $c'_P(d_i) > 0$ (the cost of participatory engagement is increasing in distance), the function c_P is strictly increasing. Therefore, condition (14) holds if and only if $d_i < \hat{d}(\alpha)$. This

means:

- For $d_i < \hat{d}(\alpha)$: the citizen chooses participatory engagement (c_P is low enough).
- For $d_i > \hat{d}(\alpha)$: the citizen chooses contentious engagement (if $b_C - \bar{c}_C > 0$) or does not engage.

A reduction in d_i moves the citizen from the contentious to the participatory region whenever d_i crosses $\hat{d}(\alpha)$ from above. For a population of citizens with a continuous distribution of distances $F(d)$:

$$\Pr(\text{participatory}) = F(\hat{d}(\alpha)), \quad \Pr(\text{contentious}) = 1 - F(\hat{d}(\alpha)) - \Pr(\text{no engagement}).$$

Since F is increasing, a reduction in the distance of an individual citizen (which shifts probability mass below $\hat{d}$) increases the probability of participatory engagement and decreases the probability of contentious engagement.

C.6 Proof of Proposition 5: Alignment Amplifies Participatory Engagement

Statement. The range of distances over which participatory engagement dominates is increasing in alignment: $\hat{d}'(\alpha) > 0$.

Proof. Differentiate the indifference condition (15) with respect to α :

$$c'_P(\hat{d}(\alpha)) \hat{d}'(\alpha) = b'_P(\alpha).$$

Solving for $\hat{d}'(\alpha)$:

$$\hat{d}'(\alpha) = \frac{b'_P(\alpha)}{c'_P(\hat{d}(\alpha))}.$$

Since $b'_P(\alpha) > 0$ (the benefit of participatory engagement is increasing in alignment, as a more responsive state makes formal engagement more rewarding) and $c'_P(\hat{d}) > 0$ (the cost of participatory engagement is increasing in distance):

$$\hat{d}'(\alpha) = \frac{\overbrace{b'_P(\alpha)}^{>0}}{\underbrace{c'_P(\hat{d}(\alpha))}_{>0}} > 0.$$

The positive sign means that higher alignment expands the set of distances for which participatory engagement is preferred. In aligned districts, citizens are willing to bear higher travel costs to engage formally because they expect the state to be more responsive to their input.

C.7 Proof of Proposition 6: Proximity Deters Corruption

Statement. There exists a distance threshold $\hat{d}_c$ such that corruption is deterred for $d_i < \hat{d}_c$ and occurs for $d_i > \hat{d}_c$.

Setup. After effort choices are made, the RDC can divert a fraction $\kappa_R \in [0, \bar{\kappa}]$ of resources for private gain. The RDC's net payoff from corruption is:

$$\Pi(\kappa_R) = b \kappa_R - \mu(d_i) P \kappa_R - \alpha \lambda \kappa_R = [b - \mu(d_i) P - \alpha \lambda] \kappa_R,$$

where $b > 0$ is the private benefit per unit diverted, $\mu(d_i)$ is the monitoring intensity with $\mu'(d_i) < 0$ and $\mu(0) = \bar{\mu}$, $P > 0$ is the penalty upon detection, and $\lambda > 0$ is the marginal welfare cost of diversion internalized at rate α .

Proof. Since $\Pi(\kappa_R)$ is linear in κ_R , the RDC's optimal strategy is a corner solution:

$$\kappa_R^* = \begin{cases} \bar{\kappa} & \text{if } b - \alpha \lambda > \mu(d_i) P, \\ 0 & \text{if } b - \alpha \lambda \leq \mu(d_i) P. \end{cases}$$

Corruption is deterred whenever $\mu(d_i) P \geq b - \alpha \lambda$, i.e., when monitoring is strong enough relative to the net temptation to divert. Assuming $b - \alpha \lambda > 0$ (corruption is tempting absent monitoring) and $\mu(0) P > b - \alpha \lambda$ (monitoring deters corruption at zero distance), there exists a unique threshold $\hat{d}_c$ solving:

$$\mu(\hat{d}_c) P = b - \alpha \lambda. \quad (16)$$

The threshold exists and is unique because $\mu(\cdot)$ is continuous and strictly decreasing. Since $\mu'(d_i) < 0$, monitoring weakens with distance, and therefore:

- For $d_i < \hat{d}_c$: $\mu(d_i) > \mu(\hat{d}_c)$, so $\mu(d_i) P > b - \alpha \lambda$, and corruption is deterred ($\kappa_R^* = 0$).
- For $d_i > \hat{d}_c$: $\mu(d_i) < \mu(\hat{d}_c)$, so $\mu(d_i) P < b - \alpha \lambda$, and corruption occurs ($\kappa_R^* = \bar{\kappa}$).

C.8 Proof of Proposition 7: Alignment Amplifies the Accountability Effect of Proximity

Statement. The corruption threshold $\hat{d}_c(\alpha)$ is increasing in alignment: $\partial \hat{d}_c / \partial \alpha > 0$.

Proof. Differentiate the threshold condition (16) with respect to α :

$$\mu'(\hat{d}_c) P \frac{\partial \hat{d}_c}{\partial \alpha} = -\lambda.$$

Solving for $\frac{\partial \hat{d}_c}{\partial \alpha}$:

$$\frac{\partial \hat{d}_c}{\partial \alpha} = \frac{-\lambda}{\mu'(\hat{d}_c) P} = \frac{\lambda}{-\mu'(\hat{d}_c) P}.$$

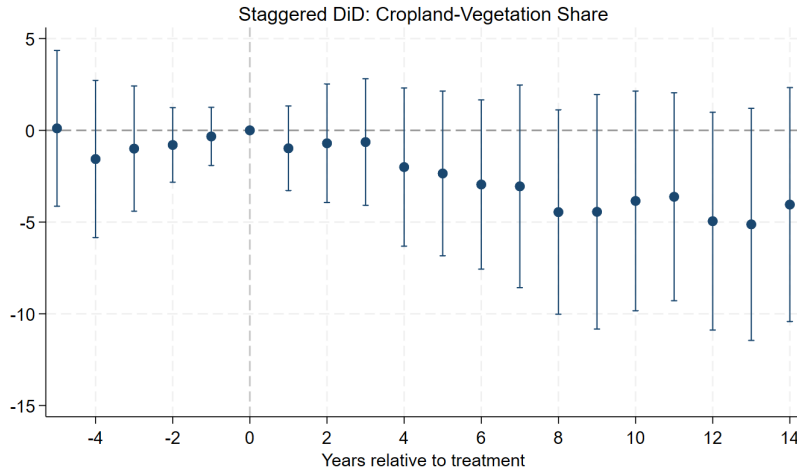
Since $\lambda > 0$, $-\mu'(\hat{d}_c) > 0$ (because $\mu'(\hat{d}_c) < 0$), and $P > 0$:

$$\frac{\partial \hat{d}_c}{\partial \alpha} = \frac{\overbrace{\lambda}^{>0}}{\underbrace{(-\mu'(\hat{d}_c))}_{>0} \underbrace{P}_{>0}} > 0.$$

A higher α increases the range of distances over which corruption is deterred. The mechanism is that an aligned RDC internalizes a larger share of the welfare cost of diversion ($\alpha\lambda$ is larger), which reduces the net temptation to engage in corruption ($b - \alpha\lambda$ falls). This means that a lower level of external monitoring is sufficient to deter corruption, extending the deterrence region to more distant households. In aligned districts, both the internal alignment cost and the external monitoring force work together to discipline administrators, generating stronger accountability effects of proximity.

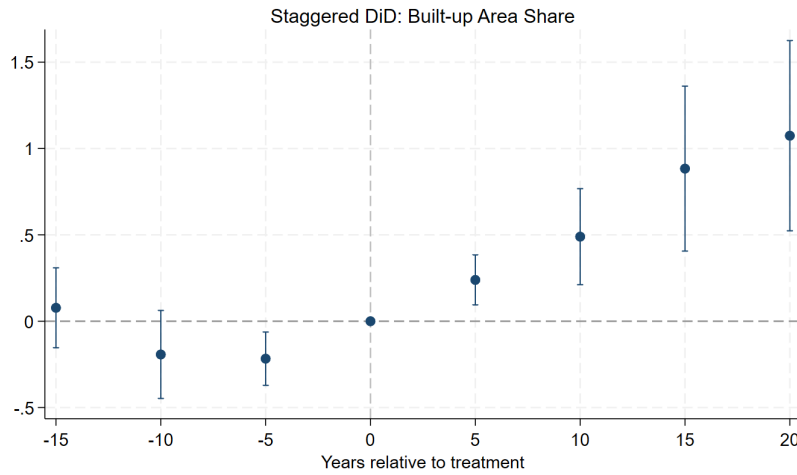
D Figures

FIGURE A1: EVENT STUDY: DISTRICT HEADQUARTER CREATION ON PARISH CROPLAND-VEGETATION SHARE



Notes: Event study graph of the estimated coefficients of the [De Chaisemartin & d'Haultfoeuille \(2020\)](#) staggered difference in difference estimator of cropland-vegetation share on headquarters creation.

FIGURE A2: EVENT STUDY: DISTRICT HEADQUARTERS CREATION ON BUILT-UP AREA SHARE



Notes: Event study of the [De Chaisemartin & d'Haultfoeuille \(2020\)](#) staggered difference-in-differences estimator of the parish built-up area share on district headquarters creation. The outcome is the share of a parish that is built-up, measured from the JRC Global Human Settlement Layer built-up surface product (GHS-BUILT-S; 100 m resolution), available in five-year epochs from 1990 to 2025. Event time is measured in five-year epochs and shown in years on the horizontal axis: the three points to the left of zero are placebo (pre-trend) estimates and the four points to the right are the post-creation effects. Vertical bars are 95 percent confidence intervals from standard errors clustered at the parish level.

E Robustness Checks

TABLE A2: FUTURE HEADQUARTERS' PLACEMENT: POPULATION DENSITY AND NIGHT LIGHTS

	New headquarter in 2014-2020			
	(1)	(2)	(3)	(4)
Population Density in 2010	-0.000231 (0.000873)	0.000458 (0.000933)		
Night Lights in 2013			-0.142 (0.346)	-0.0745 (0.345)
Mean of Dependent Variable	0.135	0.135	0.135	0.135
Observations	178	178	178	178
District FE	No	Yes	No	Yes

Notes: * p<0.1, ** p<0.05, *** p<0.01. Logistic regression of future district headquarters' placement (2014-2020) on population density as of 2010 and night lights as of 2013. Each observation corresponds to one sub-county of the 24 newly created districts between 2014 and 2020. Robust standard errors clustered at the district level in parenthesis.

TABLE A3: FIRST STAGE: DISTANCE HOUSEHOLD - LARGE TOWN

	Log SR Proximity Household - District Headquarters Improved water source				
	(1)	(2)	(3)	(4)	(5)
Log Proximity HH-DC × New District.91.20	0.429*** (0.080)	0.383*** (0.081)	0.437*** (0.082)	0.425*** (0.082)	0.373*** (0.085)
Log Proximity HH - District's centroid	-0.051 (0.063)	-0.002 (0.065)	-0.057 (0.064)	-0.047 (0.065)	0.007 (0.069)
New district between 1991-2020	1.696*** (0.220)	1.565*** (0.221)	1.712*** (0.222)	1.677*** (0.227)	1.516*** (0.236)
Log Distance Household - Closest Large City		-0.090*** (0.027)			-0.099*** (0.029)
Log Distance Household - Closest Large City within Region			-0.063** (0.030)		-0.057* (0.032)
Log Distance Household - Kampala				0.065 (0.073)	0.174** (0.073)
Constant	-0.857*** (0.270)	-0.622** (0.266)	-0.760*** (0.264)	-1.086*** (0.321)	-1.126*** (0.307)
Mean of Dependent Variable	-2.686	-2.686	-2.686	-2.686	-2.686
Observations	8,417	8,417	8,417	8,417	8,417
Number of Sub-counties	967	967	967	967	967
IV F-stat Kleibergen-Paap rk Wald	28.745	22.546	28.509	27.138	19.266
Controls	Yes	Yes	Yes	Yes	Yes

Notes: * p<0.1, ** p<0.05, *** p<0.01. Controls include: district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's distance to district's capital as of 1991 (in log.), district contains a lake other than Lake Victoria, sub-county mean elevation (in m.a.s.l.), four major regions. Standard errors are clustered at the Parish level.

TABLE A4: Placebo Outcomes and Proximity to District Headquarters (2SLS)

	Cookstove (1)	Air transport (2)	Minerals (3)
Log Proximity to District HQ	-0.016 (0.033)	-0.002 (0.007)	-0.022 (0.018)
Observations	8415	8417	8415
Number of clusters	967	967	967
Mean of DV	0.929	0.006	0.019
IV F-stat (KP rk Wald)	28.749	28.745	28.683

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of placebo variables on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

TABLE A5: Contact with Local Government Institutions and Proximity to District Headquarters (2SLS)

	Knows LCI (1)	Distance LCI (2)	Knows LCII (3)	Distance LCII (4)	Knows LCIII (5)	Distance LCIII (6)
<i>Panel A: State presence</i>						
Log Proximity to District HQ	0.022 (0.036)	0.112 (0.160)	0.107 (0.104)	-0.851** (0.404)	0.065 (0.090)	-0.065 (0.090)
p-val	[0.528]	[0.482]	[0.304]	[0.035]	[0.471]	[0.471]
q-val	{0.785}	{0.785}	{0.785}	{0.268}	{0.785}	{0.785}
Observations	8414	8044	8413	4286	8413	8413
Number of clusters	967	966	967	901	967	967
Mean of DV	0.956	0.651	0.509	1.945	0.485	1.515
IV F-stat (KP rk Wald)	28.597	25.897	28.592	27.234	28.592	28.592
	HH decision involvement (1)	Ever in LC1 (2)	LC1 meeting freq (3)	Attends LC1 (4)	LC1 performance (5)	LC1 quality change (6)
<i>Panel B: Citizen participation</i>						
Log Proximity to District HQ	0.136* (0.070)	0.009 (0.045)	-0.129* (0.075)	-0.013 (0.076)	0.064 (0.069)	0.011 (0.025)
p-val	[0.050]	[0.843]	[0.085]	[0.865]	[0.357]	[0.658]
q-val	{0.344}	{1.000}	{0.344}	{1.000}	{0.909}	{1.000}
Observations	8414	8417	8417	8417	8217	8143
Number of clusters	967	967	967	967	967	967
Mean of DV	0.471	0.097	0.190	0.306	0.835	0.962
IV F-stat (KP rk Wald)	28.655	28.745	28.745	28.745	29.514	29.509

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of citizens' contact with local government institutions on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis. Unadjusted p-values are in brackets. Anderson's sharpened False Discovery Rate (FDR) q-values, computed within each panel, are in curly brackets.

TABLE A6: Effects on Services — PA Distance Instrument (2SLS)

	Schooling (1)	Health (2)	Water (3)	Roads (4)	Agr inputs (5)
<i>Panel A: District services</i>					
Log Proximity to District HQ	0.222 (0.156)	0.204* (0.115)	0.329** (0.137)	0.019 (0.102)	-0.042 (0.122)
Observations	7170	7627	8529	8529	8529
Number of clusters	974	966	984	984	984
Mean of DV	-0.002	-0.020	-0.005	0.006	0.027
IV F-stat (KP rk Wald)	25.481	26.367	24.385	24.385	24.385
	Schooling (1)	Health (2)	Water (3)	Roads (4)	Electricity (5)
<i>Panel B: Central govt services</i>					
Log Proximity to District HQ	0.004 (0.188)	0.264 (0.216)	0.307*** (0.119)	0.886*** (0.229)	0.353*** (0.113)
Observations	5582	8363	8529	8529	8529
Number of clusters	967	963	984	984	984
Mean of DV	-0.001	0.016	-0.015	-0.061	-0.038
IV F-stat (KP rk Wald)	16.993	23.989	24.385	24.385	24.385

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Coefficients from regressions of outcome index variables on self-reported household's distance to the district's capital. The indices are constructed to have a mean of 0 and a standard deviation of 1. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's parish centroid to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

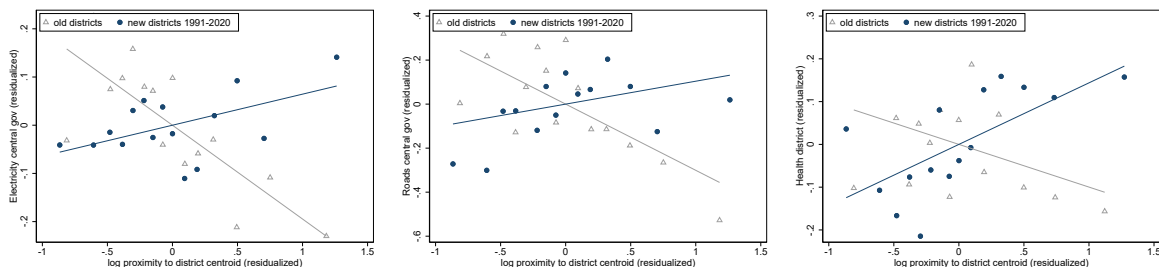


FIGURE A3: Reduced forms: binned scatters of the residualized electricity (left), roads — central government (centre), and health — district (right) indices against residualized proximity to the district centroid, separately for households in new districts (1991–2020) and old districts. Construction as in Figure 3.

TABLE A7: Effects on Services — New Districts Only (2SLS)

	Schooling (1)	Health (2)	Water (3)	Roads (4)	Agr inputs (5)
<i>Panel A: District services</i>					
Log Proximity to District HQ	0.155 (0.113)	0.317*** (0.093)	0.049 (0.080)	0.143* (0.081)	0.094 (0.079)
Observations	3692	3973	4242	4242	4242
Number of clusters	472	470	474	474	474
Mean of DV	-0.045	-0.073	-0.067	0.063	0.126
IV F-stat (KP rk Wald)	49.487	47.274	45.888	45.888	45.888
	Schooling (1)	Health (2)	Water (3)	Roads (4)	Electricity (5)
<i>Panel B: Central govt services</i>					
Log Proximity to District HQ	0.255* (0.132)	0.156 (0.127)	0.080 (0.073)	0.246 (0.165)	0.149** (0.066)
Observations	2857	4242	4242	4242	4242
Number of clusters	470	474	474	474	474
Mean of DV	-0.015	-0.234	-0.080	-0.288	-0.171
IV F-stat (KP rk Wald)	28.925	45.888	45.888	45.888	45.888

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Coefficients from regressions of outcome index variables on self-reported household's distance to the district's capital. The indices are constructed to have a mean of 0 and a standard deviation of 1. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid. Standard errors clustered at the Parish level are in parenthesis.

TABLE A8: Donut Specifications: Excluding Households Near District Headquarters

	Baseline (1)	Excl. HQ parish (2)	Excl. <2 km (3)	Excl. <5 km (4)
<i>Panel A: Electricity (central government)</i>				
Log Proximity to District HQ	0.428*** (0.137)	0.413*** (0.136)	0.453*** (0.164)	0.464** (0.188)
IV F-stat (KP rk Wald)	28.75	29.88	22.32	21.77
Observations	8,417	8,152	7,502	6,123
<i>Panel B: Roads (central government)</i>				
Log Proximity to District HQ	0.706*** (0.261)	0.689*** (0.262)	0.984*** (0.305)	1.115*** (0.357)
IV F-stat (KP rk Wald)	28.75	29.88	22.32	21.77
Observations	8,417	8,152	7,502	6,123
<i>Panel C: Health (district)</i>				
Log Proximity to District HQ	0.387*** (0.121)	0.376*** (0.120)	0.377*** (0.136)	0.301* (0.156)
IV F-stat (KP rk Wald)	33.35	34.48	27.14	22.79
Observations	7,595	7,378	6,838	5,700

Notes: *p<0.1, **p<0.05, ***p<0.01. Each column re-estimates the main 2SLS specification excluding households near their district headquarters: Column 2 drops enumeration areas located in the same parish as the district headquarters point (35 enumeration areas); Columns 3 and 4 drop enumeration areas within 2 and 5 kilometres of the headquarters (125 and 304 enumeration areas, respectively). All regressions use the main specification's controls, weights, and clustering; Kampala and Wakiso are excluded as in the main tables. Standard errors clustered at the enumeration area level are in parenthesis.

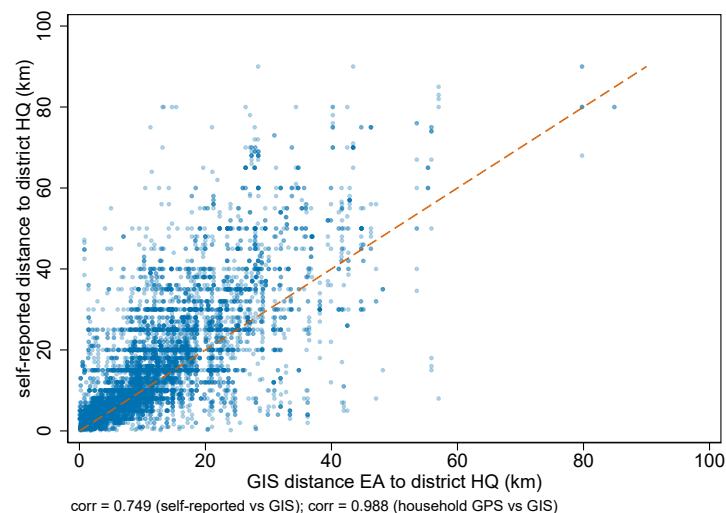


FIGURE A4: Calibration of the self-reported distance measure: household self-reported distance to the district headquarters against the GIS straight-line distance from the household's enumeration area to the headquarters point (in kilometres). The dashed line is the 45-degree line. The correlation between the two measures is 0.75; the correlation between household GPS-based and enumeration area GIS distances is 0.99, indicating that the gap reflects self-reporting rather than geocoding.

TABLE A9: Alternative Endogenous Variable: Self-Reported vs. GIS Proximity to District Headquarters

	2SLS Self-reported (1)	2SLS GIS (2)	OLS Self-reported (3)	OLS GIS (4)
<i>Panel A: Electricity (central government)</i>				
Log Proximity to District HQ	0.428*** (0.137)	0.557** (0.240)	0.102*** (0.020)	0.121*** (0.028)
IV F-stat (KP rk Wald)	28.75	21.19	—	—
Observations	8,417	9,313	8,417	9,313
<i>Panel B: Roads (central government)</i>				
Log Proximity to District HQ	0.706*** (0.261)	0.924** (0.401)	0.222*** (0.035)	0.165*** (0.051)
IV F-stat (KP rk Wald)	28.75	21.18	—	—
Observations	8,417	9,303	8,417	9,303
<i>Panel C: Health (district)</i>				
Log Proximity to District HQ	0.387*** (0.121)	0.593*** (0.205)	0.111*** (0.020)	0.128*** (0.033)
IV F-stat (KP rk Wald)	33.35	20.10	—	—
Observations	7,595	8,320	7,595	8,320

Notes: *p<0.1, **p<0.05, ***p<0.01. Columns 1 and 3 use the household's self-reported proximity to the district headquarters as the variable of interest; Columns 2 and 4 replace it with the GIS straight-line proximity from the household's enumeration area to the headquarters point. The instrument, controls, weights, and clustering follow the main specification; Kampala and Wakiso are excluded as in the main tables. The GIS measure removes self-reporting error but provides less variation, as it does not reflect transportation networks; first-stage F-statistics nevertheless remain above 20 for both measures. Standard errors clustered at the enumeration area level are in parenthesis.

TABLE A10: OLS and 2SLS Estimates: Effects on Services

	School (1)	Health (2)	Water (3)	Roads (4)	Agr inputs (5)
<i>Panel A: NSDS 2021 District services</i>					
OLS	0.060** (0.028)	0.111*** (0.020)	0.054*** (0.021)	0.003 (0.016)	-0.005 (0.019)
2SLS	0.282 (0.214)	0.387*** (0.121)	0.279* (0.157)	0.089 (0.110)	-0.085 (0.145)
IV F-stat (KP rk Wald)	22.24	33.35	28.75	28.75	28.75
Observations	7,137	7,595	8,417	8,417	8,417
	School (1)	Health (2)	Water (3)	Roads (4)	Electricity (5)
<i>Panel B: NSDS 2021 Central govt services</i>					
OLS	0.090*** (0.026)	0.215*** (0.033)	0.073*** (0.019)	0.222*** (0.035)	0.102*** (0.020)
2SLS	0.151 (0.192)	0.233 (0.201)	0.372** (0.146)	0.706*** (0.261)	0.428*** (0.137)
IV F-stat (KP rk Wald)	22.62	28.75	28.75	28.75	28.75
Observations	5,553	8,417	8,417	8,417	8,417

Notes: *p<0.1, **p<0.05, ***p<0.01. Each column reports the coefficient on the household's (log) self-reported proximity to the district headquarters from two estimators: OLS, and the paper's main 2SLS specification instrumenting proximity with the interaction between the enumeration area's log proximity to the district centroid and the new-district indicator. The 2SLS rows reproduce the main results tables. Both estimators use the same controls, sampling weights, and clustering; Kampala and Wakiso are excluded as in the main tables. Standard errors clustered at the enumeration area level are in parenthesis.

TABLE A11: Weak-Instrument-Robust Inference in the Alignment Subsamples

	Electricity (1)	Roads (central) (2)	Health (district) (3)
<i>Panel A: Aligned districts</i>			
Log Proximity to District HQ	0.424** (0.193)	1.013*** (0.384)	0.324* (0.167)
AR 95% confidence set	[0.05, 0.96]	[0.31, 2.12]	[0.00, 0.75]
IV F-stat (KP rk Wald)	14.42	14.42	16.99
Observations	6,266	6,266	5,639
<i>Panel B: Non-aligned districts</i>			
Log Proximity to District HQ	0.254 (0.167)	0.202 (0.274)	0.383** (0.157)
AR 95% confidence set	[−0.07, 0.78]	[−0.39, 0.93]	[0.11, 0.91]
IV F-stat (KP rk Wald)	12.49	12.49	11.81
Observations	2,151	2,151	1,956

Notes: *p<0.1, **p<0.05, ***p<0.01. 2SLS estimates of the main specification in the subsamples of districts where the elected LCV and the appointed RDC share political affiliation (Panel A) and where they do not (Panel B). Because the first-stage F-statistics fall below conventional thresholds in these subsamples, the table reports weak-instrument-robust Anderson–Rubin 95 percent confidence sets (cluster-robust) alongside the conventional estimates. All regressions use the main specification’s controls, weights, and clustering; Kampala and Wakiso are excluded as in the main tables. Standard errors clustered at the enumeration area level are in parenthesis.

F Recentered Instrument and Randomization Inference

The instrument interacts an arguably exogenous shock, the placement of district headquarters following district creation, with non random exposure: a household's fixed location relative to the geometry of its mother district. Even if the realized partition is as good as random among the feasible partitions, households located near the geometric interior of any feasible fragment mechanically receive high instrument values under every counterfactual configuration (Borusyak & Hull, 2023). This appendix implements the correction proposed by Borusyak & Hull (2023). I compute the expected instrument μ_i across counterfactual redistricting configurations, control for it directly, construct the recentered instrument $z_i - \mu_i$, and conduct randomization inference over placebo redistrictings.

Counterfactual redistricting design

The building blocks are the 1991 county borders,²² which were fixed before the entire redistricting wave and therefore cannot respond to it. Holding fixed which mothers split and into how many terminal 2020 districts, I compute, for every splitting mother, all feasible partitions of its counties into the realized number of contiguous fragments. The feasible sets range from a single partition (eleven mothers whose splits had no spatial alternative at the county level) to 5,822 partitions for Mbarara. The computation of the counterfactuals is exhaustive everywhere, so μ_i is an exact expectation rather than a simulation average. In each counterfactual, the fragment containing the 1991 district headquarters remains the mother district, all other fragments are new districts, and the instrument is recomputed exactly as in the actual data: the logarithm of the household's proximity to its fragment's centroid, interacted with the new-district indicator.²³ Figure A5 illustrates the design for Mbarara, the mother district with the richest counterfactual set.

The counterfactual redistrictings substantially reshuffle households. Among enumeration areas in mother districts that split, 47 percent are assigned to a different fragment under at least one feasible alternative partition. As a result, the recentered instrument retains about two thirds of the standard deviation of the raw instrument, leaving meaningful variation to identify the effect. At the same time, a large share of the raw instrument is predictable from a household's fixed position in the geometry of the mother district, with the correlation between z_i and the expected instrument μ_i reaching 0.74.

²²IPUMS shapefile for the 1991 Uganda census (156 counties).

²³As in the main analysis, centroids falling outside half-moon shaped fragments are forced to lie within the polygon. Randomization inference draws one partition per mother per configuration, using independent random streams from a single seed, so that each of the 1,000 draws constitutes one national counterfactual configuration.

Redrawing Mbarara: actual vs counterfactual county partitions (5,822 feasible partitions)

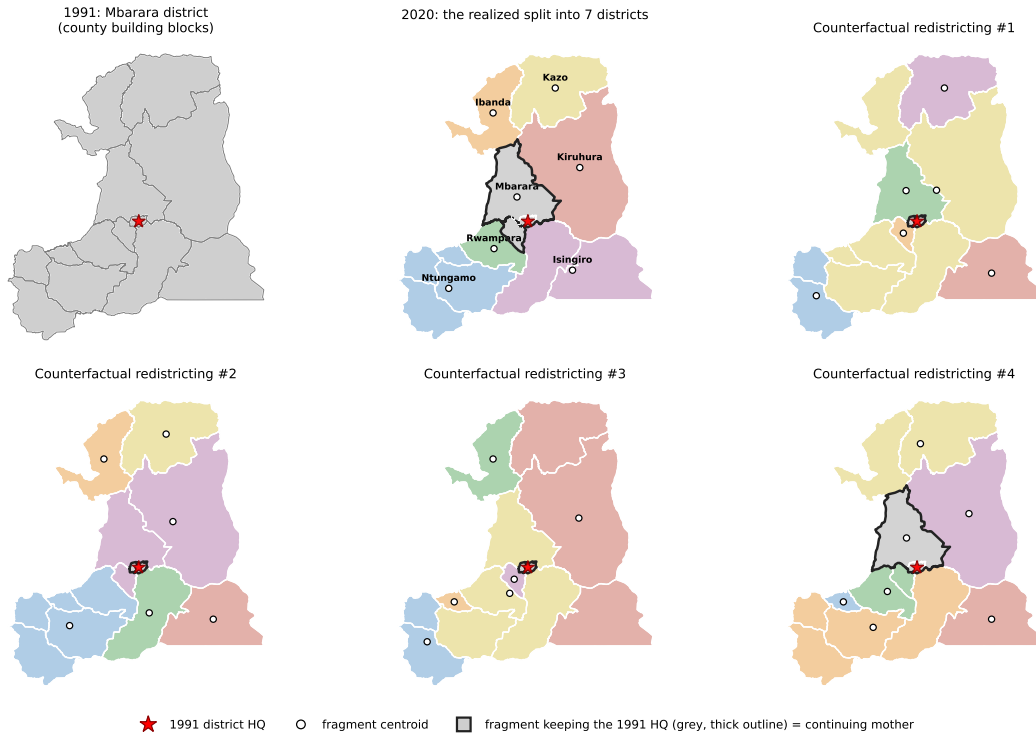


FIGURE A5: Redrawing Mbarara: actual and counterfactual county partitions. The 1991 district (top left) is composed of its county building blocks; the realized 2020 configuration (top centre) splits it into seven districts. The remaining panels show four of the 5,822 feasible contiguous partitions into seven fragments. The fragment retaining the 1991 headquarters (red star) remains the mother district (grey, thick outline); white dots mark fragment centroids, from which the instrument is computed.

Results

Table A12 compares three specifications for each service index: the baseline 2SLS estimate, an expected-instrument-adjusted specification that adds μ_z , μ_{prox} , and μ_{new} as controls,²⁴ and a specification that uses the pure recentered instrument $z - \mu_z$. Adding the expected-instrument controls leaves the results qualitatively unchanged and generally increases the estimated effects. For example, the coefficient rises from 0.43 to 0.80 for electricity, from 0.39 to 0.52 for district health, and from 0.28 to 0.50 for district water. The corresponding first-stage F -statistics range from approximately 10 to 25. These results indicate that the main findings are not driven only by households' predictable geographical exposure to the redistricting process.

The pure recentered instrument, by contrast, has a weak first stage, with Kleibergen–Paap F -statistics of around 4 or below, so I do not interpret its 2SLS magnitudes. A weak first stage

²⁴Three regressors respond to how the mother is partitioned, so each enters with its counterfactual expectation. For a given household these are the average, over all feasible partitions of its mother, of the excluded instrument (μ_z), of proximity to the fragment centroid (μ_{prox}), and of the new-district indicator (μ_{new} , the share of partitions placing the household in a new district rather than the mother rump). Controlling for all three, rather than μ_z alone, prevents predictable geometry from re-entering through the proximity and new-district controls.

also renders Anderson–Rubin confidence sets constructed from the recentered instrument itself uninformative, as they widen toward the entire parameter space. I therefore base inference on two complementary procedures, each robust to weak identification but in a different way. First, I report Anderson–Rubin 95 percent confidence sets for the μ -control specification, whose first stage is strong enough (Kleibergen–Paap F between roughly 10 and 25) for these sets to be bounded and informative while retaining correct size irrespective of instrument strength. They exclude zero for health (district) [0.26, 0.90], health (central government) [0.08, 1.17], water (district) [0.17, 0.97], water (central government) [0.18, 0.88], roads (central government) [0.37, 2.15], and electricity [0.47, 1.45]. Second, for the recentered instrument I conduct randomization inference, whose validity is design-based and therefore does not depend on the strength of the first stage at all. For each of 1,000 counterfactual national redistricting configurations I recompute the reduced-form coefficient of each outcome on the corresponding recentered placebo instrument and compare the realized estimate with this placebo distribution (Figure A6). The realized first-stage configuration is itself an extreme draw, with an omnibus randomization p -value of 0.001. Reading the reduced forms with attention to sign, seven of the ten service indices are both significant and positively signed (randomization p -values at or below 0.02, most at 0.001), matching the main tables, and these seven include all six headline outcomes, namely health and water at both tiers, roads (central government), and electricity. District roads is indistinguishable from the placebo distribution. The remaining two indices are extreme but negatively signed. Agricultural inputs (district) is also negative in the main 2SLS and plausibly reflects that input delivery is not organized around the headquarters town, while schools (central government) is a precise zero in the main specification whose recentered reduced form is unstable given its weak first stage.

Randomization inference: reduced form across 1,000 counterfactual redistricting configurations
(recentered instrument, μ -control specification; orange line = actual estimate)

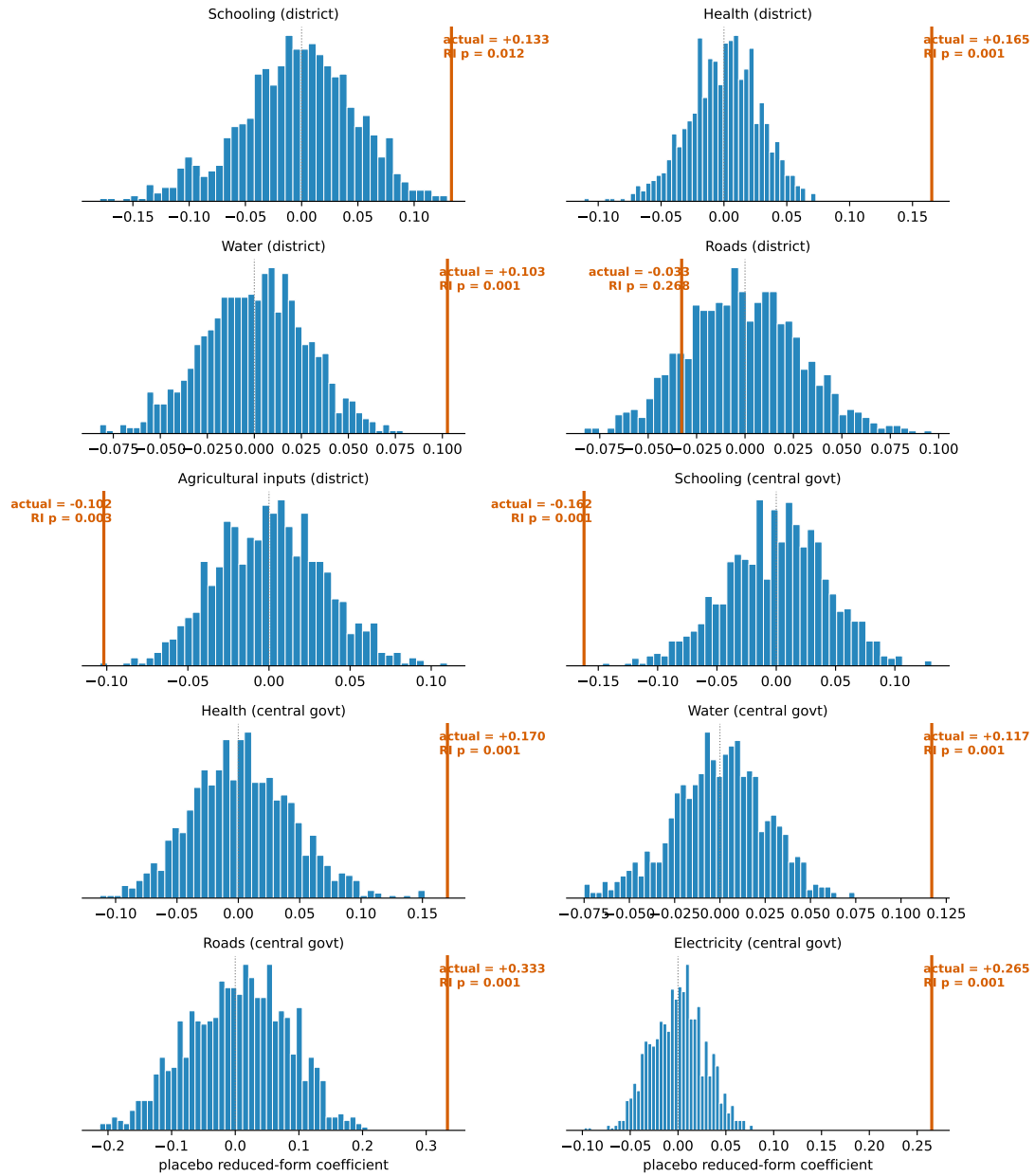


FIGURE A6: Randomization inference. Each panel shows the distribution of reduced-form coefficients on the recentered instrument across 1,000 counterfactual national redistricting configurations (μ -control specification); the orange line marks the estimate under the realized configuration, with its two-sided randomization p-value.

TABLE A12: Recentered Instrument: Effects on Services (2SLS)

	School (1)	Health (2)	Water (3)	Roads (4)	Agr inputs (5)
<i>Panel A: NSDS 2021 District services</i>					
Baseline 2SLS	0.282 (0.214)	0.387*** (0.121)	0.279* (0.157)	0.089 (0.110)	-0.085 (0.145)
2SLS, μ controls	0.464* (0.249)	0.516*** (0.148)	0.496*** (0.176)	-0.012 (0.129)	-0.291 (0.180)
2SLS, recentered IV ($z - \mu$)	1.420 (0.935)	-0.155 (0.349)	1.079** (0.533)	-0.858* (0.498)	-0.048 (0.378)
KP F (baseline)	22.24	33.35	28.75	28.75	28.75
KP F (μ controls)	17.58	24.51	17.46	17.46	17.46
KP F (recentered)	2.79	4.44	4.17	4.17	4.17
RI p (μ controls, $S=1,000$)	[0.002]	[0.001]	[0.001]	[0.081]	[0.001]
AR 95% set (μ controls)	[-0.00, 1.11]	[0.26, 0.90]	[0.17, 0.97]	[-0.31, 0.26]	[-0.75, 0.06]
Observations	7,137	7,595	8,417	8,417	8,417
	School (1)	Health (2)	Water (3)	Roads (4)	Electricity (5)
<i>Panel B: NSDS 2021 Central govt services</i>					
Baseline 2SLS	0.151 (0.192)	0.233 (0.201)	0.372** (0.146)	0.706*** (0.261)	0.428*** (0.137)
2SLS, μ controls	0.058 (0.261)	0.573** (0.244)	0.449*** (0.154)	0.970** (0.377)	0.795*** (0.205)
2SLS, recentered IV ($z - \mu$)	3.454 (4.219)	0.048 (0.479)	0.620* (0.345)	2.146* (1.222)	0.572 (0.387)
KP F (baseline)	22.62	28.75	28.75	28.75	28.75
KP F (μ controls)	9.86	17.46	17.46	17.46	17.46
KP F (recentered)	0.67	4.17	4.17	4.17	4.17
RI p (μ controls, $S=1,000$)	[0.020]	[0.001]	[0.001]	[0.001]	[0.001]
AR 95% set (μ controls)	[-0.60, 0.68]	[0.08, 1.17]	[0.18, 0.88]	[0.37, 2.15]	[0.47, 1.45]
Observations	5,553	8,417	8,417	8,417	8,417

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ (cluster-robust). Borusyak–Hull (2023) recentering exercise. The expected instrument μ_i is the exact mean of the instrument over all feasible contiguous re-partitions of each splitting 1991 mother district into its realized number of 2020 districts, holding fixed which mothers split and into how many units; building blocks are 1991-vintage counties (IPUMS), and every mother’s feasible set is enumerated exhaustively. Column blocks: the paper’s 2SLS specification; adding ($\mu_z, \mu_{prox}, \mu_{new}$) as controls; and the recentered excluded instrument $z - \mu$. All regressions use the paper’s controls, weights, and clustering (NSDS: EA; AB: town/village); Kampala and Wakiso excluded as in the main tables. RI p : two-sided randomization-inference p-value of the reduced form on $z - \mu$ across $S=1,000$ national placebo redistricting configurations (one independent draw per mother per configuration, seed 20260706); first-stage omnibus RI $p = 0.001$ (NSDS) and 0.003 (AB). AR 95% set: weak-instrument-robust Anderson–Rubin confidence set (cluster-robust). Recentered-column magnitudes are estimated off within-mother reassignment variation with a different complier mix and should not be compared to baseline magnitudes.

G Additional results

TABLE A13: Service Availability and Proximity to District HQ (2SLS)

	School (1)	Health clinic (2)	Piped water (3)	Electricity (4)	Paved road (5)	Police station (6)
<i>Panel A: Afrobarometer R10</i>						
Log Proximity to District HQ	0.006 (0.165)	0.166 (0.201)	0.495** (0.224)	0.034 (0.203)	0.358** (0.162)	0.504** (0.209)
Observations	2184	2168	2168	2176	2168	2176
Number of clusters	271	269	269	271	269	270
Mean of DV	0.788	0.613	0.306	0.540	0.214	0.312
IV F-stat (KP rk Wald)	19.446	19.883	17.858	19.591	19.418	19.464

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of citizens' access to public goods and services on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

TABLE A14: Schooling and Proximity to District Headquarters (2SLS)

	School attendance (1)	Distance to school (2)	Travel time (3)	Teacher quality (4)	Facility quality (5)	Audited school (6)
<i>Panel A: District services</i>						
Log Proximity to District HQ	-0.128* (0.078)	-1.165** (0.536)	-19.130*** (6.807)	0.021 (0.067)	0.046 (0.065)	0.343 (0.267)
Observations	4566	3329	3329	4563	4564	5581
Number of clusters	951	916	916	951	951	638
Mean of DV	0.714	1.710	31.421	0.804	0.809	0.559
IV F-stat (KP rk Wald)	26.298	22.676	22.676	26.298	26.244	9.399
	School attendance (1)	Distance to school (2)	Travel time (3)	Teacher quality (4)	Facility quality (5)	Audited school (6)
<i>Panel B: Central govt services</i>						
Log Proximity to District HQ	-0.169 (0.105)	-0.606 (0.689)	-10.457 (7.829)	0.104 (0.087)	0.067 (0.082)	0.127 (0.225)
Observations	3546	2132	2132	3539	3541	3659
Number of clusters	943	827	827	943	943	420
Mean of DV	0.700	2.007	34.856	0.769	0.780	0.695
IV F-stat (KP rk Wald)	20.727	16.442	16.442	20.948	20.697	15.054

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of citizens' access to schooling services on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis. Column 6 of each panel is based on information from the community questionnaire, collected once per enumeration area and assigned to all households in the enumeration area; observation counts refer to households.

TABLE A15: Health Services and Proximity to District Headquarters (2SLS)

	Public hospital (1)	Injury (2)	Injury travel time (3)	Health facility score (4)	HC absenteeism (5)	HC ambulance (6)	District HC supervision (7)
<i>Panel A: District services</i>							
Log Proximity to District HQ	-0.013 (0.090)	-0.040 (0.034)	-14.329* (8.670)	29.451*** (9.847)	-0.265* (0.138)	0.265* (0.160)	0.162 (0.161)
Observations	2291	8415	3252	6948	6821	6909	6619
Number of clusters	826	967	869	791	778	786	750
Mean of DV	0.865	0.146	36.098	26.347	0.215	0.303	0.787
IV F-stat (KP rk Wald)	26.663	28.803	23.237	33.461	30.568	30.178	29.761
	HC type 4 (1)	Distance gov hospital (2)	PHC insufficient (3)				
<i>Panel B: Central govt services</i>							
Log Proximity to District HQ	0.146** (0.066)	0.049 (0.099)	-0.029 (0.157)				
Observations	6979	8417	6819				
Number of clusters	795	967	776				
Mean of DV	0.075	0.240	0.233				
IV F-stat (KP rk Wald)	33.260	28.745	32.624				

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of citizens' access to health services on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis. Columns 4–7 of Panel A and all columns of Panel B are based on information from the community questionnaire, collected once per enumeration area and assigned to all households in the enumeration area; observation counts refer to households.

TABLE A16: Water Services and Proximity to District Headquarters (2SLS)

	Water source dry (1)	Water distance dry (2)	Water source wet (3)	Piped water (4)	Borehole (5)	Protected well (6)			
<i>Panel A: District services</i>									
Log Proximity to District HQ	0.128 (0.111)	-0.225 (0.194)	-0.029 (0.089)	0.212 (0.136)	0.184 (0.158)	0.293 (0.188)			
Observations	8417	7229	8417	8371	8371	8374			
Number of clusters	967	952	967	961	961	961			
Mean of DV	0.834	0.786	0.889	0.331	0.636	0.302			
IV F-stat (KP rk Wald)	28.745	30.837	28.745	28.094	28.094	27.880			
	Water src dry outside (1)	Water dist dry outside (2)	Water src wet outside (3)	Water dist wet outside (4)	WAP crops (5)	WAP cattle (6)	WAP fish (7)		
<i>Panel B: Central govt services</i>									
Log Proximity to District HQ	0.145 (0.110)	-0.141 (0.228)	0.039 (0.075)	-0.065 (0.166)	0.042 (0.030)	0.084** (0.042)	0.123* (0.063)		
Observations	8417	7177	8416	5603	7815	8356	7762		
Number of clusters	967	949	967	894	901	959	895		
Mean of DV	0.784	0.814	0.855	0.746	0.993	0.969	0.961		
IV F-stat (KP rk Wald)	28.745	28.853	28.682	28.733	25.727	27.845	25.966		

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of citizens' access to water services on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis. Columns 4–6 of Panel A and Columns 5–7 of Panel B are based on information from the community questionnaire, collected once per enumeration area and assigned to all households in the enumeration area; observation counts refer to households.

TABLE A17: Road Infrastructure and Proximity to District Headquarters (2SLS)

	Trunk road (1)	Feeder road (2)	Communal road (3)	Murram road quality (4)	Feeder road quality (5)	Communal road quality (6)	Murram maintenance (7)	Feeder maintenance (8)	Communal maintenance (9)
<i>Panel A: District services</i>									
Log Proximity to District HQ	-0.119 (0.110)	0.058 (0.062)	-0.134** (0.058)	-0.005 (0.152)	0.149 (0.143)	0.268** (0.117)	-0.314** (0.138)	0.075 (0.143)	-0.034 (0.052)
Observations	8416	8416	8415	6220	7814	7758	8336	8364	8372
Number of clusters	967	967	967	708	894	888	957	960	961
Mean of DV	0.634	0.883	0.914	0.288	0.205	0.099	0.157	0.751	0.026
IV F-stat (KP rk Wald)	28.702	28.702	28.674	21.068	27.638	27.202	27.662	27.655	27.642
	Trunk road (1)	Tarmac road quality (2)	Tarmac maintenance (3)						
<i>Panel B: Central govt services</i>									
Log Proximity to District HQ	0.262** (0.118)	0.672** (0.317)	0.117 (0.146)						
Observations	8416	6236	8362						
Number of clusters	967	721	960						
Mean of DV	0.436	0.775	0.866						
IV F-stat (KP rk Wald)	28.702	13.092	28.125						

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of citizens' access to roads on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis. Columns 4-9 of Panel A and Columns 2-3 of Panel B are based on information from the community questionnaire, collected once per enumeration area and assigned to all households in the enumeration area; observation counts refer to households.

TABLE A18: Agricultural Inputs (District) and Proximity to District Headquarters (2SLS)

	Hybrid seeds (1)	Planting material (2)	Pesticides (3)	Fertilizer (4)
<i>Panel A: Inputs Availability</i>				
Log Proximity to District HQ	-0.080 (0.073)	0.005 (0.096)	-0.125 (0.094)	-0.070 (0.072)
Observations	8417	8417	8417	8417
Number of clusters	967	967	967	967
Mean of DV	0.205	0.325	0.238	0.106
IV F-stat (KP rk Wald)	28.745	28.745	28.745	28.745
	Hybrid seeds (1)	Planting material (2)	Pesticides (3)	Fertilizer (4)
<i>Panel B: Inputs Use</i>				
Log Proximity to District HQ	-0.115** (0.059)	0.108 (0.101)	0.043 (0.062)	0.006 (0.044)
Observations	8417	8417	8417	8417
Number of clusters	967	967	967	967
Mean of DV	0.137	0.351	0.151	0.053
IV F-stat (KP rk Wald)	28.745	28.745	28.745	28.745

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Coefficients from regressions of the reported availability (Panel A) and household use (Panel B) of agricultural inputs on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

TABLE A19: Electricity (Central Government) and Proximity to District Headquarters (2SLS)

	HH electricity (1)	Connected grid (2)	Interested grid (3)	Electricity availability (4)
<i>Panel A: Central govt services</i>				
Log Proximity to District HQ	0.276*** (0.101)	0.197** (0.091)	0.009 (0.037)	0.237* (0.127)
Observations	8417	8417	8417	8417
Number of clusters	967	967	967	967
Mean of DV	0.463	0.175	0.055	0.824
IV F-stat (KP rk Wald)	28.745	28.745	28.745	28.745

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of citizens' access to electricity on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis. Column 4 is based on information from the community questionnaire, collected once per enumeration area and assigned to all households in the enumeration area; observation counts refer to households.

TABLE A20: Community Top Priorities and Proximity to District HQ (2SLS)

	New school (1)	Classroom (2)	Health facility (3)	Water (4)	Electricity (5)	Roads bridges (6)
<i>Panel A: NSDS 2021</i>						
Log Proximity to District HQ	0.662*** (0.200)	0.237 (0.159)	0.133 (0.191)	0.254 (0.184)	0.008 (0.238)	-0.136 (0.294)
Observations	5200	4435	5675	6477	5353	3808
Number of clusters	590	511	647	737	612	432
Mean of DV	0.559	0.709	0.504	0.360	0.502	0.539
IV F-stat (KP rk Wald)	23.016	22.200	25.330	21.934	13.846	9.744

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of indicators equal to one if the household's community received a public investment in a sector the household ranks among its top three priorities on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis. All outcomes are based on information from the community questionnaire, collected once per enumeration area and assigned to all households in the enumeration area; observation counts refer to households.

TABLE A21: Trust in Others and Social Capital and Proximity to District HQ (2SLS)

	Trust relatives (1)	Trust neighbours (2)	Trust citizens (3)	Trust other religion (4)	Trust other ethnicity (5)
<i>Panel A: Afrobarometer R10</i>					
Log Proximity to District HQ	-0.013 (0.077)	0.058 (0.089)	0.131 (0.108)	0.192* (0.109)	0.257** (0.126)
Observations	2176	2179	2177	2179	2167
Number of clusters	271	271	271	271	271
Mean of DV	0.800	0.652	0.537	0.587	0.503
IV F-stat (KP rk Wald)	19.564	19.299	19.038	19.340	19.179
	Empowerment (1)	Collective action (2)	Bridging groups (3)	General trust (4)	
<i>Panel B: NSDS 2021</i>					
Log Proximity to District HQ	0.084 (0.074)	0.179** (0.080)	0.055 (0.047)	-0.050 (0.075)	
Observations	8416	8416	4334	8416	
Number of clusters	967	967	926	967	
Mean of DV	0.637	0.591	0.080	0.441	
IV F-stat (KP rk Wald)	28.702	28.702	26.874	28.702	

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of citizens' trust in others and social capital on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

TABLE A22: Trust in Government and Proximity to District HQ: 2017 Afrobarometer (2SLS)

	Trust localgov (1)	Trust president (2)	Trust parliament (3)	Trust ruling party (4)	Trust opposition (5)	Trust courts (6)
<i>Panel A: Afrobarometer R7 (2017)</i>						
Log Proximity to District HQ	-0.085 (0.100)	0.018 (0.099)	-0.142 (0.106)	0.047 (0.083)	-0.092 (0.130)	-0.025 (0.080)
Observations	995	1014	1009	1025	995	987
Number of clusters	129	129	129	129	129	129
Mean of DV	0.587	0.711	0.554	0.642	0.361	0.626
IV F-stat (KP rk Wald)	10.770	10.062	9.688	10.029	10.004	10.106

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of trust in government variables from the 2017 Afrobarometer (Round 7) on self-reported household's distance to the district's capital. Each variable indicates that the respondent trusts the corresponding institution somewhat or a lot. All regressions control for district ever split, lake Victoria district, districts' area in Km² as of 1991, household's distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of the household's distance to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the enumeration area (town/village) level are in parenthesis. The first-stage Kleibergen-Paap LM test is close to the conventional threshold of 10, so magnitudes should be read with caution; the estimates are reported to show that the absence of trust effects documented in the 2024 Afrobarometer holds in 2017.

TABLE A23: Service Availability and Proximity to District HQ Across Afrobarometer Rounds (2SLS)

	School (1)	Health clinic (2)	Piped water (3)	Electricity grid (4)	Paved road (5)	Police station (6)
<i>Panel A: Afrobarometer R7 (2017)</i>						
Log Proximity to District HQ	-0.070 (0.174)	-0.293 (0.242)	0.268* (0.153)	0.155 (0.228)	-0.105 (0.285)	-0.059 (0.260)
Observations	1048	1040	1048	1048	1048	1048
Number of clusters	129	128	129	129	129	129
Mean of DV	0.809	0.692	0.191	0.275	0.489	0.366
IV F-stat (KP rk Wald)	9.621	9.806	9.621	9.621	9.621	9.621
	School (1)	Health clinic (2)	Piped water (3)	Electricity grid (4)	Paved road (5)	Police station (6)
<i>Panel B: Afrobarometer R10 (2024)</i>						
Log Proximity to District HQ	0.006 (0.165)	0.166 (0.201)	0.495** (0.224)	0.034 (0.203)	0.358** (0.162)	0.504** (0.209)
Observations	2184	2168	2168	2176	2168	2176
Number of clusters	271	269	269	271	269	270
Mean of DV	0.788	0.613	0.306	0.540	0.214	0.312
IV F-stat (KP rk Wald)	19.446	19.883	17.858	19.591	19.418	19.464

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Coefficients from regressions of enumeration-area service availability indicators, recorded by Afrobarometer e (not self-reported), on self-reported household's distance to the district's capital. Outcomes are identical across the two rounds and indicate whether the e area has a school, a health clinic, a piped water system, an electricity grid, a paved road surface over the last five kilometers, and a police station. All control for district ever split, lake Victoria district, districts' area in Km^2 as of 1991, household's distance to district's capital as of 1991 (in log.), distr a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the distr is the interaction of the household's distance to the district's centroid and an indicator for new districts, defined over 1991-2016 with 2017 boundari A and over 1991-2020 with 2020 boundaries in Panel B. Standard errors clustered at the enumeration area (town/village) level are in parenthesis. The Kleibergen-Paap F statistic in the 2017 round is close to the conventional threshold of 10, so Panel A magnitudes should be read with caution.

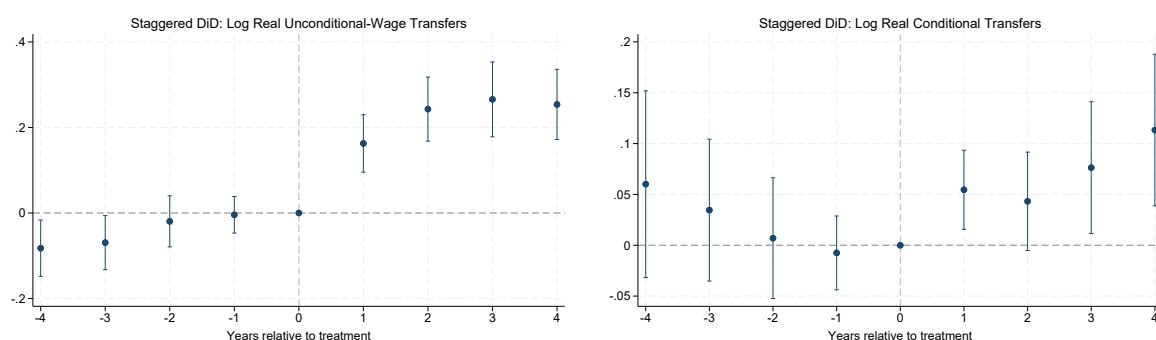


FIGURE A7: Event study: composition of the territory transfer response. Same specification and sample as Figure 5, for the wage component of the unconditional block grant (left, the duplicated administrative payroll) and conditional service-delivery grants (right), in log real FY2017/18 shillings. The joint placebo test has a p-value of 0.02 for the unconditional-wage component (with individually small placebo estimates) and 0.34 for conditional grants.

TABLE A24: Effects on Services and Proximity to District HQ (2SLS) – Aligned Districts

	School (1)	Health (2)	Water (3)	Roads (4)	Agr inputs (5)
<i>Panel A: NSDS 2021 District services</i>					
Log Proximity to District HQ	0.413 (0.307)	0.324* (0.167)	0.187 (0.218)	0.101 (0.157)	0.105 (0.195)
Observations	5291	5639	6266	6266	6266
Number of clusters	708	701	714	714	714
Mean of DV	0.004	-0.004	-0.015	-0.006	0.034
IV F-stat (KP rk Wald)	9.894	16.988	14.417	14.417	14.417
	School (1)	Health (2)	Water (3)	Roads (4)	Electricity (5)
<i>Panel B: NSDS 2021 Central govt services</i>					
Log Proximity to District HQ	-0.026 (0.309)	0.157 (0.268)	0.241 (0.184)	1.013*** (0.384)	0.424** (0.193)
Observations	3999	6266	6266	6266	6266
Number of clusters	703	714	714	714	714
Mean of DV	-0.012	0.006	-0.010	-0.074	-0.049
IV F-stat (KP rk Wald)	9.404	14.417	14.417	14.417	14.417

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Coefficients from regressions of outcome index variables on self-reported household's distance to the district's capital. The indices are constructed to have a mean of 0 and a standard deviation of 1. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis. Unadjusted p-values are in brackets. Anderson's sharpened False Discovery Rate (FDR) q-values are in curly brackets.

TABLE A25: Citizen Engagement and Proximity to District HQ (2SLS) – Aligned Districts

	Involved LC5 (1)	Met RDC (2)	RDC Baraza (3)
<i>Panel A: NSDS 2021</i>			
Log Proximity to District HQ	0.179* (0.097)	0.072 (0.066)	0.065 (0.042)
Observations	6266	6266	6266
Number of clusters	714	714	714
Mean of DV	0.241	0.105	0.051
IV F-stat (KP rk Wald)	14.417	14.417	14.417
	Collective civic (1)	Protest (2)	Contacted media (3)
<i>Panel B: Afrobarometer R10</i>			
Log Proximity to District HQ	-0.498** (0.249)	-0.164** (0.076)	-0.195* (0.108)
Observations	1782	1784	1784
Number of clusters	221	221	221
Mean of DV	0.362	0.037	0.091
IV F-stat (KP rk Wald)	6.391	6.408	6.408

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of citizens' involvement with the government variables on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

TABLE A26: Institutional Knowledge and State-Traditional Substitution (2SLS) – Aligned Districts

	Knows magistrate court (1)	Knows police (2)	Knows land office (3)	Knows OAG (4)	Knows judiciary (5)
<i>Panel A: NSDS 2021</i>					
Log Proximity to District HQ	0.385*** (0.123)	0.120 (0.077)	0.119 (0.086)	0.052 (0.050)	0.149 (0.102)
Observations	6263	6264	6263	6266	6266
Number of clusters	714	714	714	714	714
Mean of DV	0.369	0.921	0.160	0.096	0.386
IV F-stat (KP rk Wald)	14.326	14.339	14.326	14.417	14.417
	Family conflict (1)	Resolved by govt (2)	Resolved by trad (3)	Contacted MP (4)	Contacted trad leader (5)
<i>Panel B: NSDS 2021 and Afrobarometer R10</i>					
Log Proximity to District HQ	0.086** (0.040)	0.007 (0.224)	-0.158 (0.195)	0.168 (0.125)	-0.275 (0.199)
Observations	6264	1739	1739	1781	1748
Number of clusters	714	221	221	221	221
Mean of DV	0.056	0.553	0.217	0.166	0.256
IV F-stat (KP rk Wald)	14.368	6.267	6.267	6.426	6.432

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of state/traditional institutions variables on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

TABLE A27: Satisfaction with Local Leaders and Proximity to District HQ (2SLS) – Aligned Districts

	LCV performance (1)	RDC satisfaction (2)	Civil servants performance (3)
<i>Panel A: NSDS 2021</i>			
Log Proximity to District HQ	0.055 (0.101)	0.208** (0.086)	0.108 (0.120)
Observations	6266	6266	5601
Number of clusters	714	714	713
Mean of DV	0.421	0.317	0.534
IV F-stat (KP rk Wald)	14.417	14.417	14.714

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of satisfaction with local leaders' variables on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

TABLE A28: Corruption and Proximity to District HQ (2SLS) – Aligned Districts

	Nepotism (1)	Embezzlement (2)	Absenteeism (3)	Victim nepotism (4)	Victim embezzlement (5)	Victim absenteeism (6)
<i>Panel A: NSDS 2021</i>						
Log Proximity to District HQ	0.221* (0.117)	0.170 (0.106)	0.063 (0.112)	-0.047 (0.030)	-0.011 (0.036)	-0.061 (0.055)
Observations	6263	6263	6263	6266	6266	6266
Number of clusters	714	714	714	714	714	714
Mean of DV	0.501	0.653	0.487	0.034	0.034	0.069
IV F-stat (KP rk Wald)	14.326	14.326	14.326	14.417	14.417	14.417
	Bribe health (1)	Bribe police (2)	Bribe avoid police (3)	Bribe ID (4)		
<i>Panel B: Afrobarometer R10</i>						
Log Proximity to District HQ	-0.483* (0.248)	-0.398 (0.385)	-0.115 (0.309)	0.284 (0.243)		
Observations	1308	277	747	311		
Number of clusters	221	156	208	167		
Mean of DV	0.398	0.740	0.497	0.492		
IV F-stat (KP rk Wald)	6.454	6.821	2.851	6.645		

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of corruption variables on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

TABLE A29: Democratic Perceptions and Proximity to District HQ (2SLS) – Aligned Districts

	Is democracy (1)	Satisf democracy (2)	Free vote choice (3)	Free join orgs (4)		
<i>Panel A: Democratic perceptions</i>						
Log Proximity to District HQ	-0.015 (0.089)	0.138 (0.161)	0.047 (0.075)	-0.099 (0.102)		
Observations	1701	1724	1765	1756		
Number of clusters	221	221	221	221		
Mean of DV	0.876	0.601	0.933	0.847		
IV F-stat (KP rk Wald)	6.356	6.400	7.134	7.767		
	President ignores law (1)	Unequal law (2)	Judges influenced (3)	Officials impunity (4)		
<i>Panel B: Rule of law perceptions</i>						
Log Proximity to District HQ	-0.176 (0.169)	0.108 (0.150)	-0.208 (0.201)	-0.087 (0.184)		
Observations	1511	1729	1613	1691		
Number of clusters	221	221	221	221		
Mean of DV	0.382	0.652	0.617	0.623		
IV F-stat (KP rk Wald)	8.769	7.373	6.714	6.818		
	Trust localgov (1)	Trust president (2)	Trust parliament (3)	Trust ruling party (4)	Trust opposition (5)	Trust courts (6)
<i>Panel C: Trust in government</i>						
Log Proximity to District HQ	0.121 (0.123)	-0.050 (0.146)	0.023 (0.155)	0.146 (0.137)	0.002 (0.146)	0.032 (0.144)
Observations	1737	1727	1784	1784	1784	1701
Number of clusters	221	221	221	221	221	221
Mean of DV	0.507	0.699	0.496	0.591	0.308	0.515
IV F-stat (KP rk Wald)	6.989	5.976	6.408	6.408	6.408	7.002

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of democratic perception variables on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

TABLE A30: Effects on Services and Proximity to District HQ (2SLS) – Non-Aligned Districts

	School (1)	Health (2)	Water (3)	Roads (4)	Agr inputs (5)
<i>Panel A: NSDS 2021 District services</i>					
Log Proximity to District HQ	-0.018 (0.275)	0.383** (0.157)	0.308 (0.203)	0.007 (0.150)	-0.277* (0.165)
Observations	1846	1956	2151	2151	2151
Number of clusters	250	250	253	253	253
Mean of DV	-0.023	-0.051	0.028	0.048	0.018
IV F-stat (KP rk Wald)	16.678	11.809	12.493	12.493	12.493
	School (1)	Health (2)	Water (3)	Roads (4)	Electricity (5)
<i>Panel B: NSDS 2021 Central govt services</i>					
Log Proximity to District HQ	-0.026 (0.244)	0.070 (0.344)	0.337 (0.213)	0.202 (0.274)	0.254 (0.167)
Observations	1554	2151	2151	2151	2151
Number of clusters	248	253	253	253	253
Mean of DV	0.025	0.034	-0.036	-0.058	-0.023
IV F-stat (KP rk Wald)	9.695	12.493	12.493	12.493	12.493

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Coefficients from regressions of outcome index variables on self-reported household's distance to the district's capital. The indices are constructed to have a mean of 0 and a standard deviation of 1. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis. Unadjusted p-values are in brackets. Anderson's sharpened False Discovery Rate (FDR) q-values are in curly brackets.

TABLE A31: Citizen Engagement and Proximity to District HQ (2SLS) – Non-Aligned Districts

	Involved LC5 (1)	Met RDC (2)	RDC Baraza (3)
<i>Panel A: NSDS 2021</i>			
Log Proximity to District HQ	-0.011 (0.083)	0.010 (0.043)	0.000 (0.040)
Observations	2151	2151	2151
Number of clusters	253	253	253
Mean of DV	0.294	0.093	0.055
IV F-stat (KP rk Wald)	12.493	12.493	12.493
	Collective civic (1)	Protest (2)	Contacted media (3)
<i>Panel B: Afrobarometer R10</i>			
Log Proximity to District HQ	0.058 (0.059)	0.024 (0.026)	-0.025 (0.049)
Observations	400	400	400
Number of clusters	50	50	50
Mean of DV	0.350	0.035	0.100
IV F-stat (KP rk Wald)	19.793	19.793	19.793

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Coefficients from regressions of citizens' involvement with the government variables on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

TABLE A32: Institutional Knowledge and State-Traditional Substitution (2SLS) – Non-Aligned Districts

	Knows magistrate court (1)	Knows police (2)	Knows land office (3)	Knows OAG (4)	Knows judiciary (5)
<i>Panel A: NSDS 2021</i>					
Log Proximity to District HQ	0.010 (0.120)	-0.054 (0.041)	0.032 (0.062)	0.125** (0.061)	-0.039 (0.078)
Observations	2150	2151	2150	2151	2151
Number of clusters	253	253	253	253	253
Mean of DV	0.392	0.956	0.142	0.079	0.377
IV F-stat (KP rk Wald)	12.440	12.493	12.440	12.493	12.493
	Family conflict (1)	Resolved by govt (2)	Resolved by trad (3)	Contacted MP (4)	Contacted trad leader (5)
<i>Panel B: NSDS 2021 and Afrobarometer R10</i>					
Log Proximity to District HQ	0.044 (0.059)	0.152* (0.085)	-0.011 (0.050)	0.043 (0.073)	-0.011 (0.073)
Observations	2150	392	392	397	398
Number of clusters	253	50	50	50	50
Mean of DV	0.060	0.571	0.184	0.174	0.369
IV F-stat (KP rk Wald)	12.440	20.733	20.733	19.764	19.820

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of state/traditional institutions variables on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

TABLE A33: Satisfaction with Local Leaders and Proximity to District HQ (2SLS) – Non-Aligned Districts

	LCV performance (1)	RDC satisfaction (2)	Civil servants performance (3)
<i>Panel A: NSDS 2021</i>			
Log Proximity to District HQ	0.237* (0.130)	-0.056 (0.084)	0.062 (0.107)
Observations	2151	2151	1850
Number of clusters	253	253	253
Mean of DV	0.450	0.364	0.586
IV F-stat (KP rk Wald)	12.493	12.493	12.557

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of satisfaction with local leaders' variables on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

TABLE A34: Corruption and Proximity to District HQ (2SLS) – Non-Aligned Districts

	Nepotism (1)	Embezzlement (2)	Absenteeism (3)	Victim nepotism (4)	Victim embezzlement (5)	Victim absenteeism (6)
<i>Panel A: NSDS 2021</i>						
Log Proximity to District HQ	-0.225** (0.104)	-0.138 (0.113)	-0.106 (0.094)	0.002 (0.022)	-0.080* (0.043)	-0.108 (0.083)
Observations	2149	2149	2150	2151	2151	2151
Number of clusters	253	253	253	253	253	253
Mean of DV	0.475	0.629	0.490	0.035	0.050	0.092
IV F-stat (KP rk Wald)	12.454	12.454	12.507	12.493	12.493	12.493
	Bribe health (1)	Bribe police (2)	Bribe avoid police (3)	Bribe ID (4)		
<i>Panel B: Afrobarometer R10</i>						
Log Proximity to District HQ	-0.019 (0.066)	-0.299** (0.121)	0.034 (0.112)	-0.034 (0.141)		
Observations	296	68	157	85		
Number of clusters	50	40	49	41		
Mean of DV	0.459	0.897	0.650	0.471		
IV F-stat (KP rk Wald)	26.108	16.760	15.963	12.900		

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of corruption variables on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

TABLE A35: Democratic Perceptions and Proximity to District HQ (2SLS) – Non-Aligned Districts

	Is democracy (1)	Satisf democracy (2)	Free vote choice (3)	Free join orgs (4)		
<i>Panel A: Democratic perceptions</i>						
Log Proximity to District HQ	-0.002 (0.038)	0.031 (0.063)	-0.113*** (0.032)	-0.033 (0.050)		
Observations	378	384	394	395		
Number of clusters	50	50	50	50		
Mean of DV	0.907	0.560	0.942	0.848		
IV F-stat (KP rk Wald)	21.059	21.506	18.921	19.641		
	President ignores law (1)	Unequal law (2)	Judges influenced (3)	Officials impunity (4)		
<i>Panel B: Rule of law perceptions</i>						
Log Proximity to District HQ	0.151** (0.067)	-0.113 (0.089)	-0.014 (0.086)	-0.014 (0.088)		
Observations	348	383	367	384		
Number of clusters	50	50	50	50		
Mean of DV	0.299	0.585	0.526	0.536		
IV F-stat (KP rk Wald)	16.684	18.516	17.712	19.476		
	Trust localgov (1)	Trust president (2)	Trust parliament (3)	Trust ruling party (4)	Trust opposition (5)	Trust courts (6)
<i>Panel C: Trust in government</i>						
Log Proximity to District HQ	-0.129* (0.077)	-0.083 (0.067)	0.012 (0.079)	-0.126* (0.074)	0.079 (0.059)	0.091 (0.056)
Observations	395	389	400	400	400	386
Number of clusters	50	50	50	50	50	50
Mean of DV	0.506	0.676	0.535	0.583	0.345	0.573
IV F-stat (KP rk Wald)	20.841	20.031	19.793	19.793	19.793	20.388

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of democratic perception variables on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

TABLE A36: Non-zero-sum fiscal accounting around district splits

	Own pre-split counterfactual	Trend-adjusted counterfactual	Matched-trend counterfactual
<i>Panel A. Net extra spending across the territory, $(M_{post} + C_{post}) - \widehat{M}_{cf}$</i>			
Mean (billions UGX)	10.11	5.37	5.60
Median (billions UGX)	7.39	2.83	3.00
Splits with positive excess	20/24	19/24	19/24
<i>Panel B. As percent of the mother's pre-split budget</i>			
Mean (%)	35.26	19.21	20.42
Median (%)	28.34	11.84	9.99
Standard deviation (%)	37.98	30.59	32.12
<i>Panel C. Underlying real budget levels (billions of FY2017/18 UGX)</i>			
Mother, pre-split (M_{pre}), median		31.05	
Mother, post-split (M_{post}), median		23.01	
Child, first two years (C_{post}), median		14.38	

Notes: The table asks whether the spending a mother district loses when it splits is recovered elsewhere, or whether the split raises total spending across the original territory. The sample is the 24 districts created within the panel window (calendar 2015–2020; approved local-government transfers FY2012/13–FY2023/24), and all amounts are in real FY2017/18 shillings. For each split at creation year T , M_{pre} is the mother's average budget over the two fiscal years ending at T , M_{post} its average over $(T+1, T+2)$, and C_{post} the child's average over its first two years. The excess spending generated by the split is $(M_{post} + C_{post}) - \widehat{M}_{cf}$, the post-split total across mother and child minus the budget the undivided territory would have had, $\widehat{M}_{cf}$. The three columns differ only in that counterfactual. Column (1) sets $\widehat{M}_{cf} = M_{pre}$, asking whether post-split spending exceeds the mother's own pre-split level. Because all local budgets grew over the period, columns (2) and (3) instead let the counterfactual grow at the pace of comparable districts that never split: $\widehat{M}_{cf} = M_{pre} \times g$, where g is the median real growth factor over the same windows of the 19 territories with no split during the panel window (column 2), or of those same territories in the mother's own quartile of pre-split budget (column 3). Kampala, Wakiso, and municipal and city councils are excluded throughout. Both trend-adjusted counterfactuals leave a positive excess in 19 of the 24 splits and an average net cost near 20 percent of the mother's pre-split budget, indicating that district creation raises total spending rather than redistributing a fixed envelope.